WaveFEM

SOLVER VALIDATION

Validation Report

WaveFEM compared against Ansys HFSS on S-parameters, far-field patterns, adaptive mesh convergence and runtime.

19 examples · Generated 02 November 2026

INTRODUCTION

About WaveFEM

WaveFEM is a three-dimensional electromagnetic simulator based on the finite element method (FEM). It is built for microwave and RF design: filters, waveguide components and antennas. Everything runs from one desktop application, from importing geometry to plotting fields, S-parameters and radiation patterns.

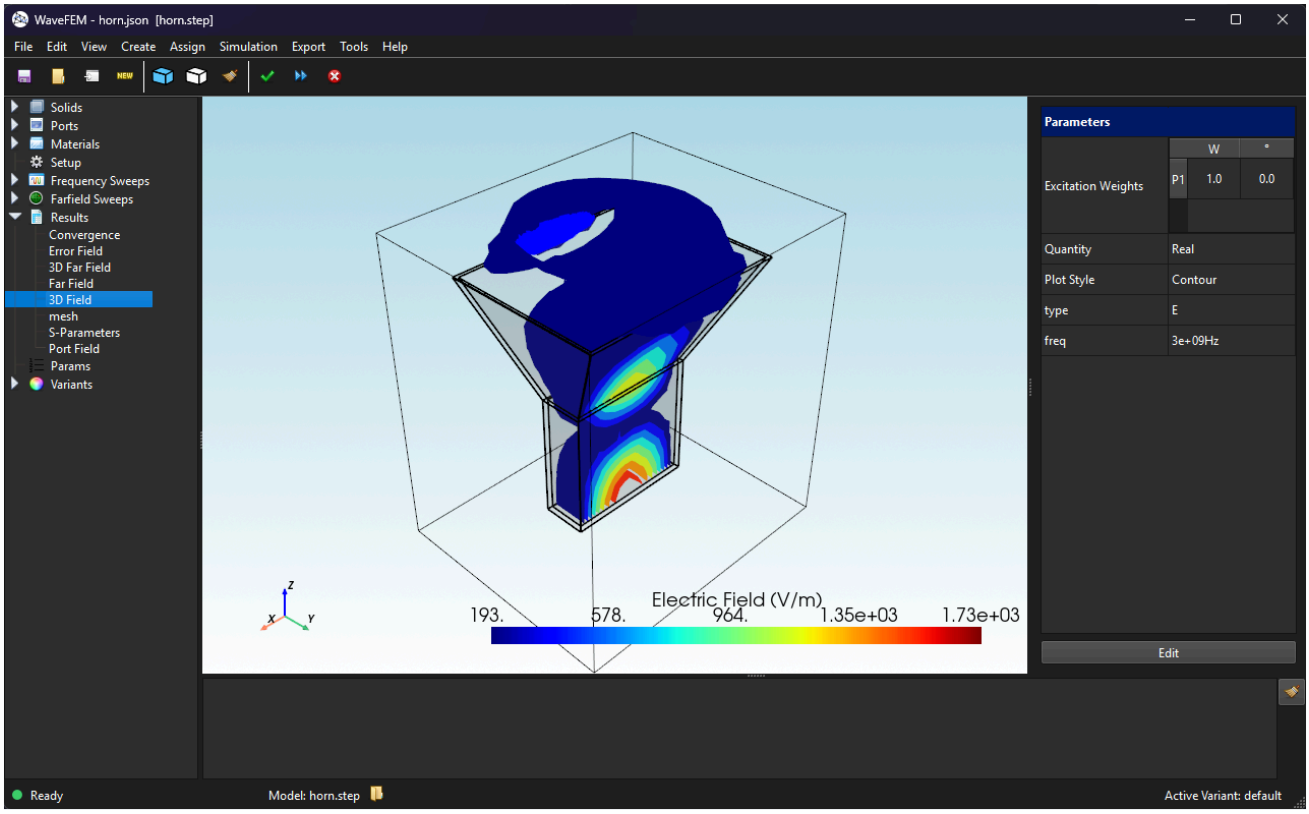


Figure 1 — Electric field inside a horn antenna

INTRODUCTION

About this document

Ansys HFSS is widely regarded as the reference tool for high-frequency electromagnetic simulation, with a mature and feature-rich environment. Its licensing costs, however, put it out of reach for many individual engineers, small teams and academic researchers. WaveFEM was developed to close this gap: a finite element solver with the accuracy and capability needed for the same class of problems, at a much lower cost.

This report presents 19 simulation examples solved in both WaveFEM and HFSS, ranging from canonical benchmark structures to practical RF and microwave designs. Each example compares S-parameters, far-field patterns, adaptive mesh convergence and runtime, so that solution accuracy, convergence behaviour and the range of supported problem types can be judged directly. Together, the examples serve as both a validation study and a practical reference for engineers evaluating WaveFEM as their primary simulation tool.

OVERVIEW

Summary of results

Each example is solved in WaveFEM and in Ansys HFSS with the same geometry, adaptive refinement target (maximum delta S) and frequency sweep. Runtimes are wall-clock times from the solver logs; element counts are for the final adaptive pass.

Example	Category	WaveFEM time	HFSS time	WaveFEM elements	HFSS elements	Page
Coupled Line Filter	Filter	00:01:03	00:01:33	29,677	17,994	4
Hairpin Filter	Filter	00:00:23	00:01:02	11,082	4,559	6
Interdigital Bandpass Filter	Filter	00:00:37	00:01:14	20,407	16,596	8
L Resonator Notch Filter	Filter	00:02:32	00:02:28	58,011	39,070	10
Stepped Impedance Filter	Filter	00:00:04	00:00:38	3,368	2,975	12
IC Filter	Filter	00:00:36	00:00:53	17,283	9,453	14
Corrugated Horn	Antenna	00:01:11	00:01:24	52,861	32,687	16
Dielectric Resonator Antenna	Antenna	00:00:04	00:00:30	4,810	3,191	19
Dipole	Antenna	00:00:24	00:00:32	19,906	9,761	22
Planar Vivaldi	Antenna	00:02:47	00:01:17	54,475	15,177	25
Probe Fed Patch	Antenna	00:00:29	00:01:14	26,213	19,159	28
Two Element Patch Array	Antenna	00:01:54	00:02:16	73,331	42,337	31
Inset Fed Patch	Antenna	00:00:31	00:01:26	23,642	24,003	34
Pyramidal Horn	Antenna	00:00:10	00:00:36	7,991	5,248	37
Exponential Horn	Antenna	00:00:11	00:01:16	14,197	23,138	40
Spiral Antenna	Antenna	00:04:21	00:05:24	40,711	36,454	43
Spiral Inductor	Planar	00:00:16	00:00:49	8,423	8,894	47
Coax to Waveguide Adapter	Transmission Line	00:00:05	00:00:38	3,486	3,732	51
Waveguide	Transmission Line	00:00:10	00:00:45	4,909	5,915	53

FILTER

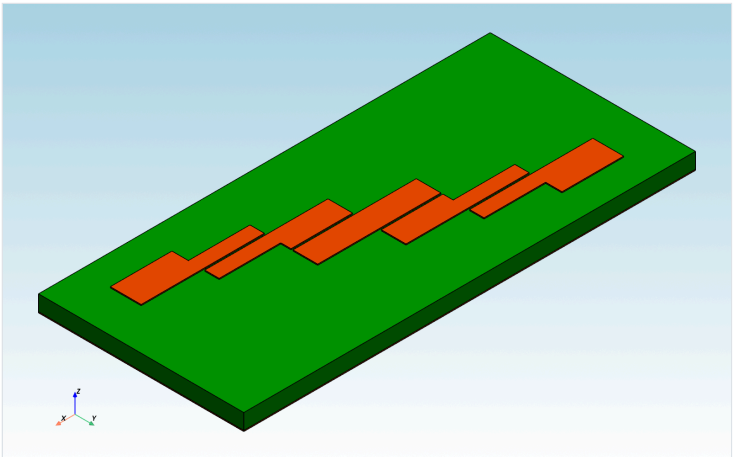
Coupled Line Filter

A 5.5-7.5 GHz bandpass filter. Using a 1.5 mm substrate with e_r=4.3

Simulation Setup

Refinement frequency	7 GHz
Delta S	0.02
Frequency range	2 - 10 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:03	00:01:33
Adaptive meshing	00:00:29	00:00:39
Frequency sweep	00:00:28	00:00:52
Adaptive passes	12	10
Final elements	29,677	17,994
Matrix size	169,338	110,368

Convergence

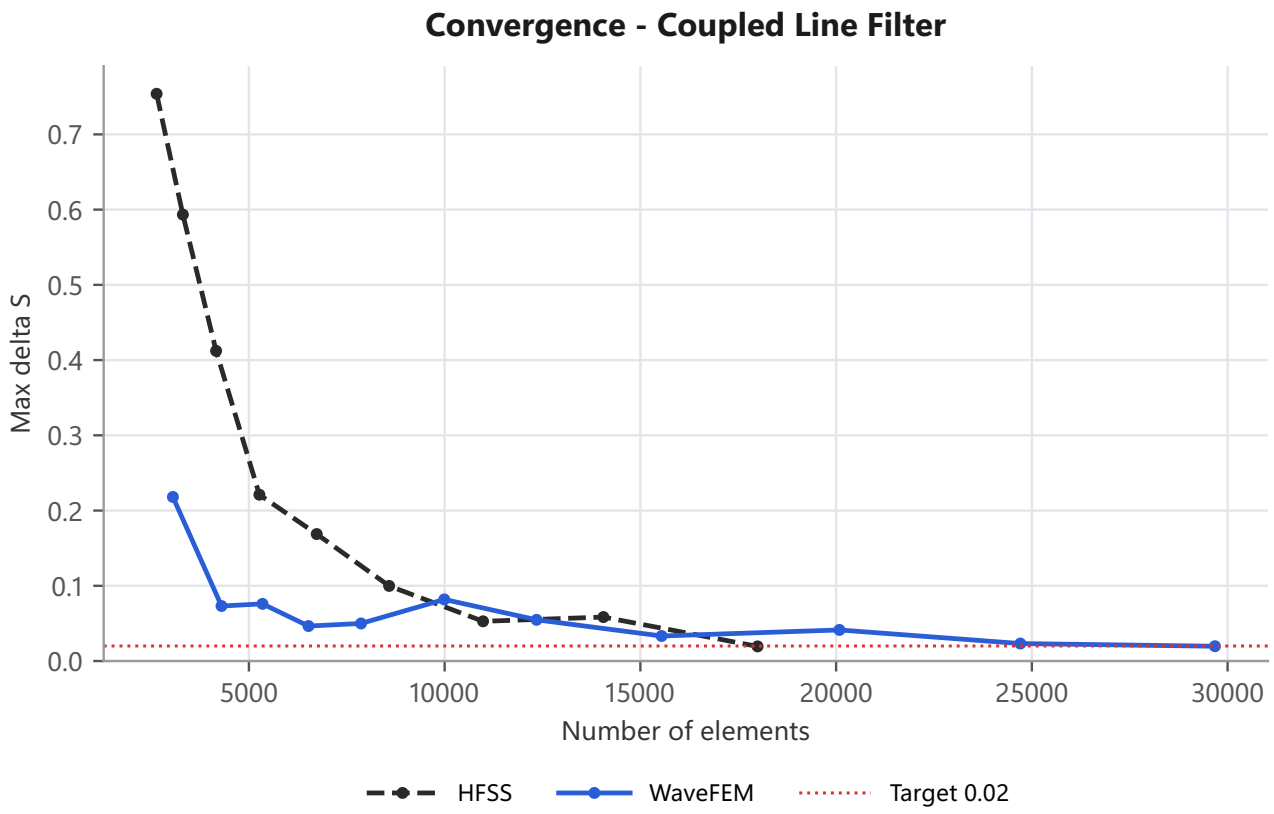


Figure 2 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

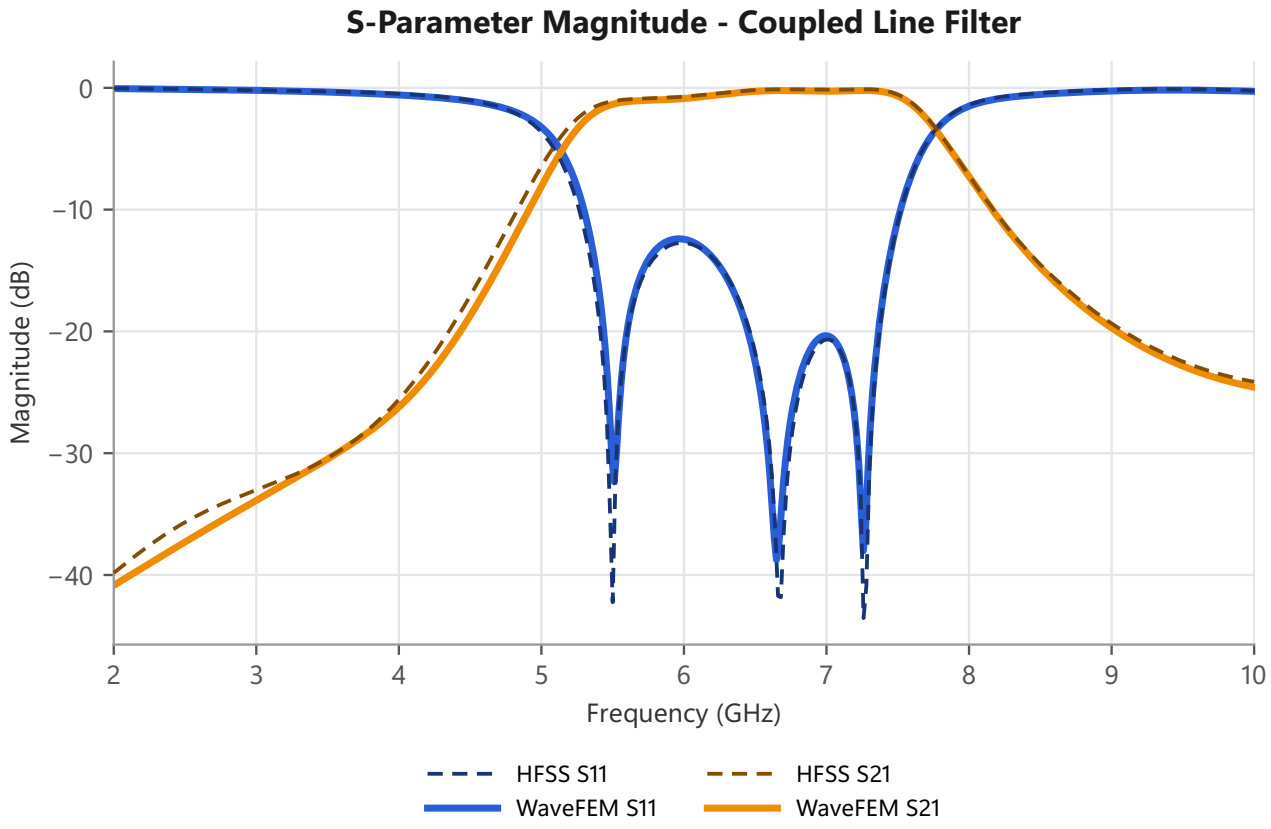


Figure 3 — S-parameter magnitude.

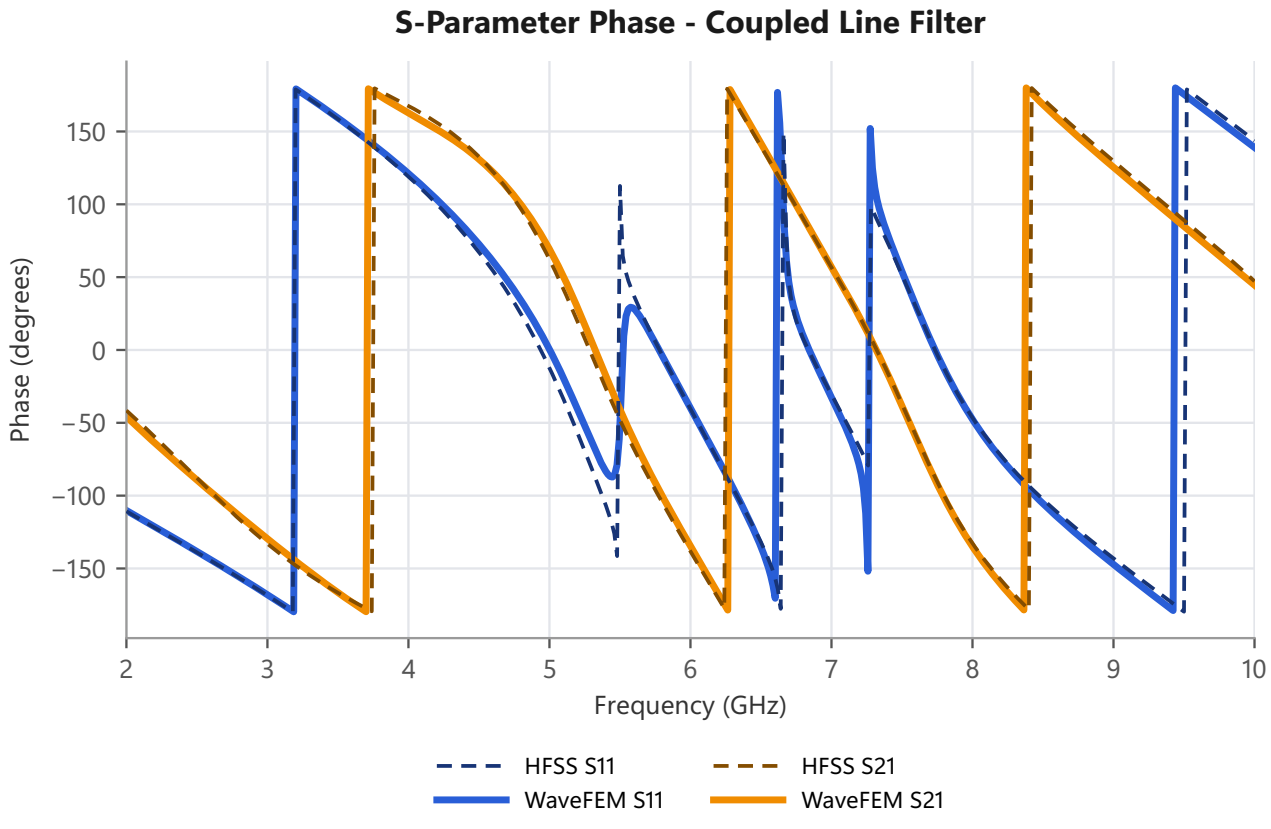


Figure 4 — S-parameter phase.

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FILTER

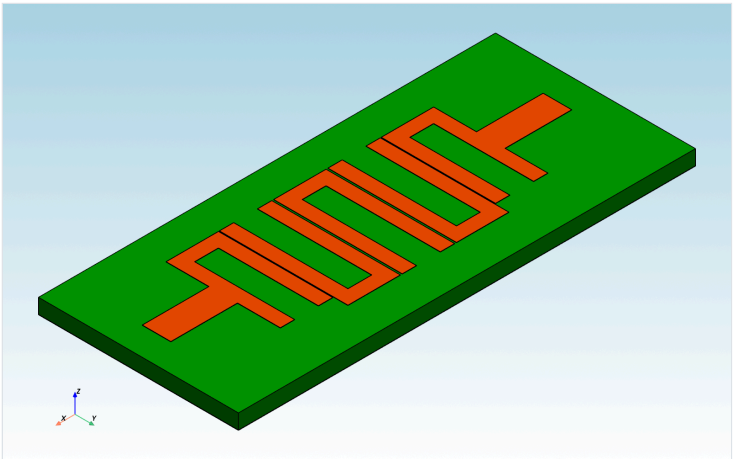
Hairpin Filter

A 2-4 GHz bandpass hairpin filter. Using a 1.55mm substrate with e_r=4.4 & tand=0.01

Simulation Setup

Refinement frequency	3.5 GHz
Delta S	0.02
Frequency range	1 - 5 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:23	00:01:02
Adaptive meshing	00:00:11	00:00:28
Frequency sweep	00:00:10	00:00:32
Adaptive passes	9	10
Final elements	11,082	4,559
Matrix size	61,046	26,618

Convergence

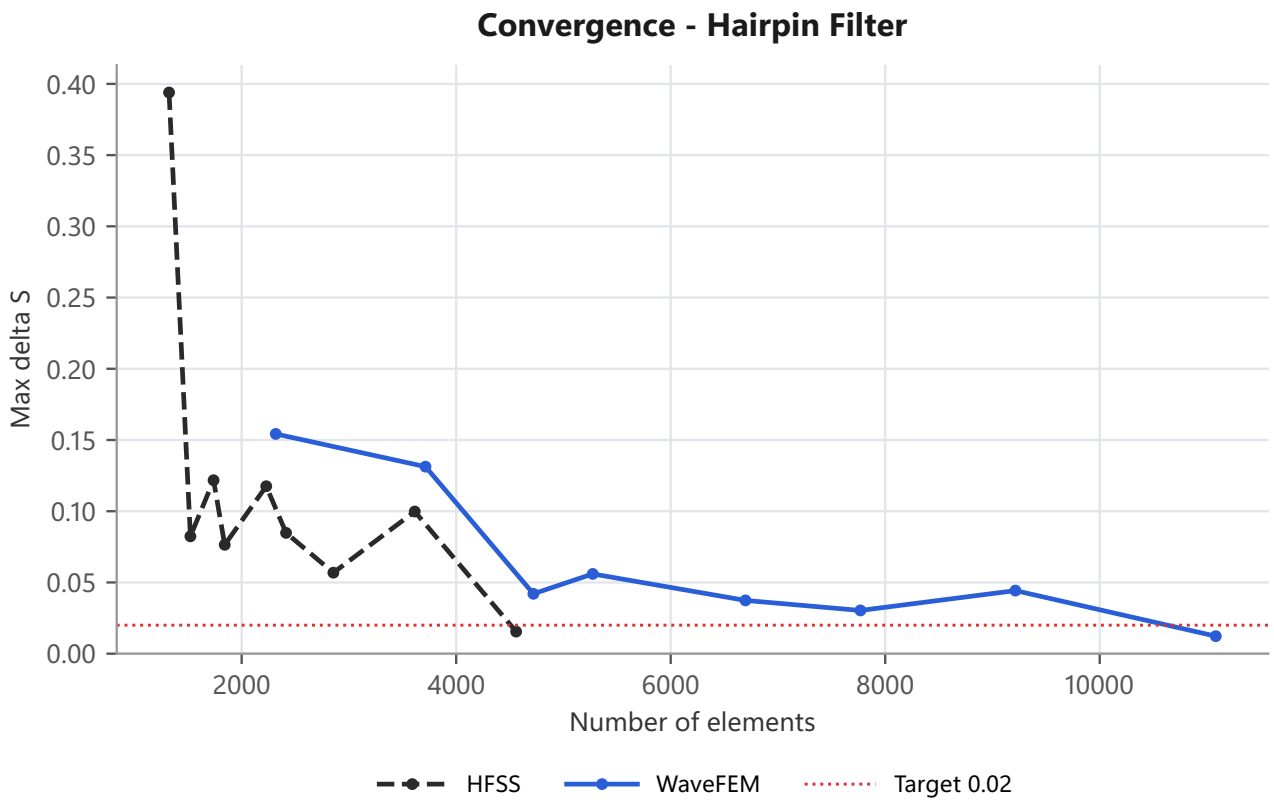


Figure 5 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

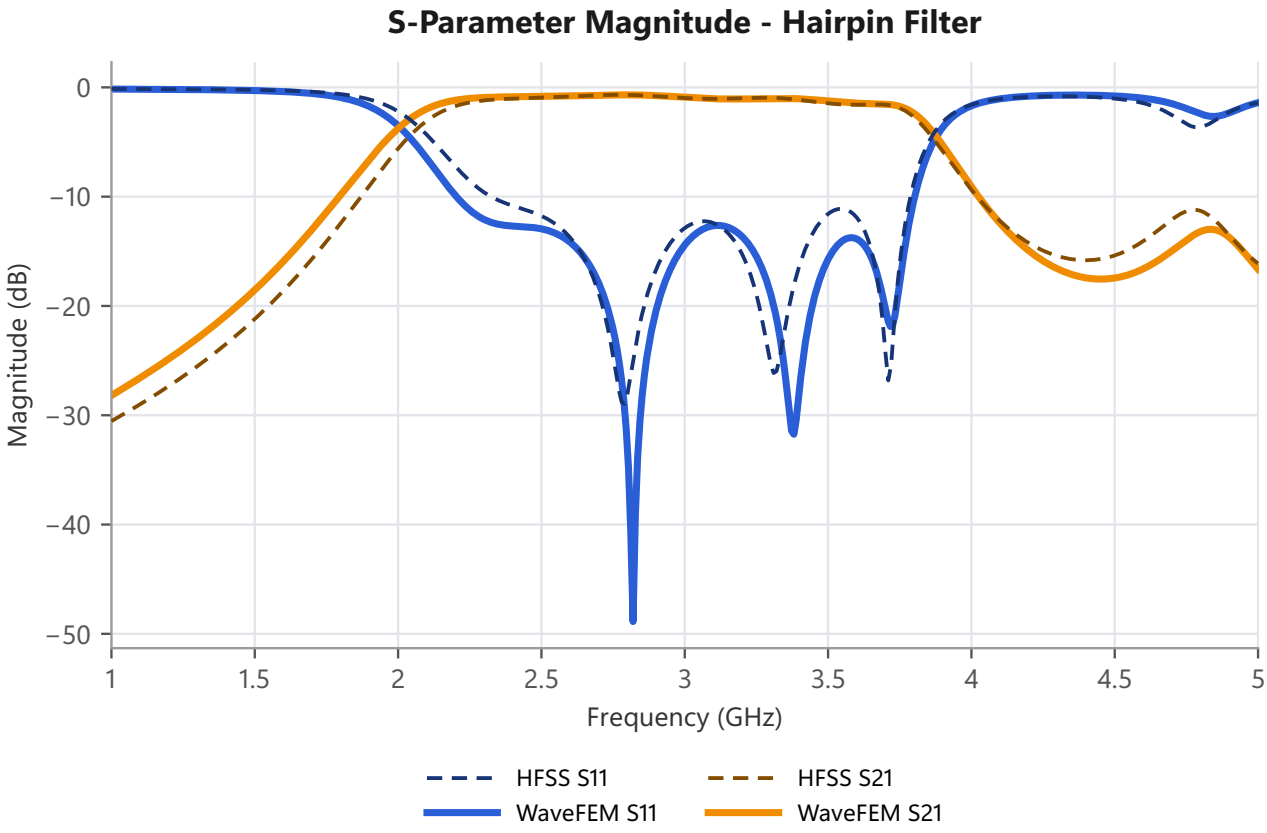


Figure 6 — S-parameter magnitude.

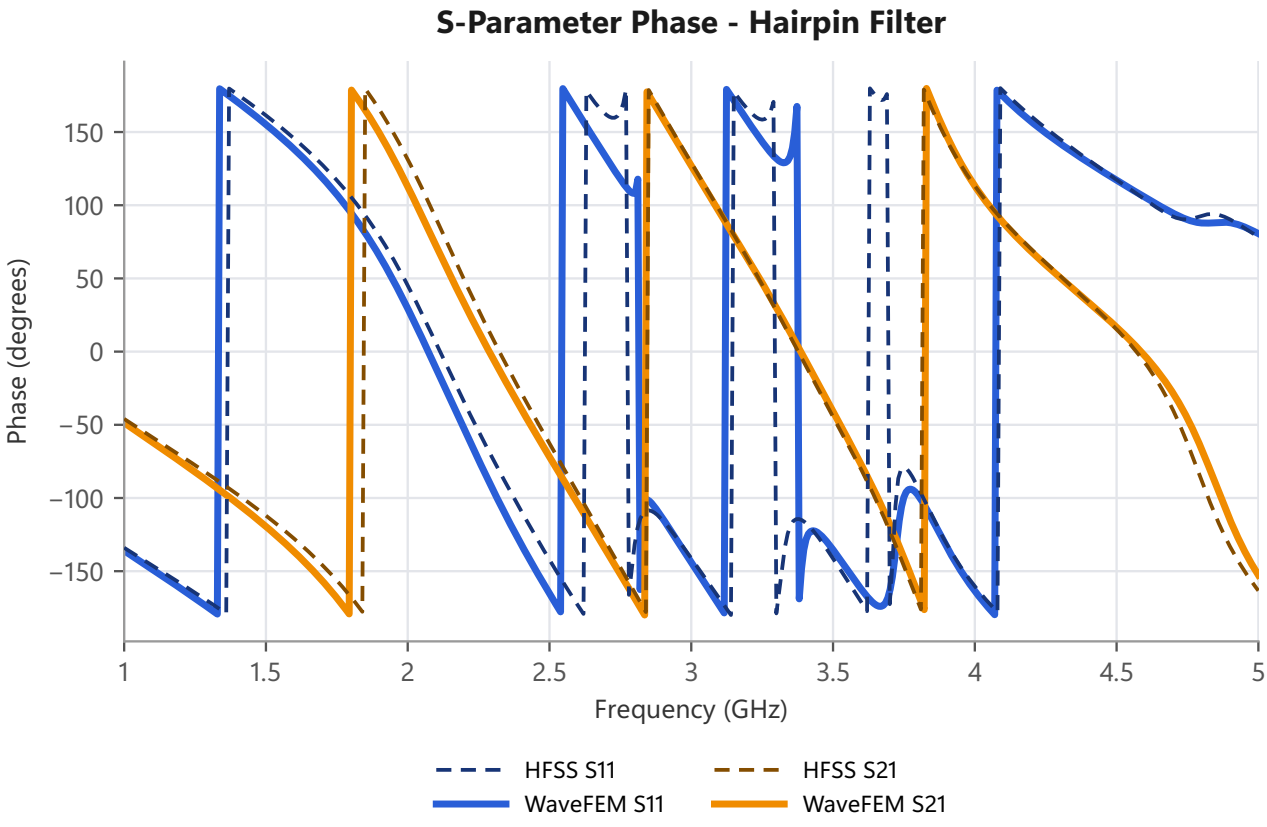


Figure 7 — S-parameter phase.

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FILTER

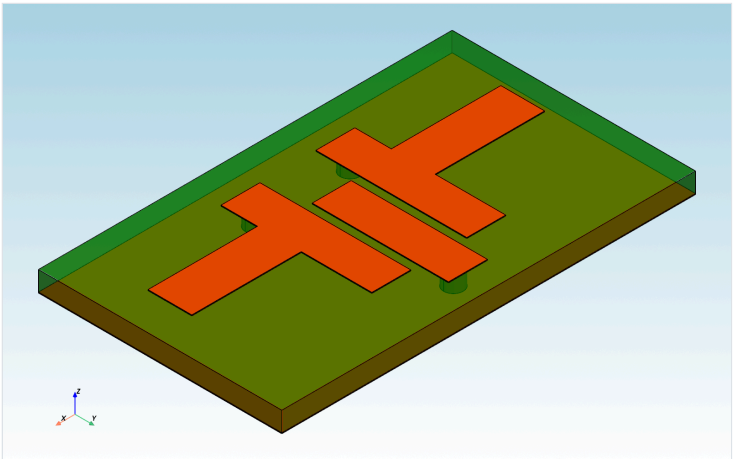
Interdigital Bandpass Filter

A 7-9 GHz bandpass filter. Using a 0.8mm substrate with e_r=3.55 tand=0.0027

Simulation Setup

Refinement frequency	8 GHz
Delta S	0.02
Frequency range	4 - 12 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:37	00:01:14
Adaptive meshing	00:00:18	00:00:27
Frequency sweep	00:00:15	00:00:41
Adaptive passes	10	8
Final elements	20,407	16,596
Matrix size	116,192	100,730

Convergence

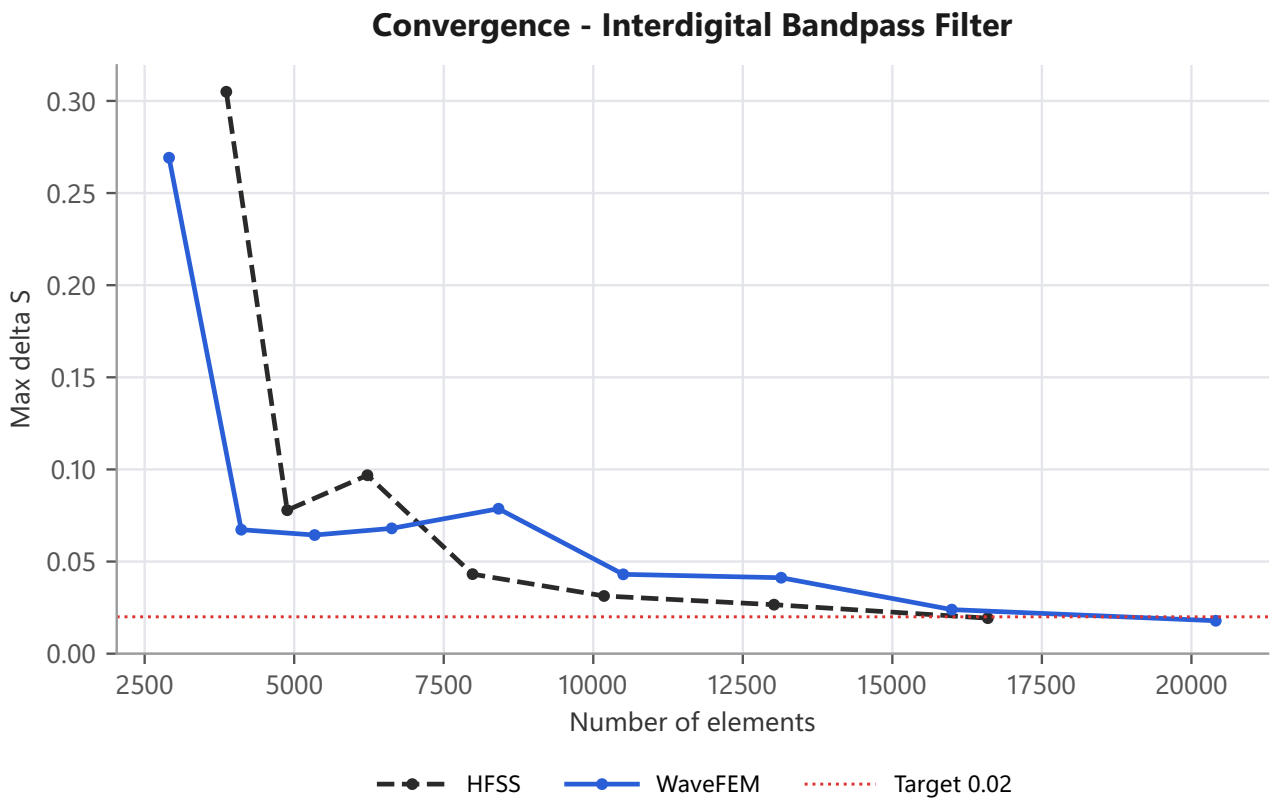


Figure 8 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

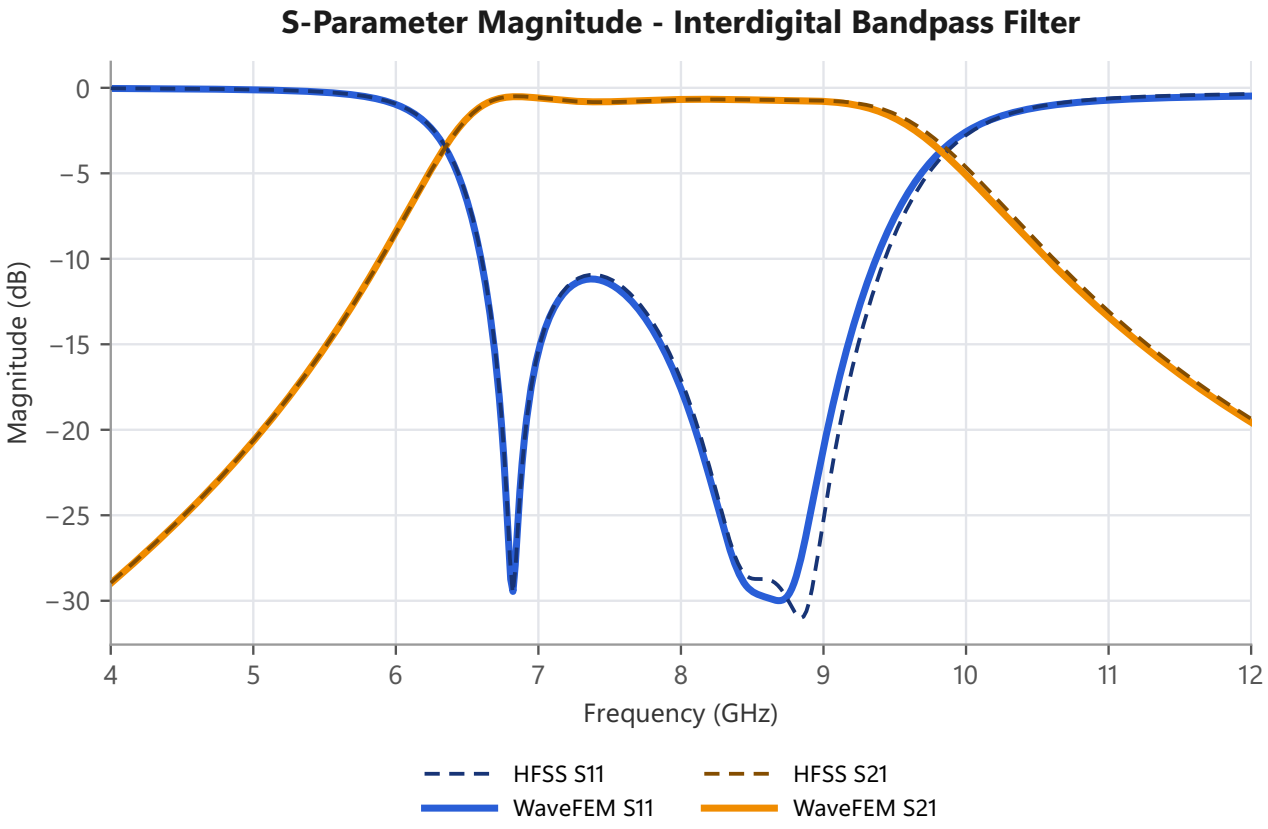


Figure 9 — S-parameter magnitude.

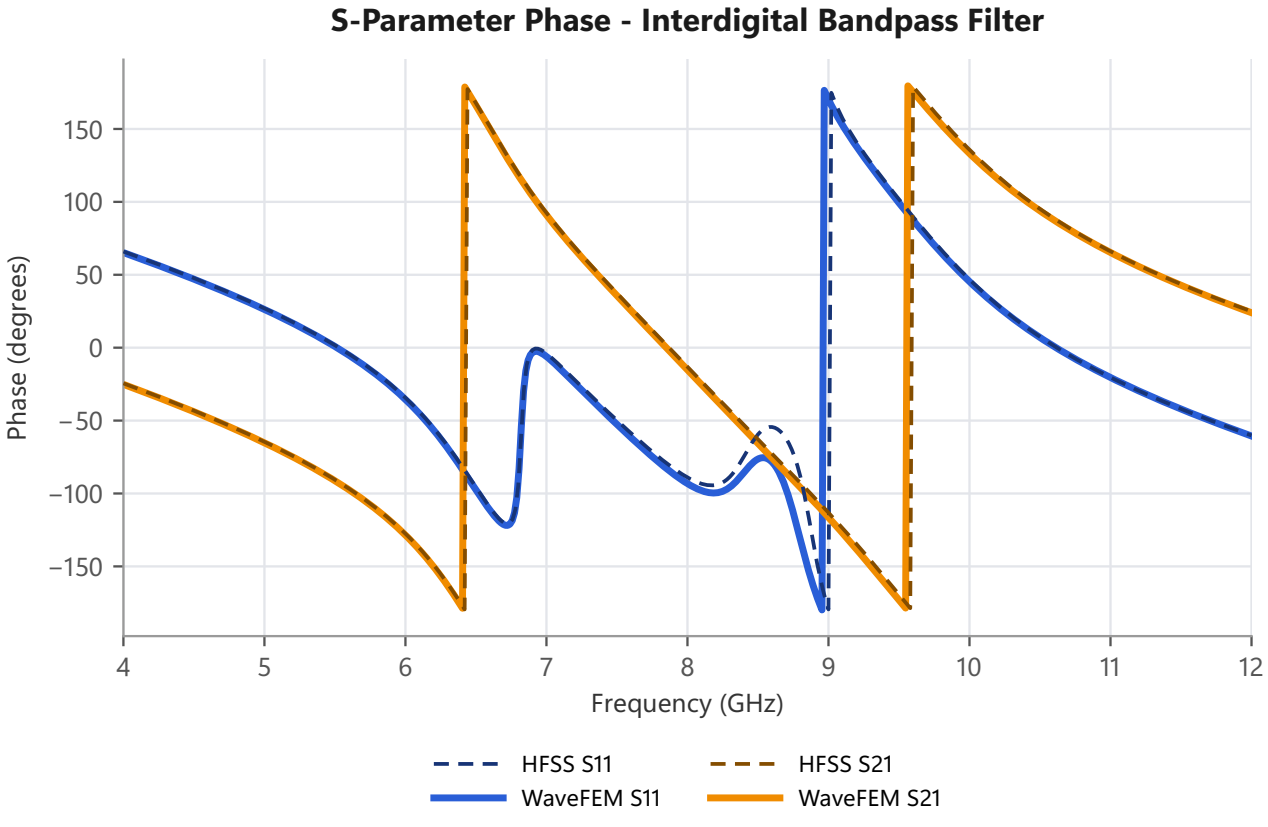


Figure 10 — S-parameter phase.

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FILTER

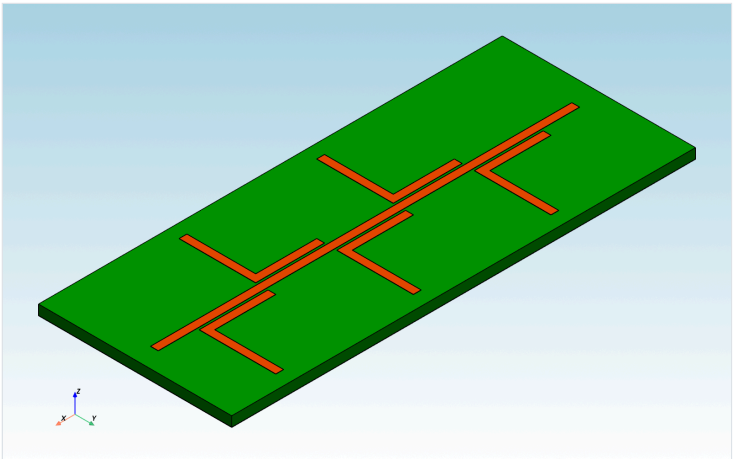
L Resonator Notch Filter

A 3.2 GHz notch filter. Using a 1.27mm substrate with e_r=10.8

Simulation Setup

Refinement frequency	3.4 GHz
Delta S	0.02
Frequency range	2 - 4 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:02:32	00:02:28
Adaptive meshing	00:01:00	00:01:02
Frequency sweep	00:01:18	00:01:24
Adaptive passes	14	15
Final elements	58,011	39,070
Matrix size	329,498	239,094

Convergence

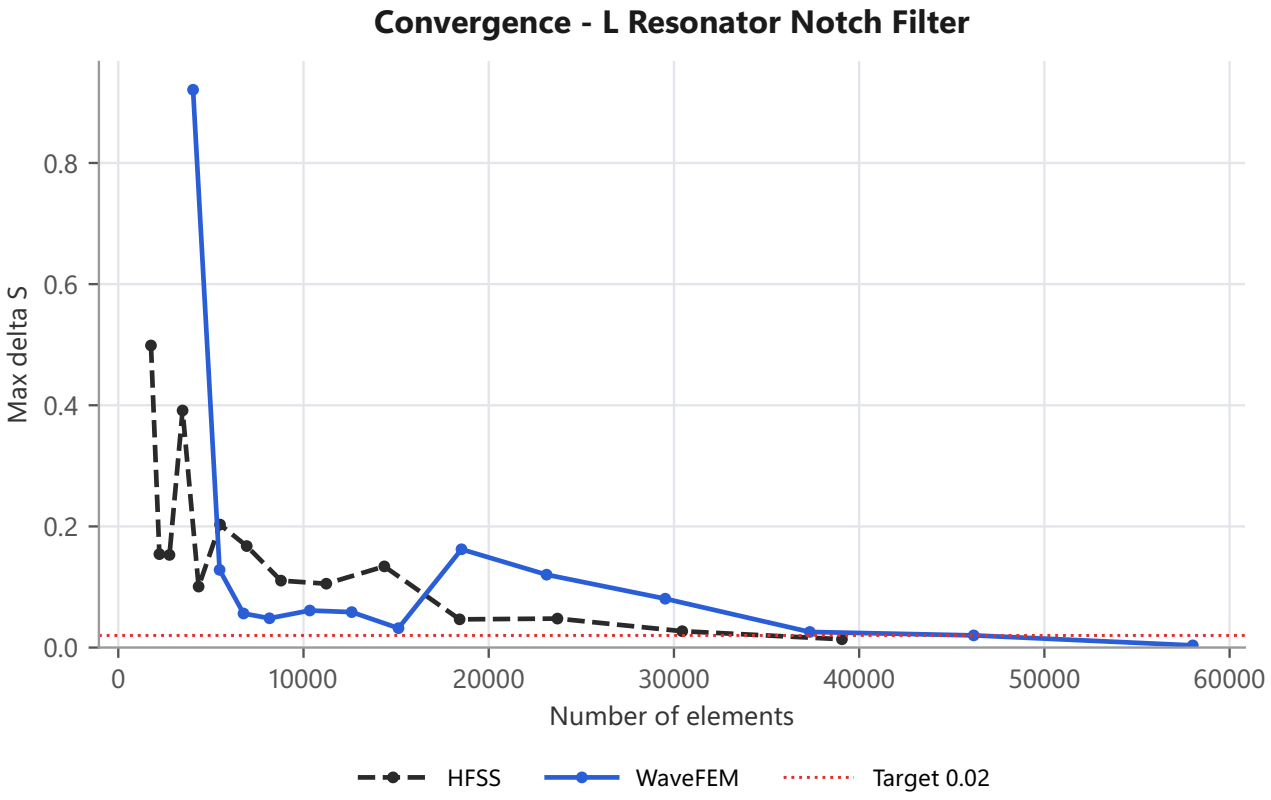


Figure 11 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

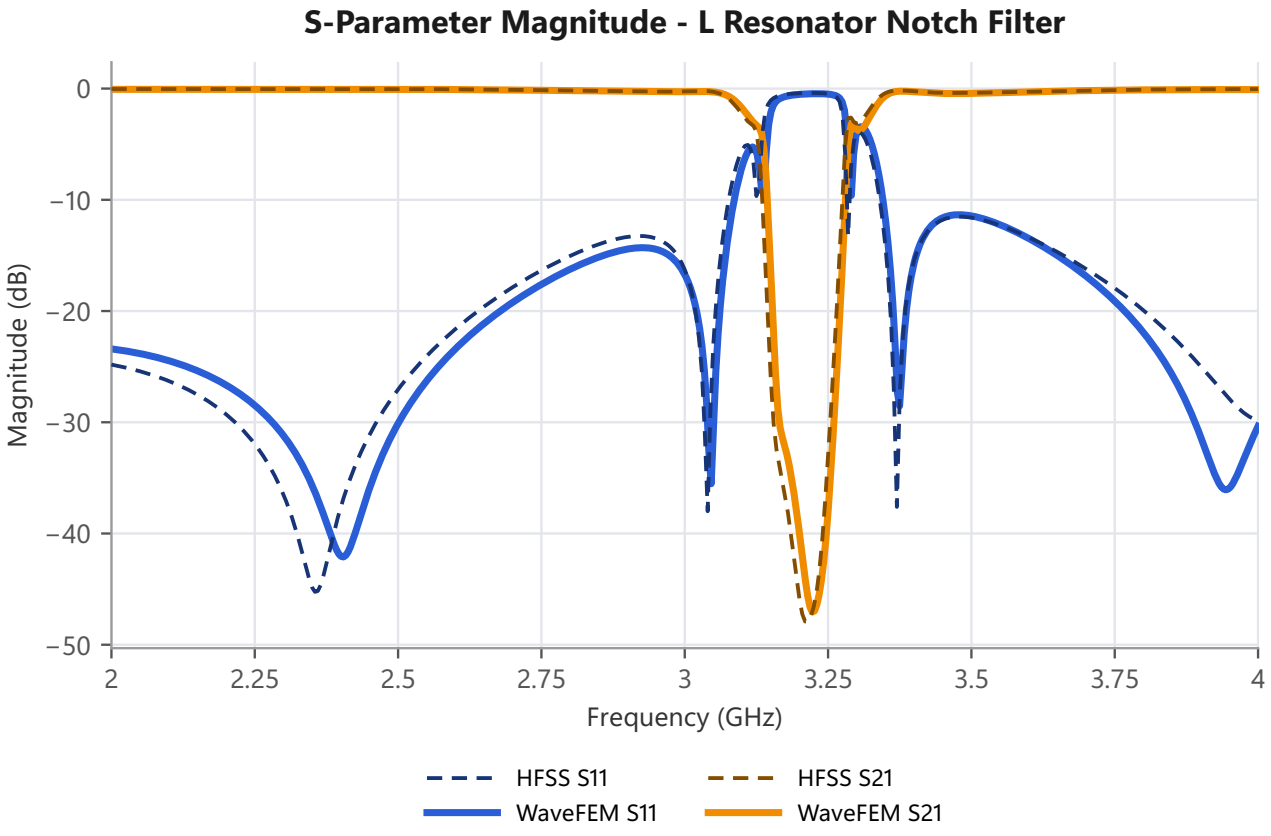


Figure 12 — S-parameter magnitude.

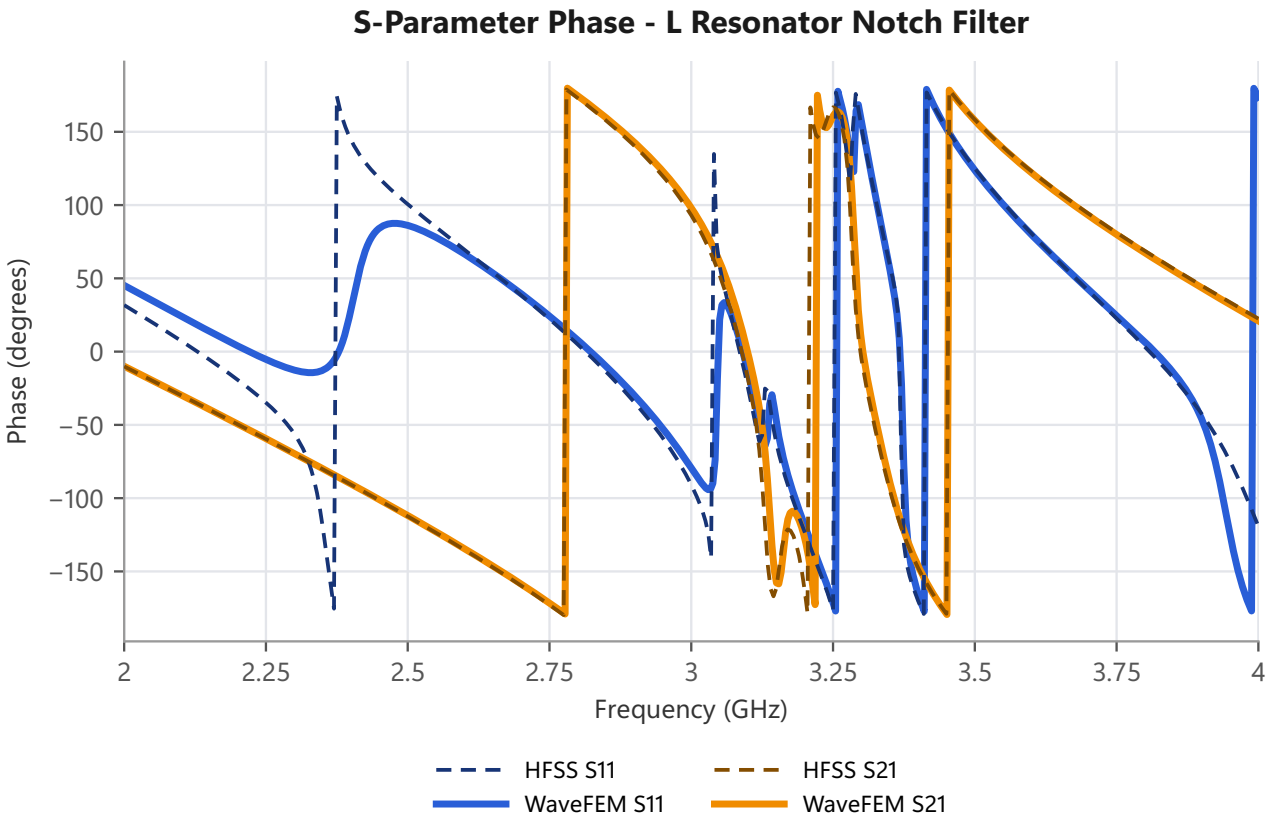


Figure 13 — S-parameter phase.

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FILTER

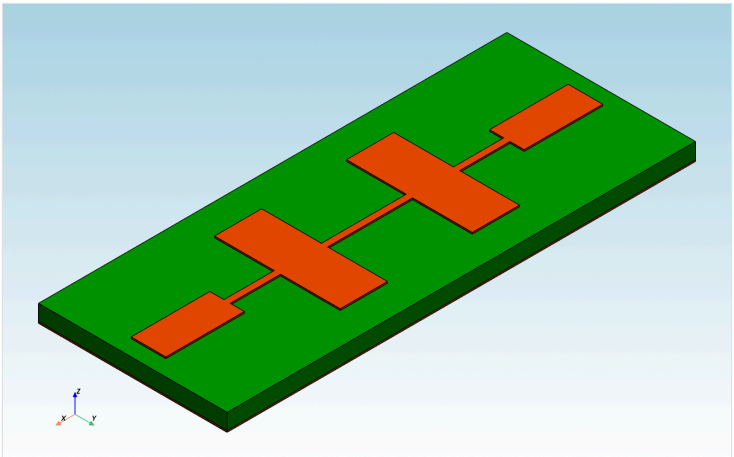
Stepped Impedance Filter

A 9 GHz lowpass filter. Using a 0.5mm substrate with ε_r=3.55 tand=0.0027

Simulation Setup

Refinement frequency	7 GHz
Delta S	0.02
Frequency range	0.1 - 20 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:04	00:00:38
Adaptive meshing	00:00:02	00:00:17
Frequency sweep	00:00:02	00:00:19
Adaptive passes	3	7
Final elements	3,368	2,975
Matrix size	19,102	17,656

Convergence

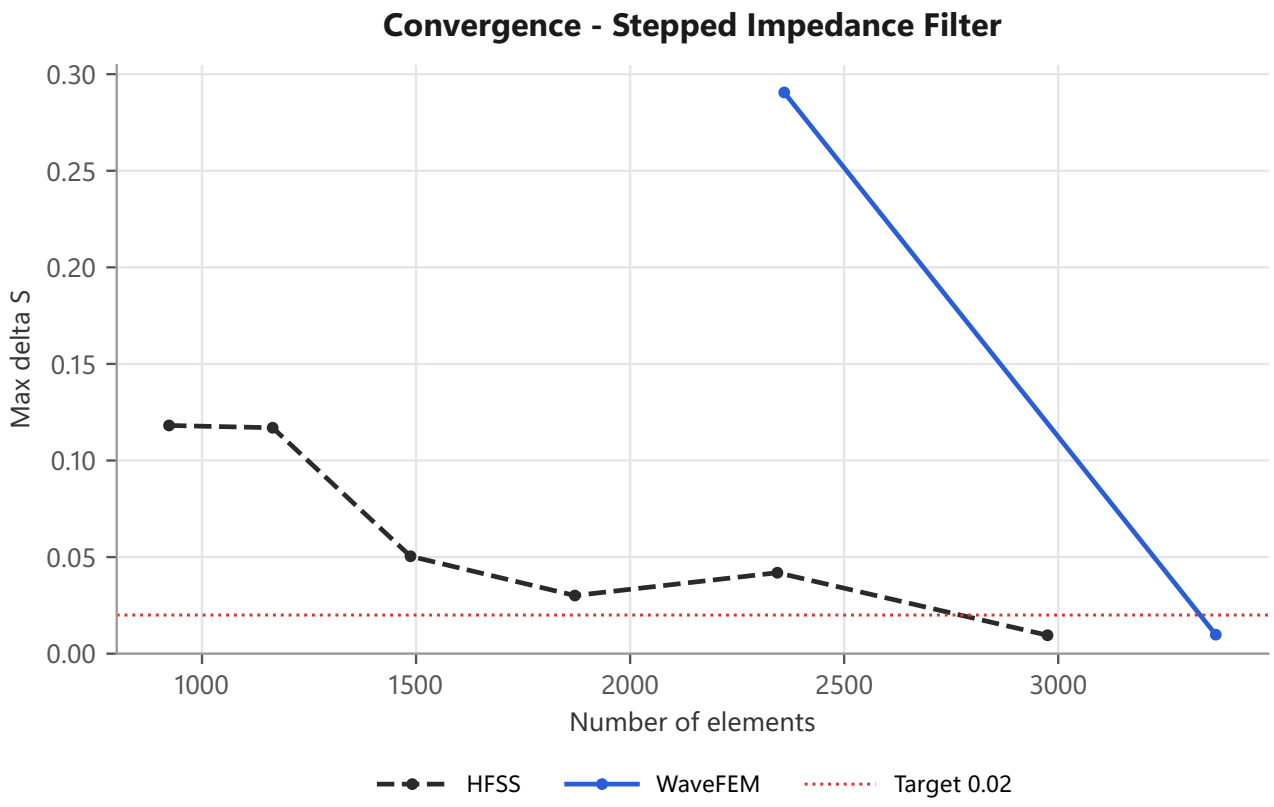


Figure 14 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

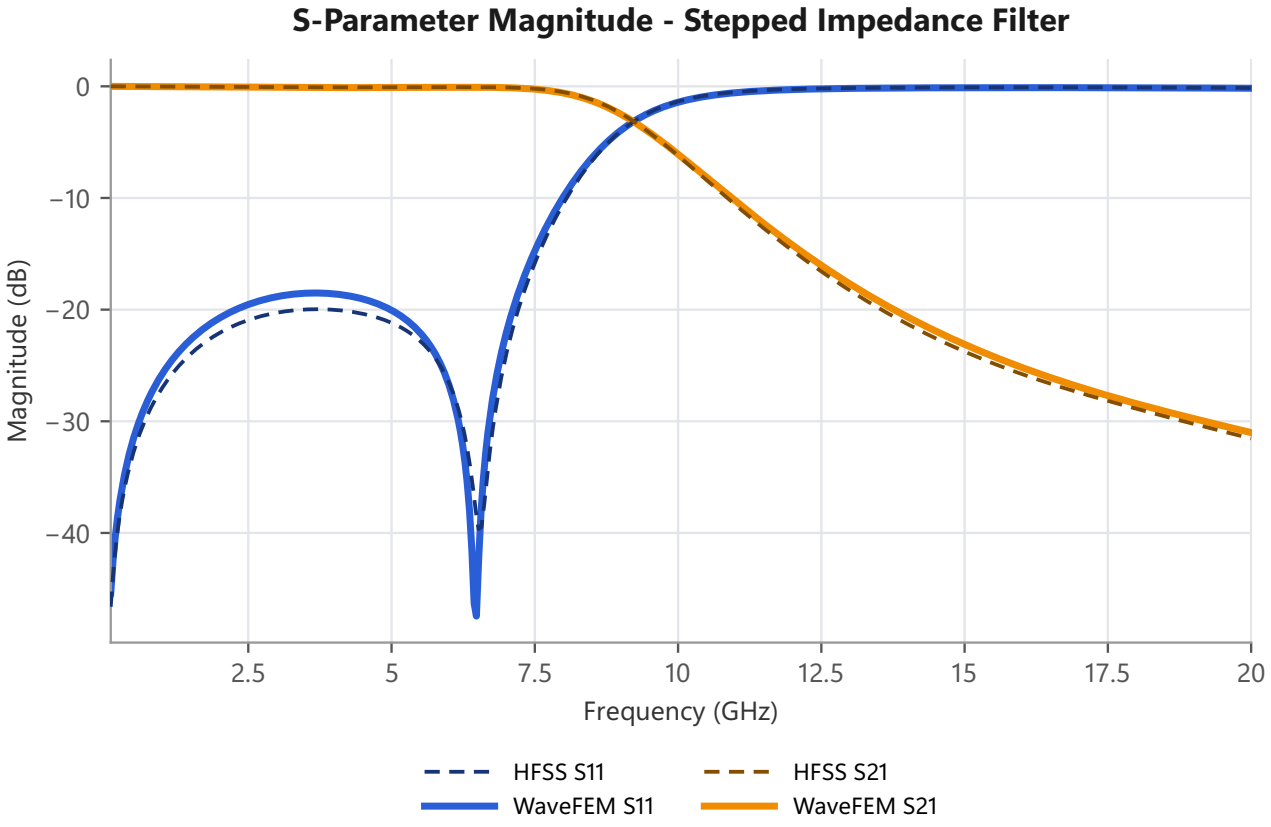


Figure 15 — S-parameter magnitude.

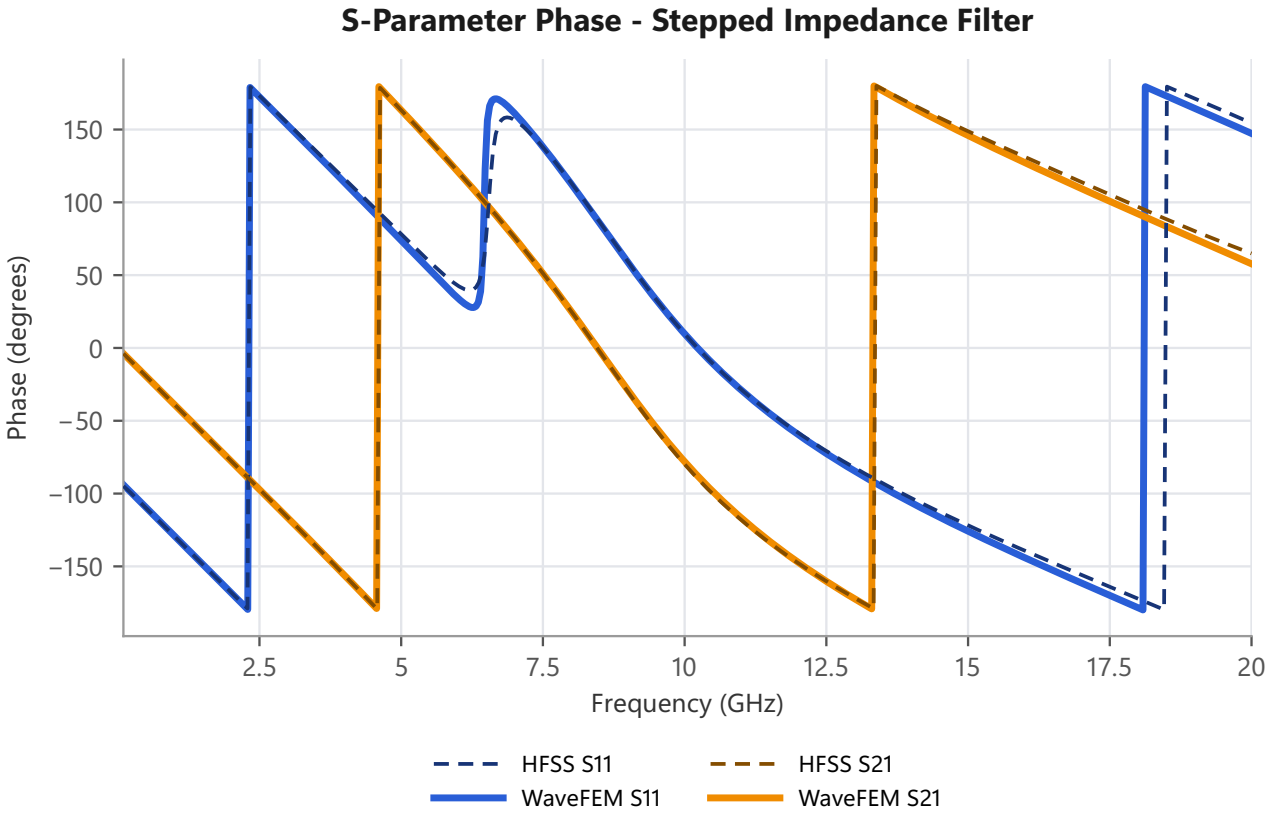


Figure 16 — S-parameter phase.

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FILTER

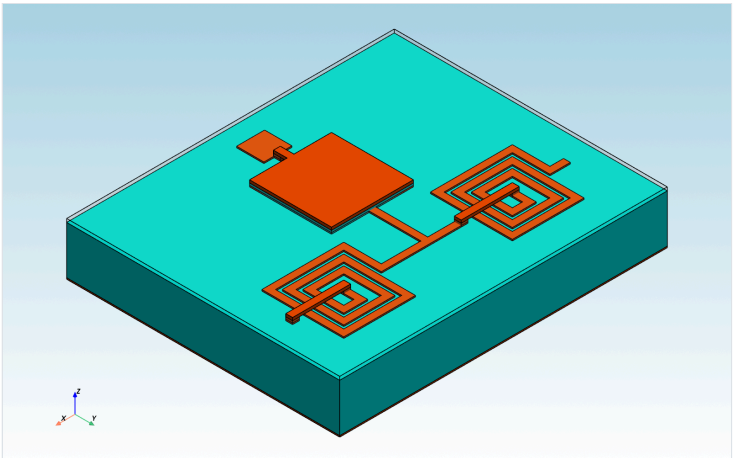
IC Filter

An MMIC-style lumped low-pass T filter: two square spiral inductors and a shunt metal-insulator-metal capacitor grounded through a via, on a 70 µm substrate.

Simulation Setup

Refinement frequency	50 GHz
Delta S	0.02
Frequency range	1 - 50 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:36	00:00:53
Adaptive meshing	00:00:20	00:00:23
Frequency sweep	00:00:14	00:00:27
Adaptive passes	7	8
Final elements	17,283	9,453
Matrix size	120,320	63,002

Convergence

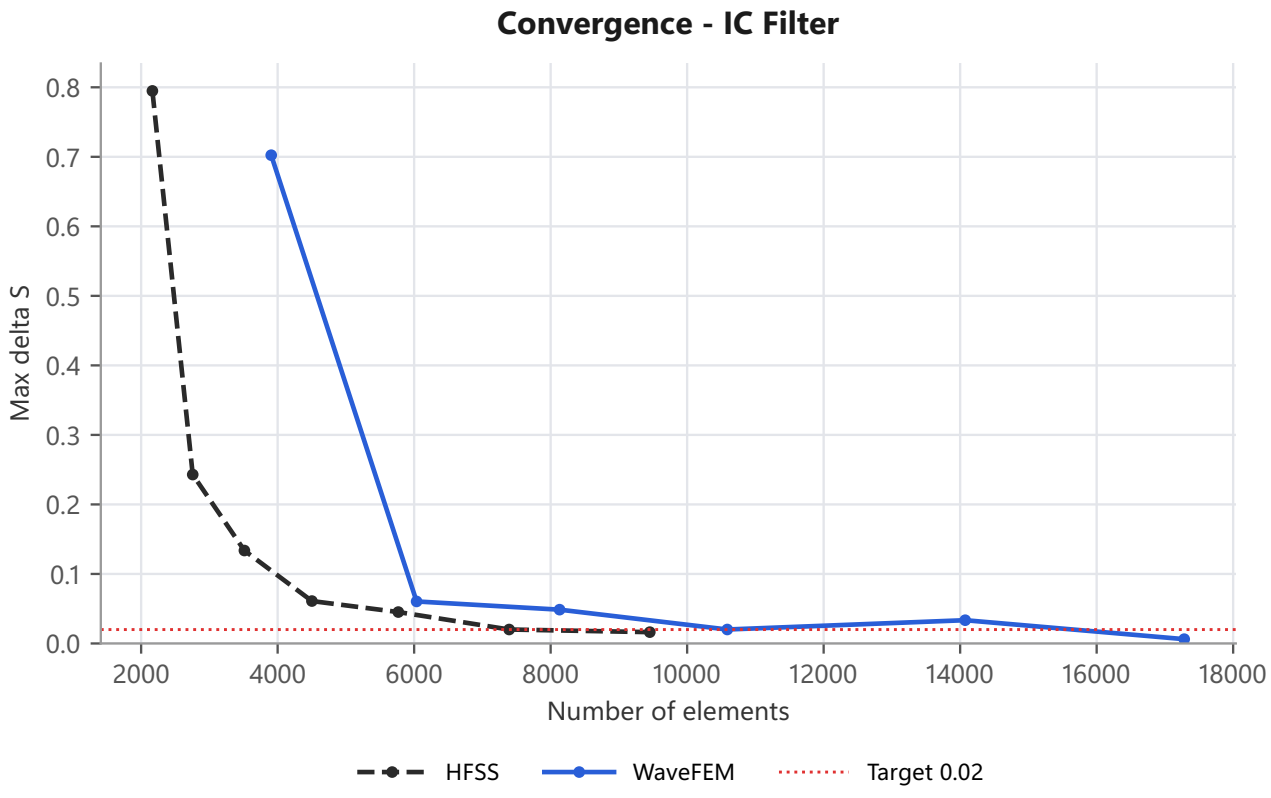


Figure 17 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

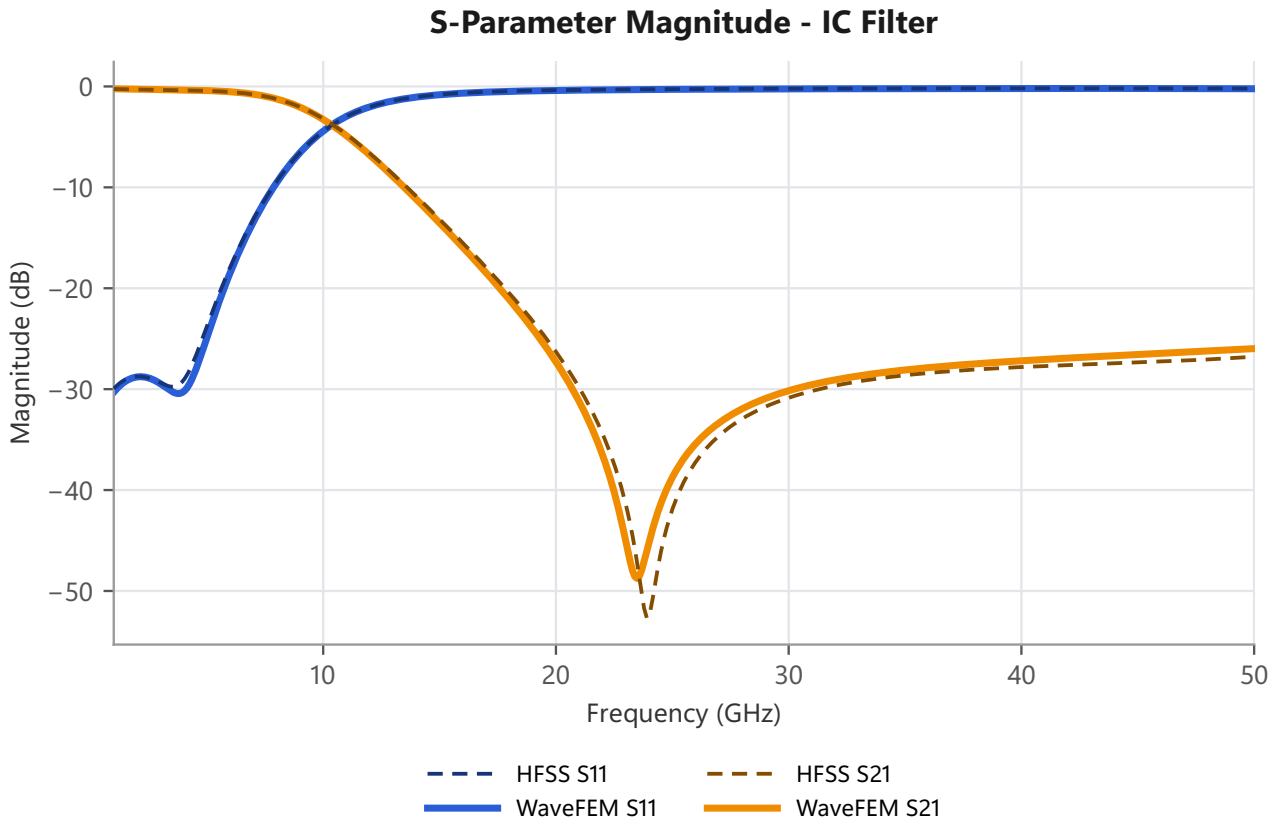


Figure 18 — S-parameter magnitude.

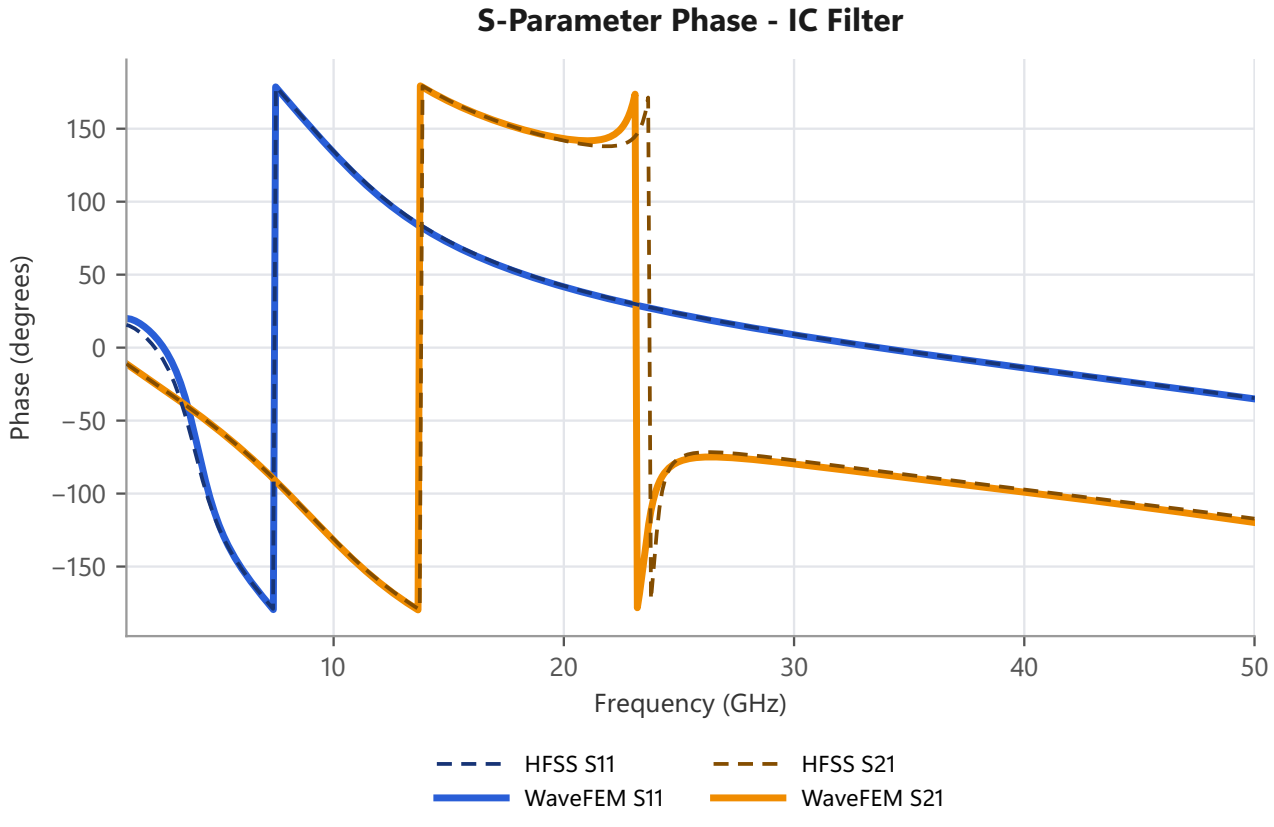


Figure 19 — S-parameter phase.

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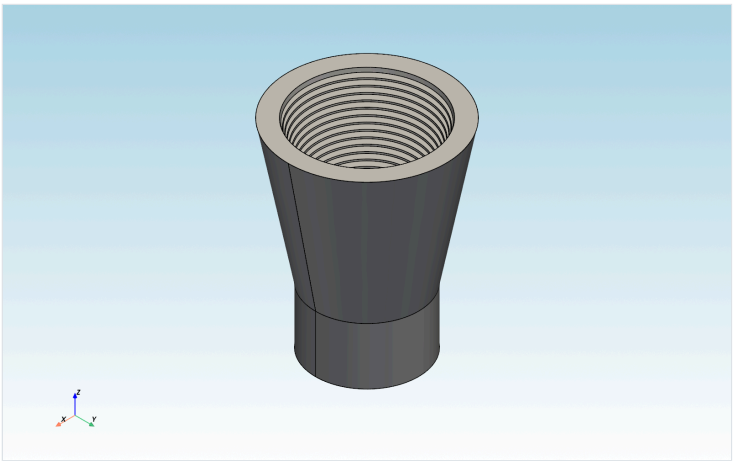
Corrugated Horn

A X-band corrugated horn antenna designed to have low side lobes

Simulation Setup

Refinement frequency	11 GHz
Delta S	0.02
Frequency range	9 - 11 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:11	00:01:24
Adaptive meshing	00:00:30	00:00:17
Frequency sweep	00:00:36	00:00:45
Adaptive passes	3	2
Final elements	52,861	32,687
Matrix size	307,740	196,483

Convergence

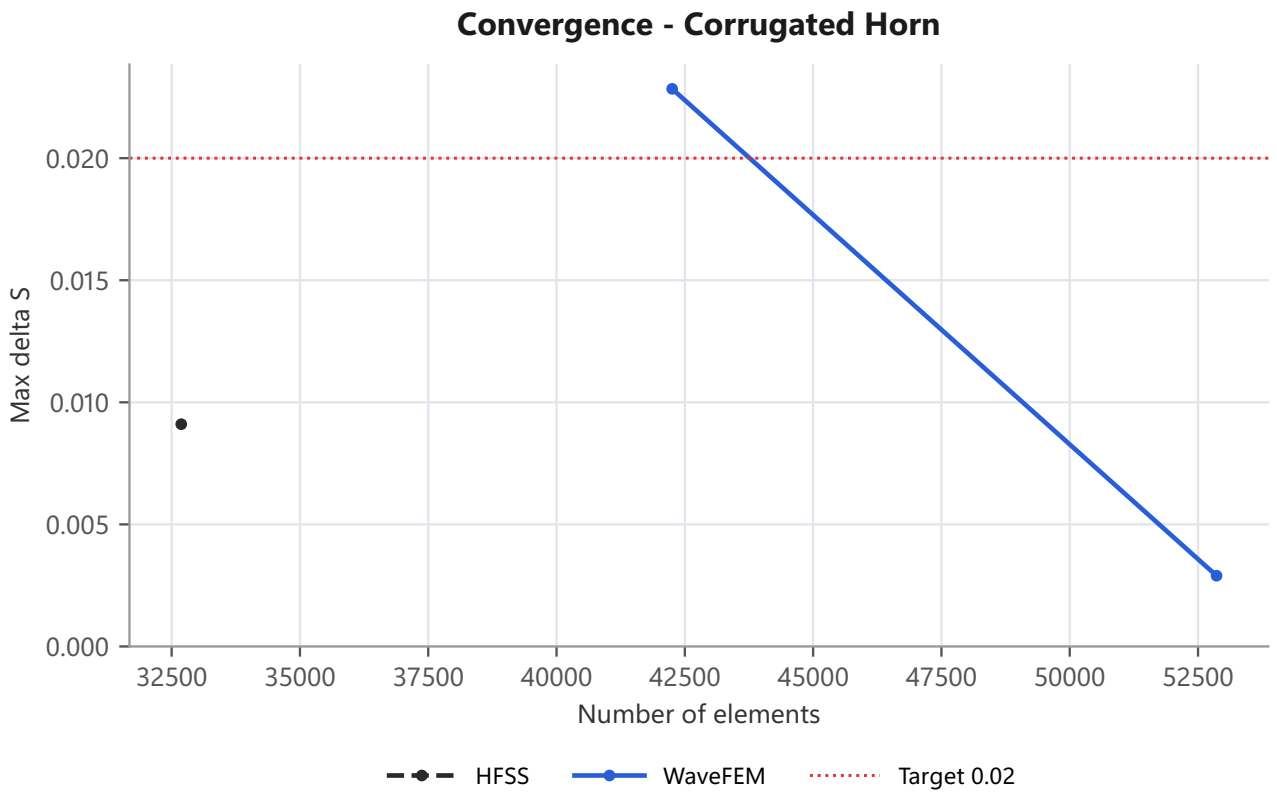


Figure 20 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

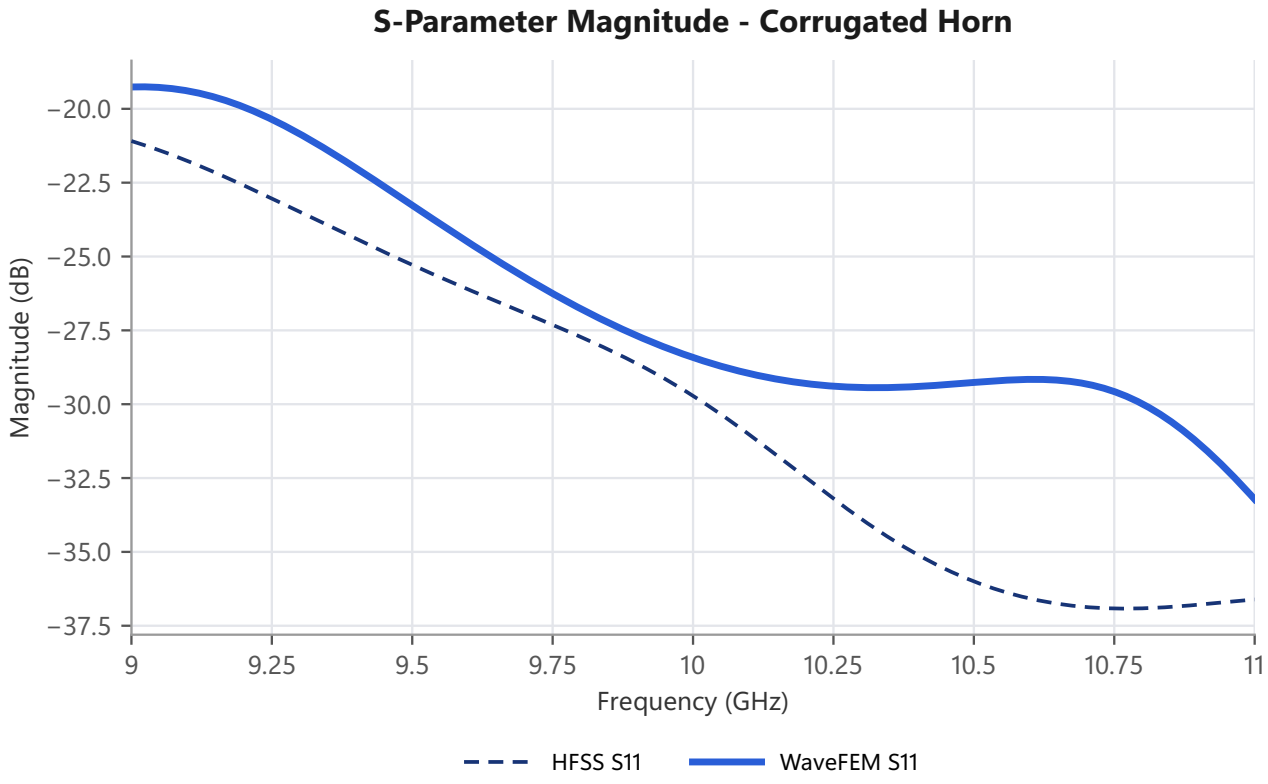


Figure 21 — S-parameter magnitude.

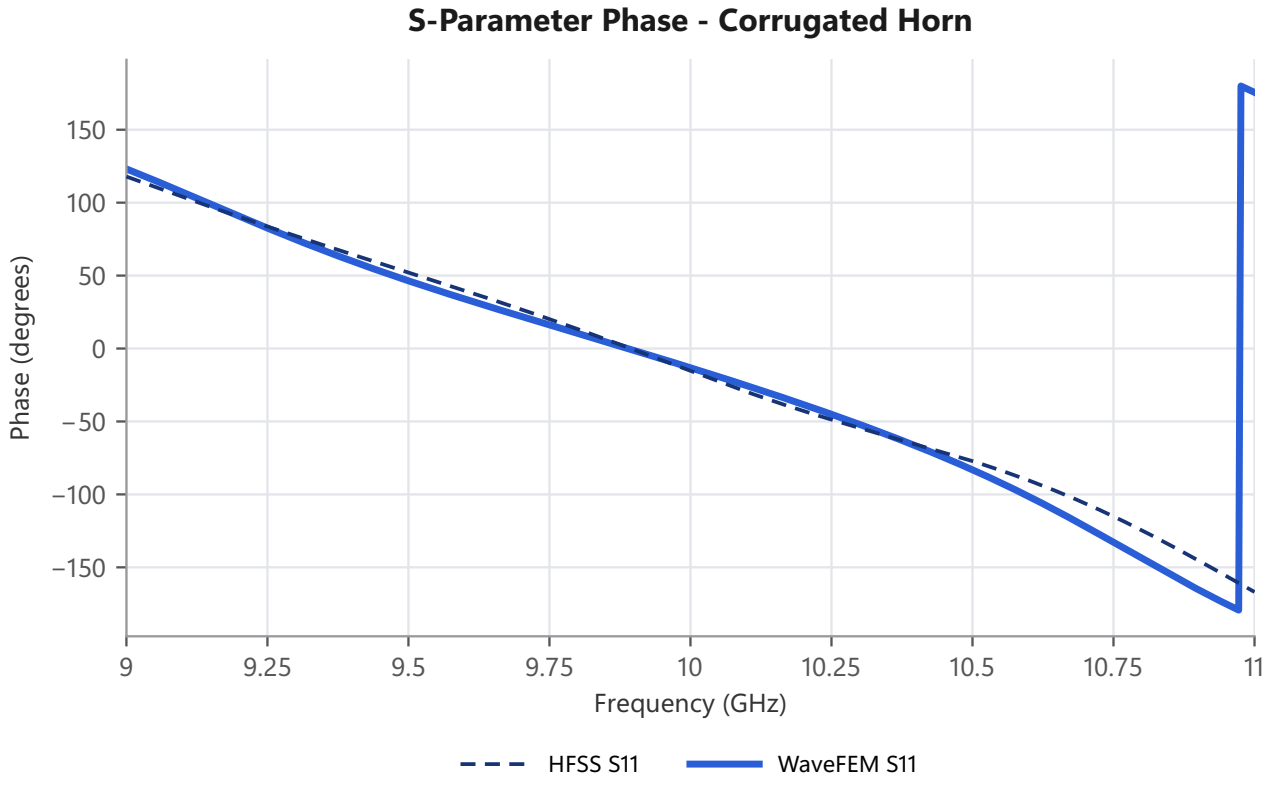


Figure 22 — S-parameter phase.

Far Field

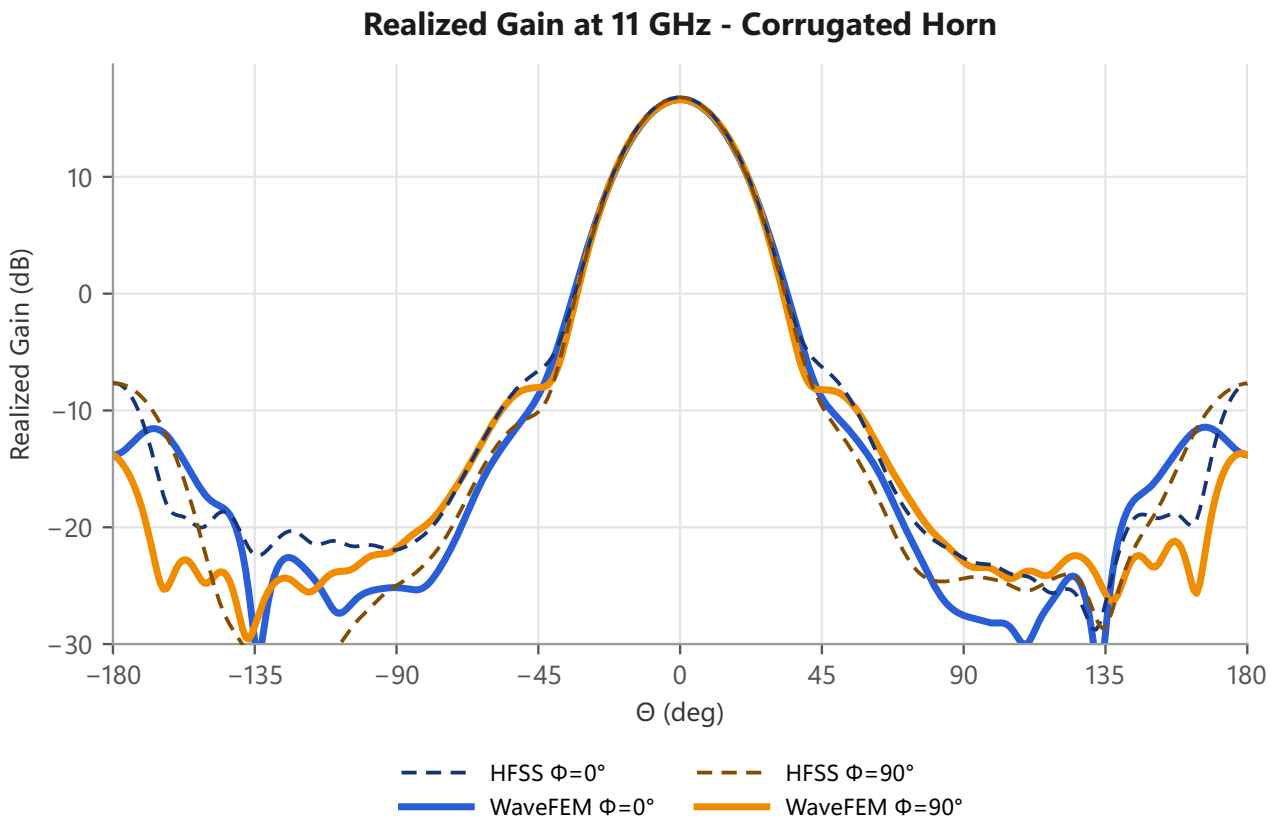


Figure 23 — Total realized gain cuts at 11 GHz.

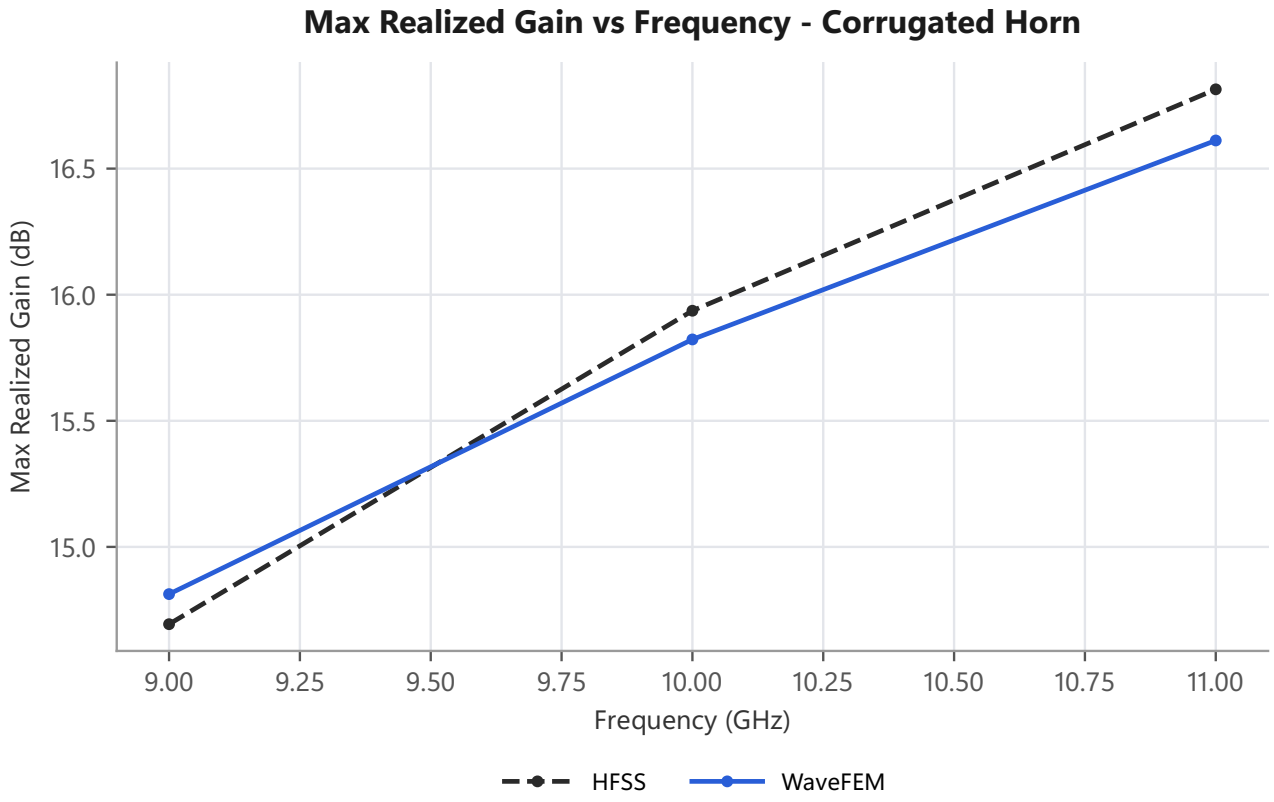


Figure 24 — Maximum realized gain against frequency.

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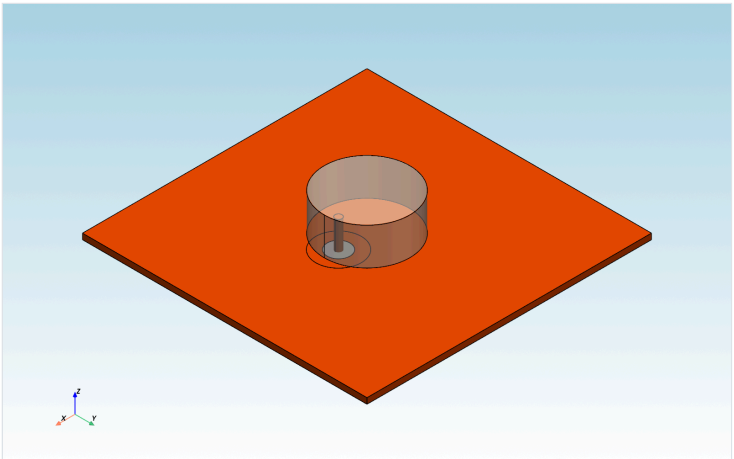
Dielectric Resonator Antenna

A 5.6 GHz probe fed dielectric resonator antenna, using ceramic with e_r=10 & tand=0.002

Simulation Setup

Refinement frequency	5.6 GHz
Delta S	0.02
Frequency range	4 - 7 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:04	00:00:30
Adaptive meshing	00:00:02	00:00:07
Frequency sweep	00:00:02	00:00:19
Adaptive passes	3	3
Final elements	4,810	3,191
Matrix size	28,820	18,923

Convergence

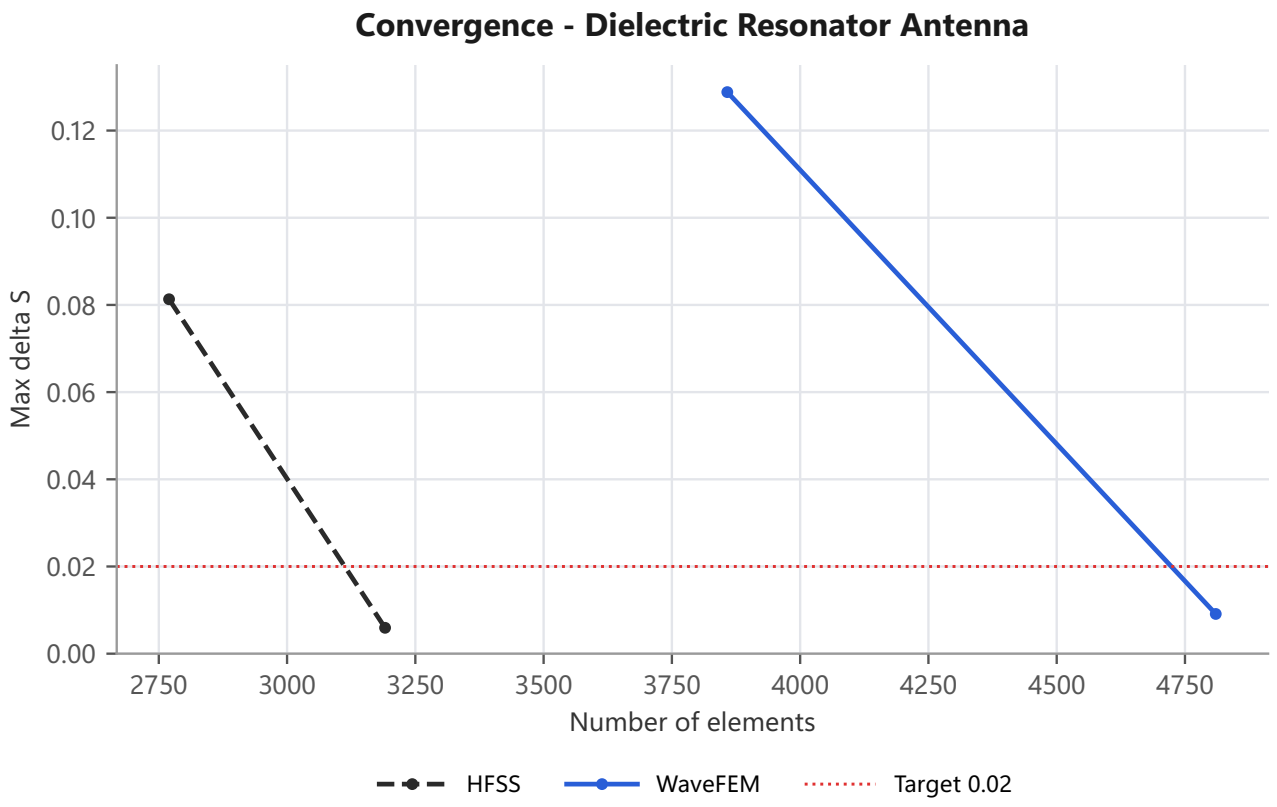


Figure 25 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

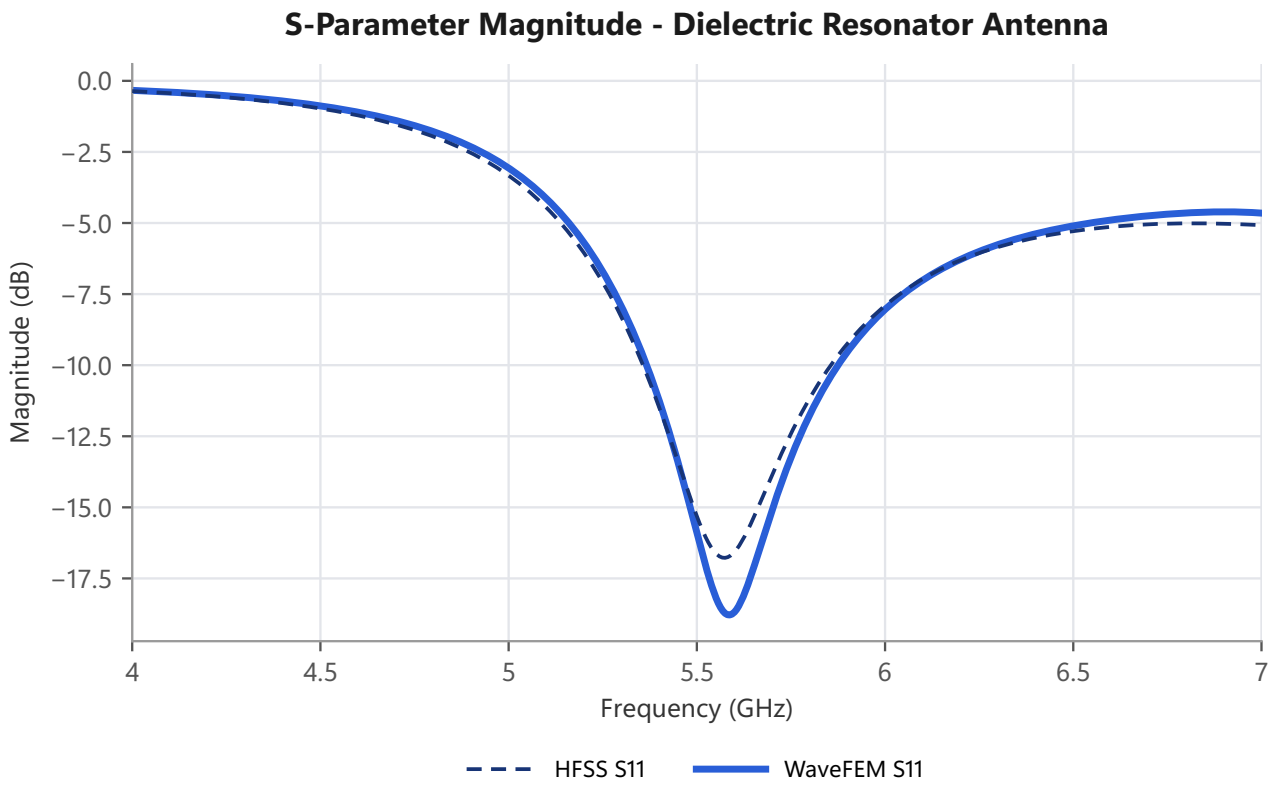


Figure 26 — S-parameter magnitude.

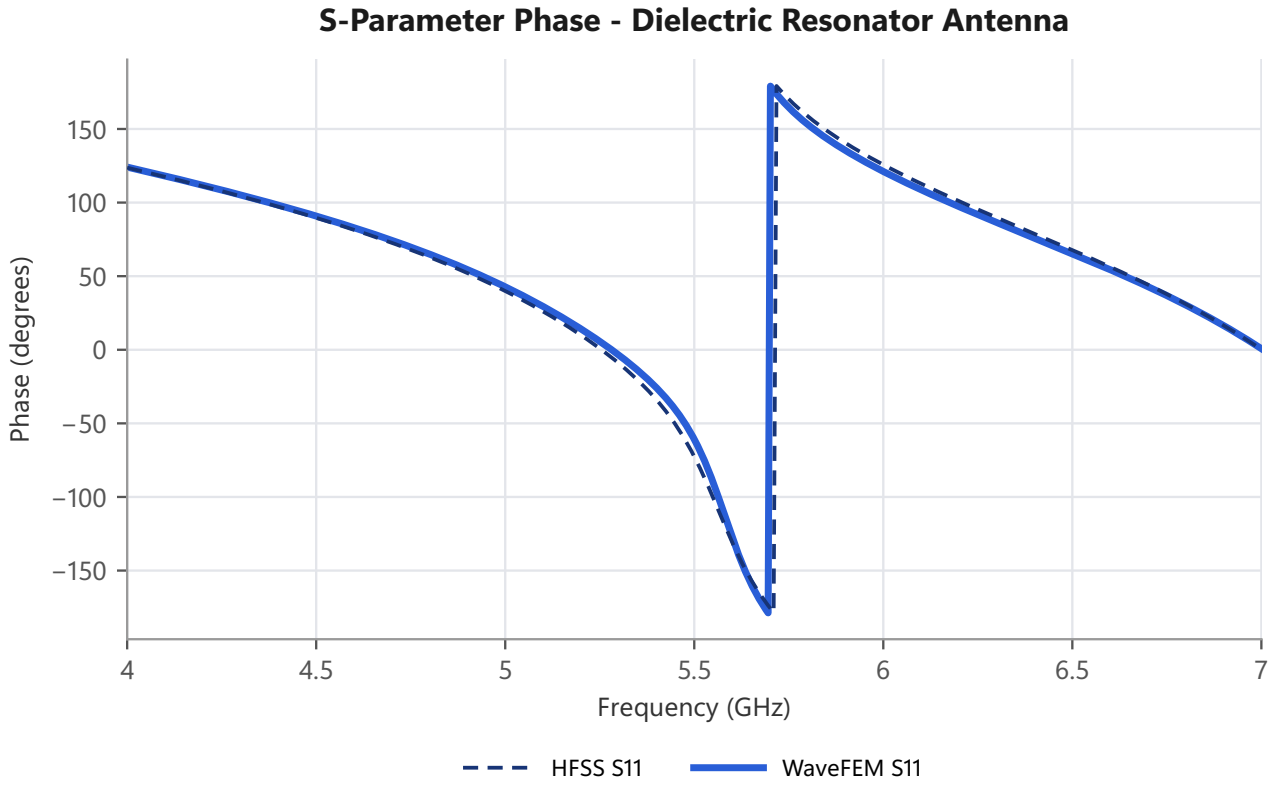


Figure 27 — S-parameter phase.

Far Field

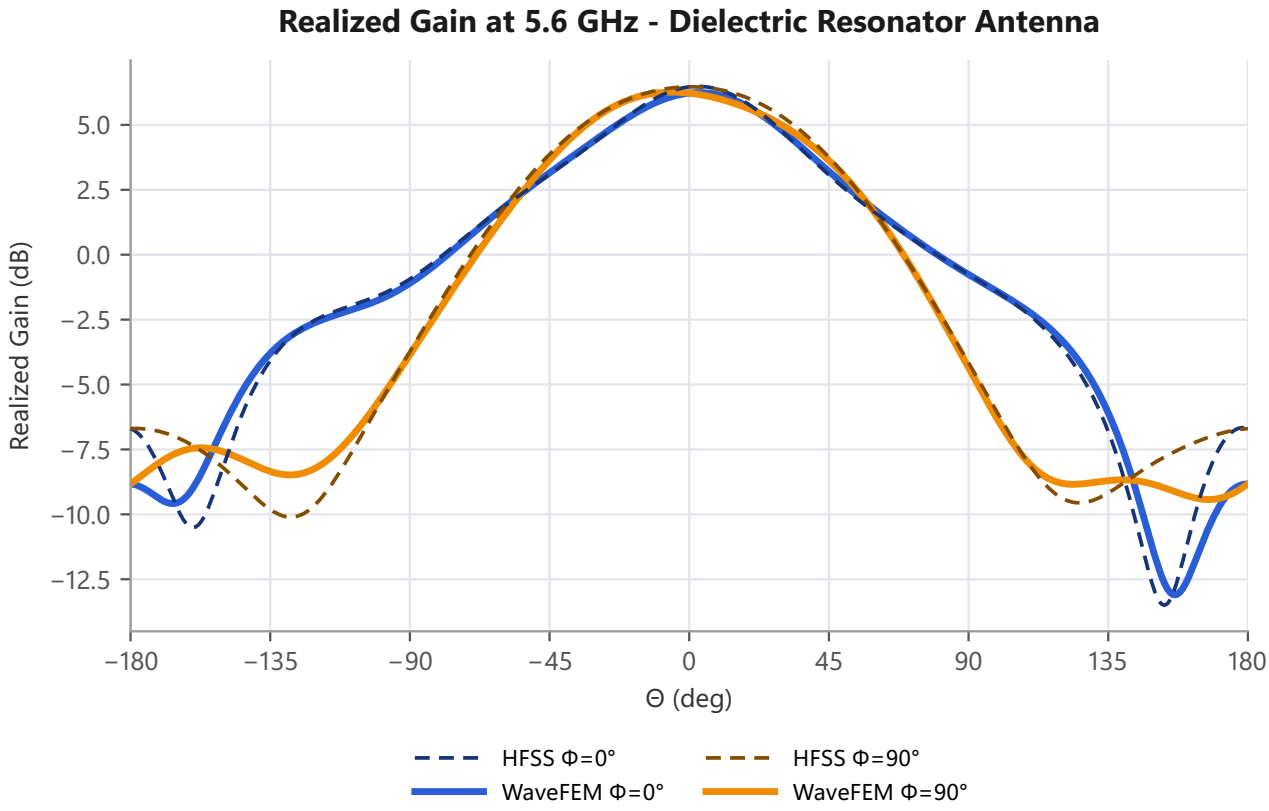


Figure 28 — Total realized gain cuts at 5.6 GHz.

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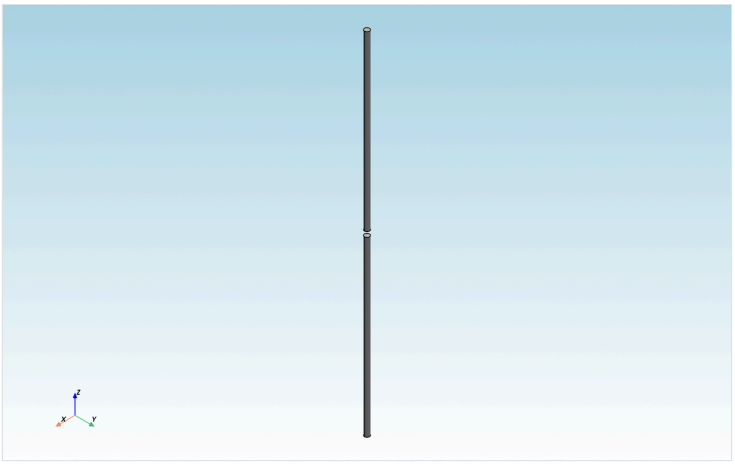
Dipole

A half-wave wire dipole along z: two PEC arms of radius 1 mm and length 69.2 mm (140.4 mm overall) across a 2 mm gap, fed by a 73 ohm lumped port and refined at 1 GHz.

Simulation Setup

Refinement frequency	1 GHz
Delta S	0.02
Frequency range	0.8 - 1.2 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:24	00:00:32
Adaptive meshing	00:00:13	00:00:21
Frequency sweep	00:00:09	00:00:08
Adaptive passes	4	8
Final elements	19,906	9,761
Matrix size	120,250	60,767

Convergence

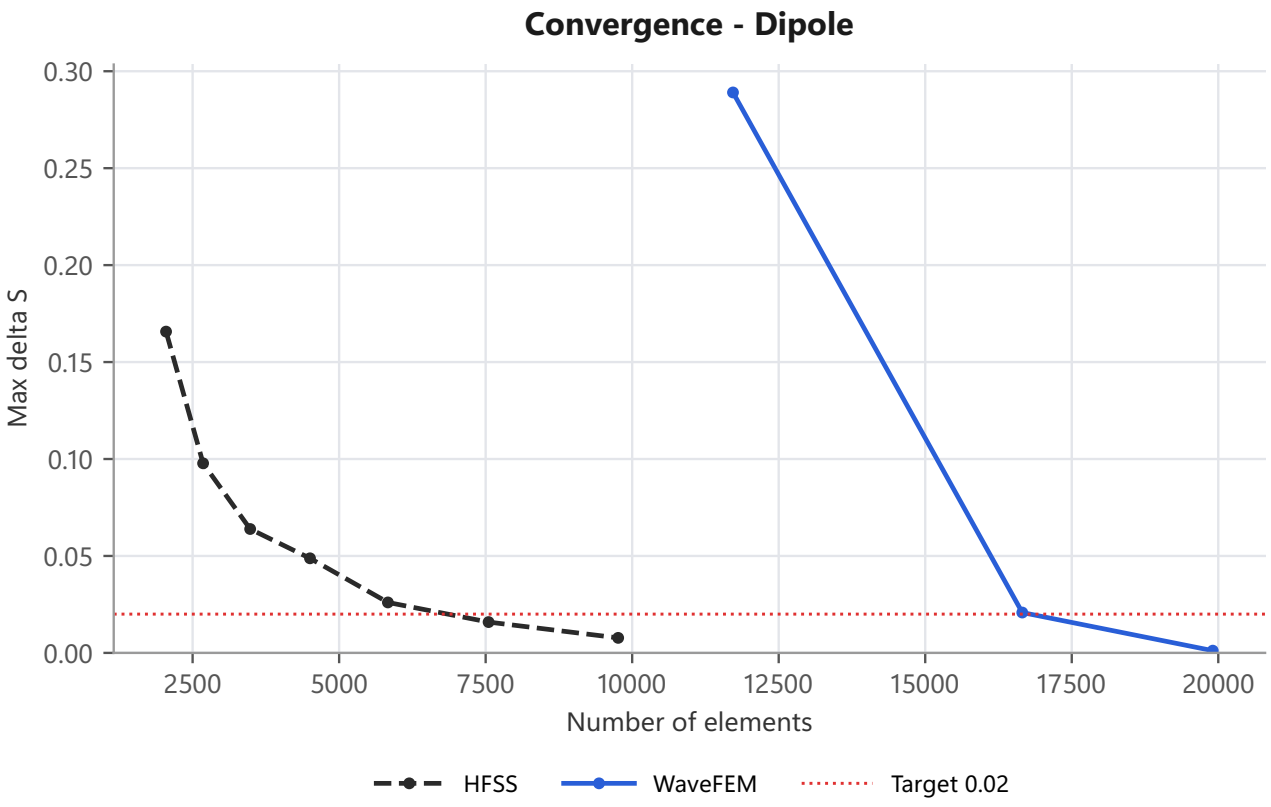


Figure 29 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

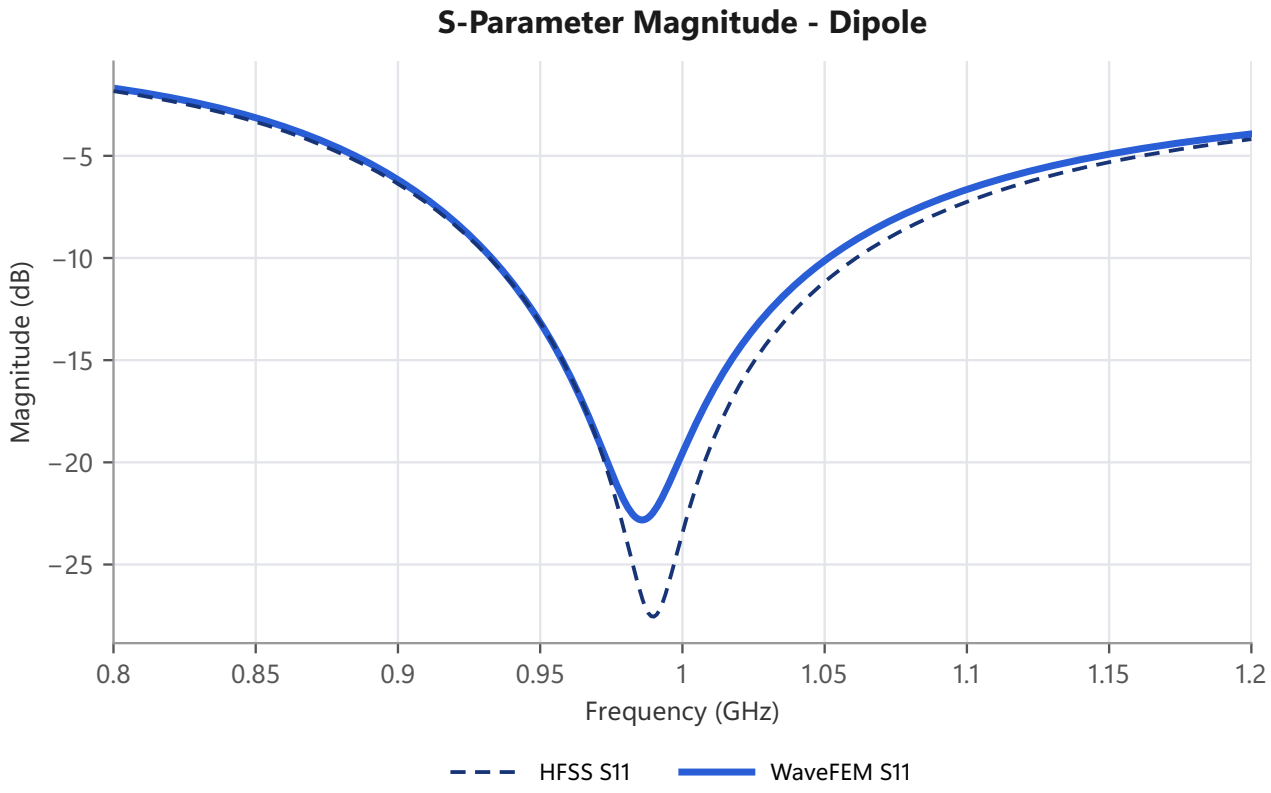


Figure 30 — S-parameter magnitude.

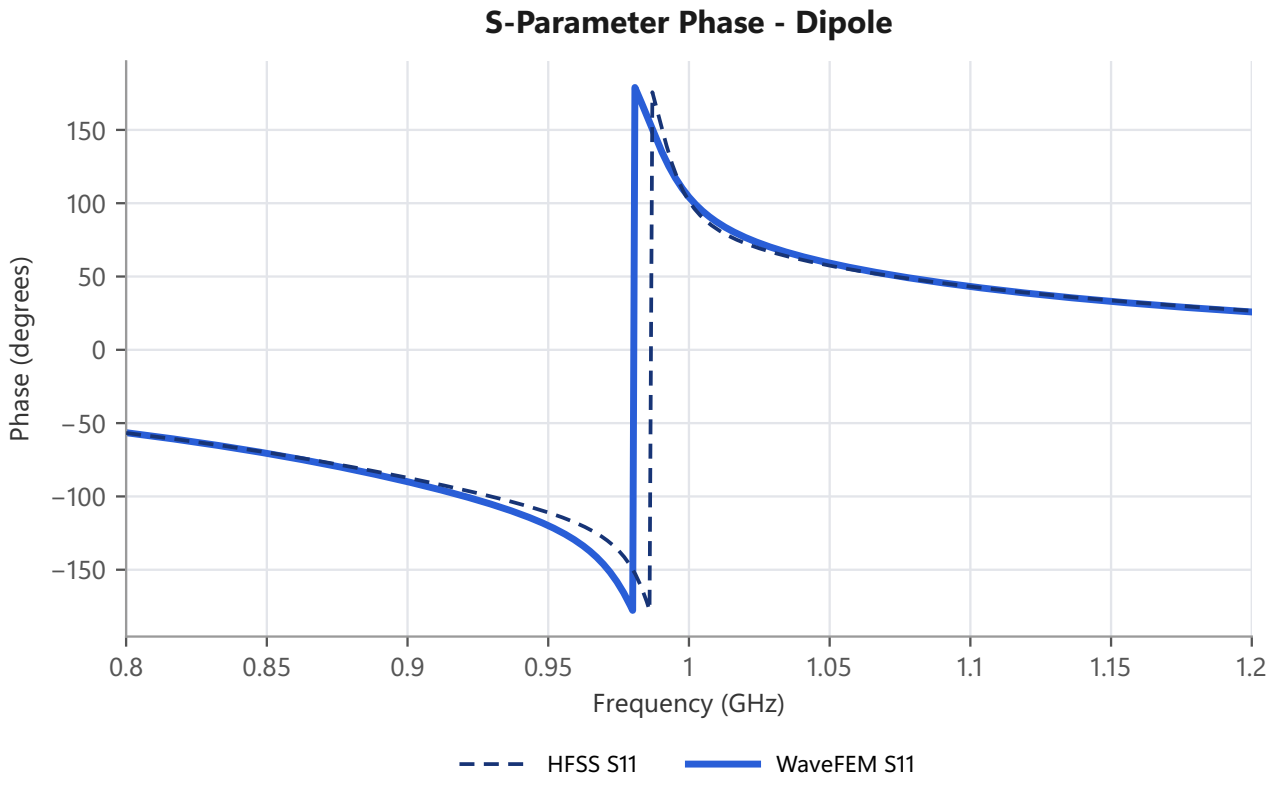


Figure 31 — S-parameter phase.

Far Field

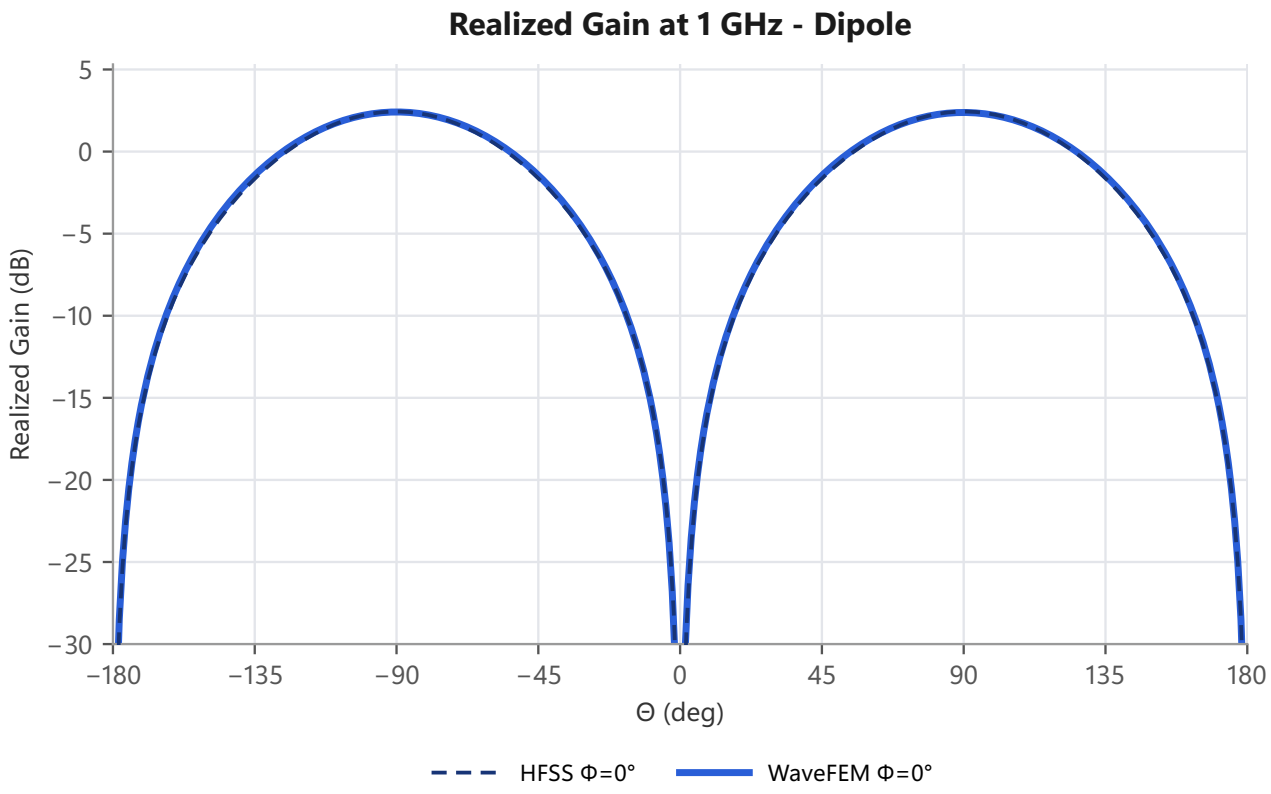


Figure 32 — Total realized gain cuts at 1 GHz.

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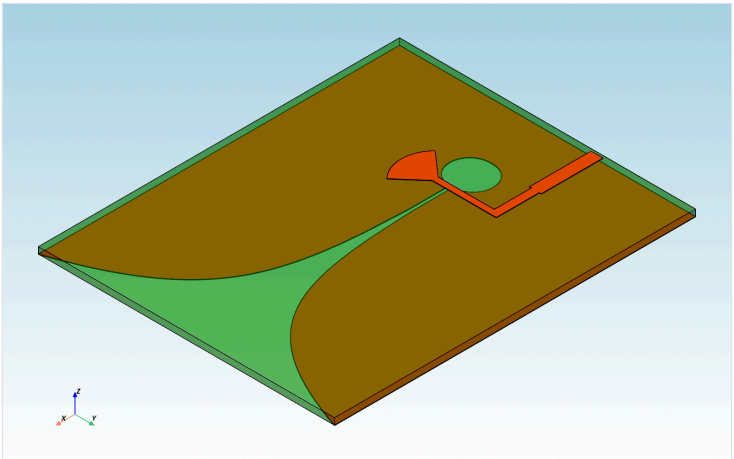
Planar Vivaldi

A wideband planar Vivaldi antenna, using radiation boundaries to terminate into free space.

Simulation Setup

Refinement frequency	11 GHz
Delta S	0.02
Frequency range	2 - 11 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:02:47	00:01:17
Adaptive meshing	00:00:46	00:00:15
Frequency sweep	00:01:55	00:00:58
Adaptive passes	7	4
Final elements	54,475	15,177
Matrix size	323,782	95,399

Convergence

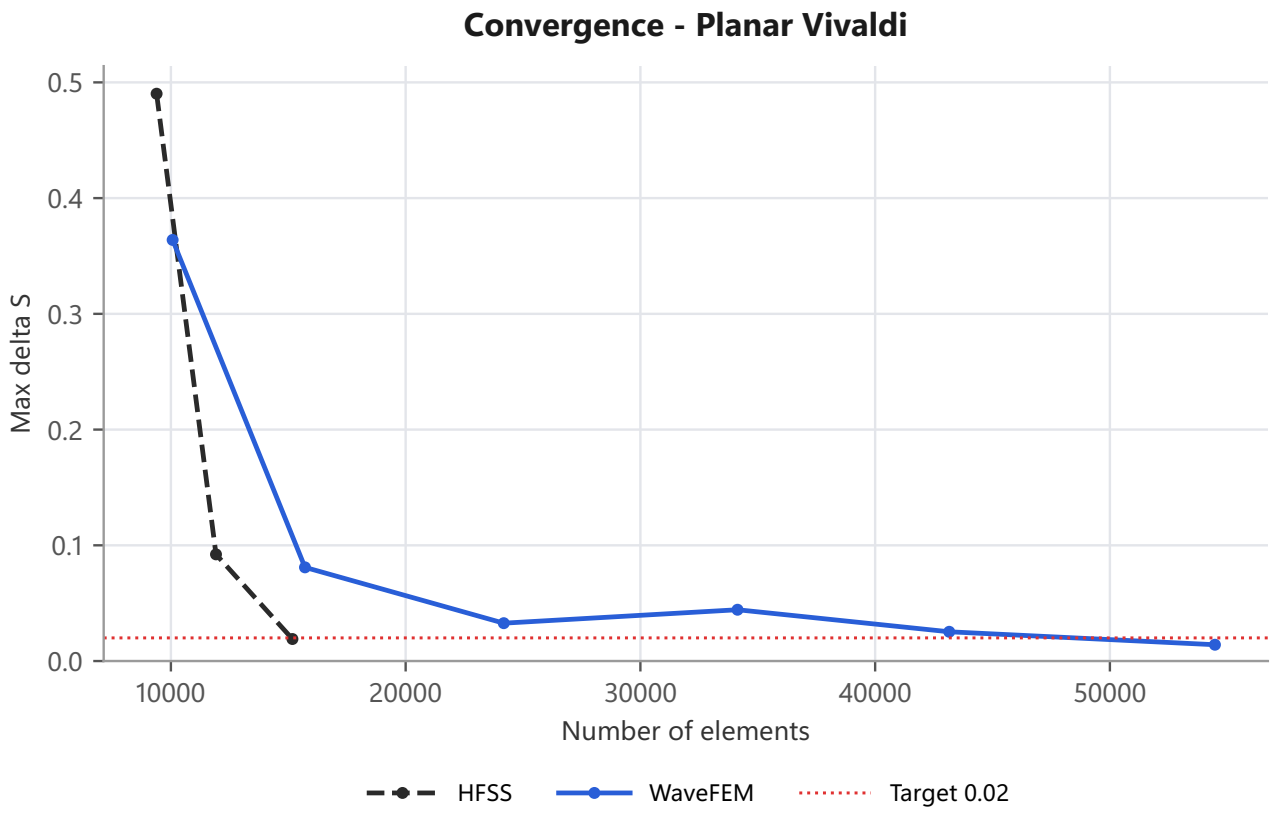


Figure 33 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

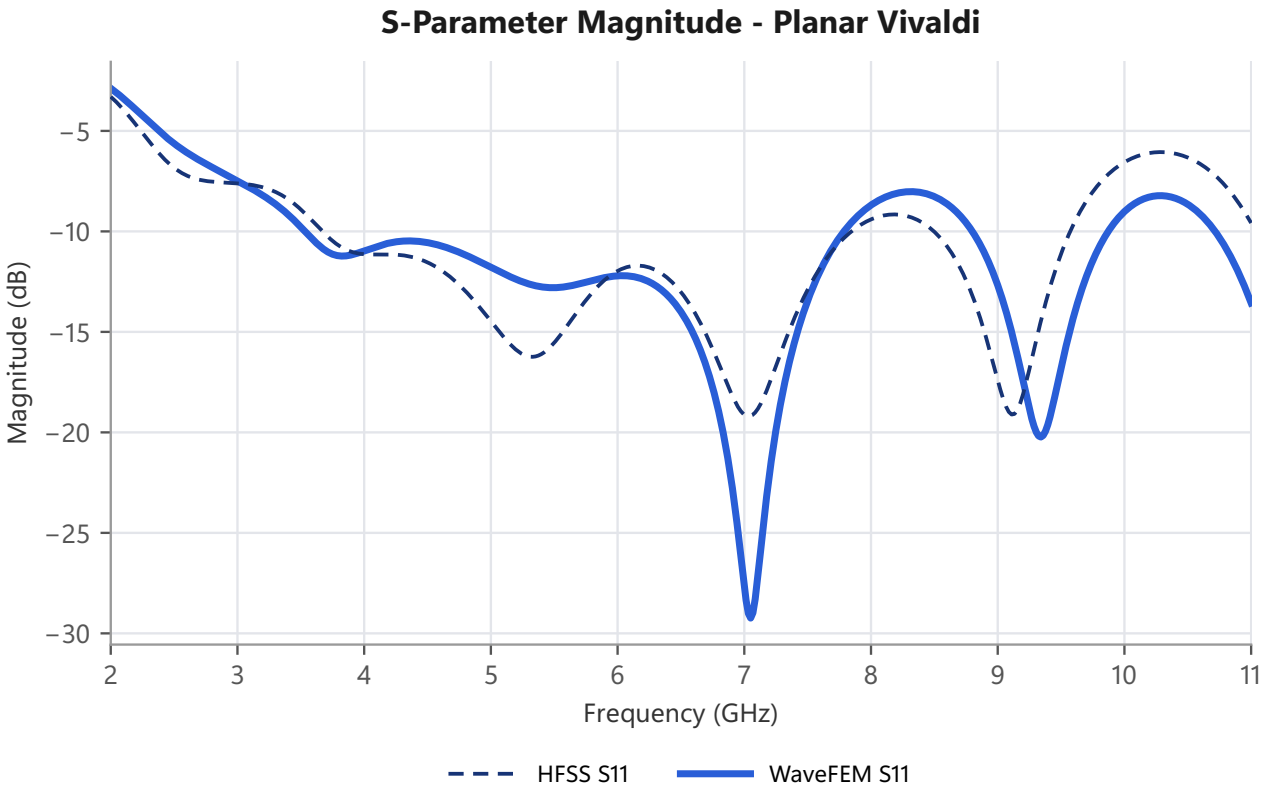


Figure 34 — S-parameter magnitude.

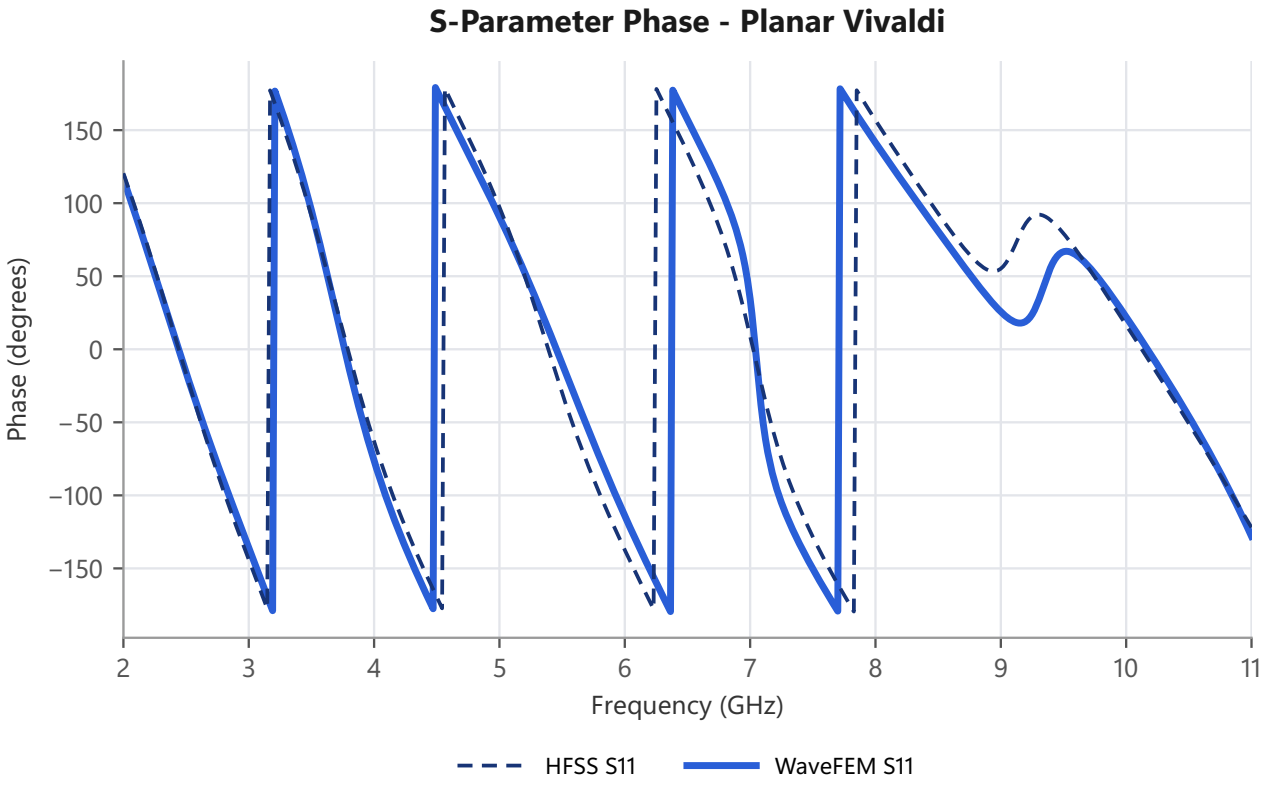


Figure 35 — S-parameter phase.

Far Field

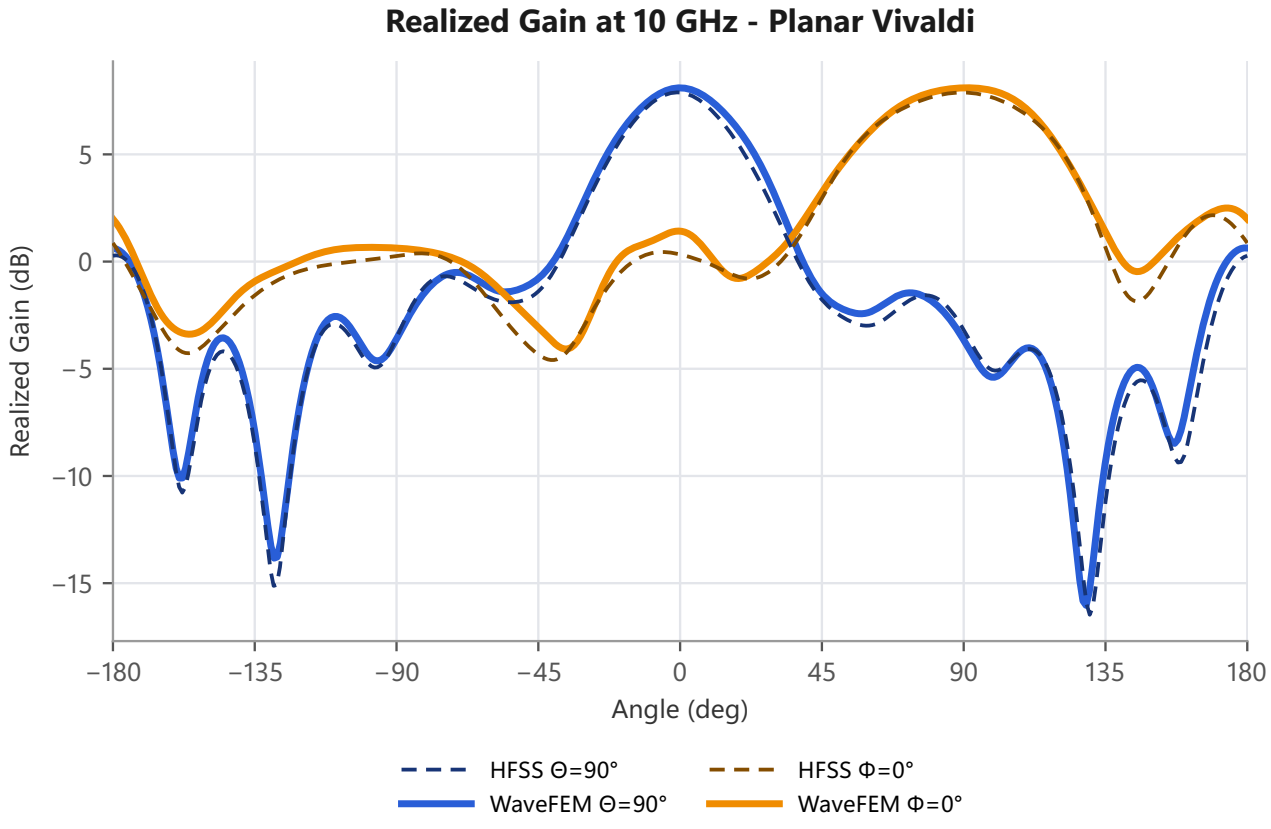


Figure 36 — Total realized gain cuts at 10 GHz.

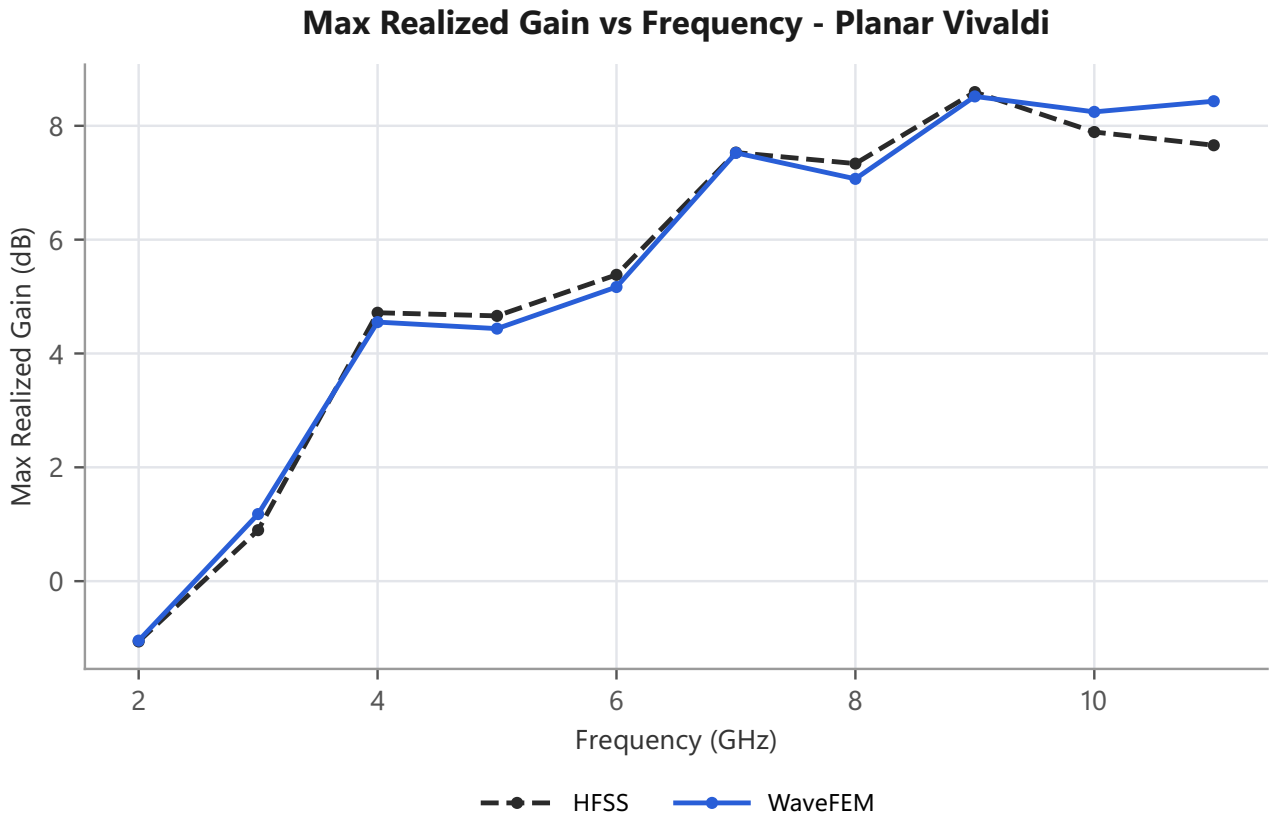


Figure 37 — Maximum realized gain against frequency.

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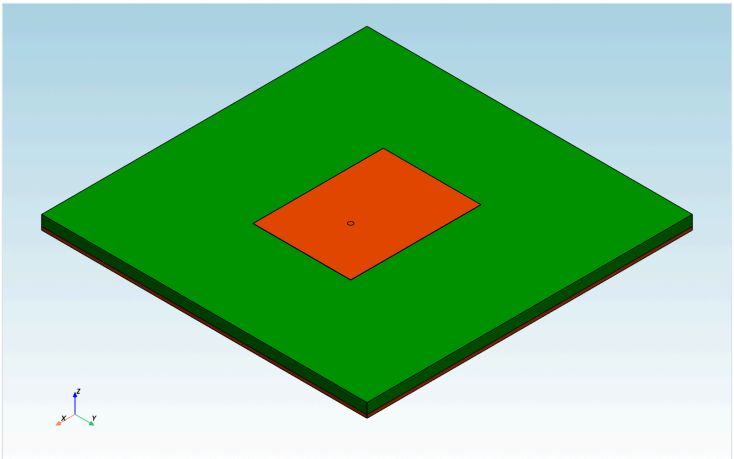
Probe Fed Patch

A coaxial probe fed rectangular patch antenna resonant at 2.4 GHz.

Simulation Setup

Refinement frequency	2.4 GHz
Delta S	0.02
Frequency range	2 - 3 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:29	00:01:14
Adaptive meshing	00:00:19	00:00:46
Frequency sweep	00:00:07	00:00:22
Adaptive passes	10	13
Final elements	26,213	19,159
Matrix size	151,810	115,581

Convergence

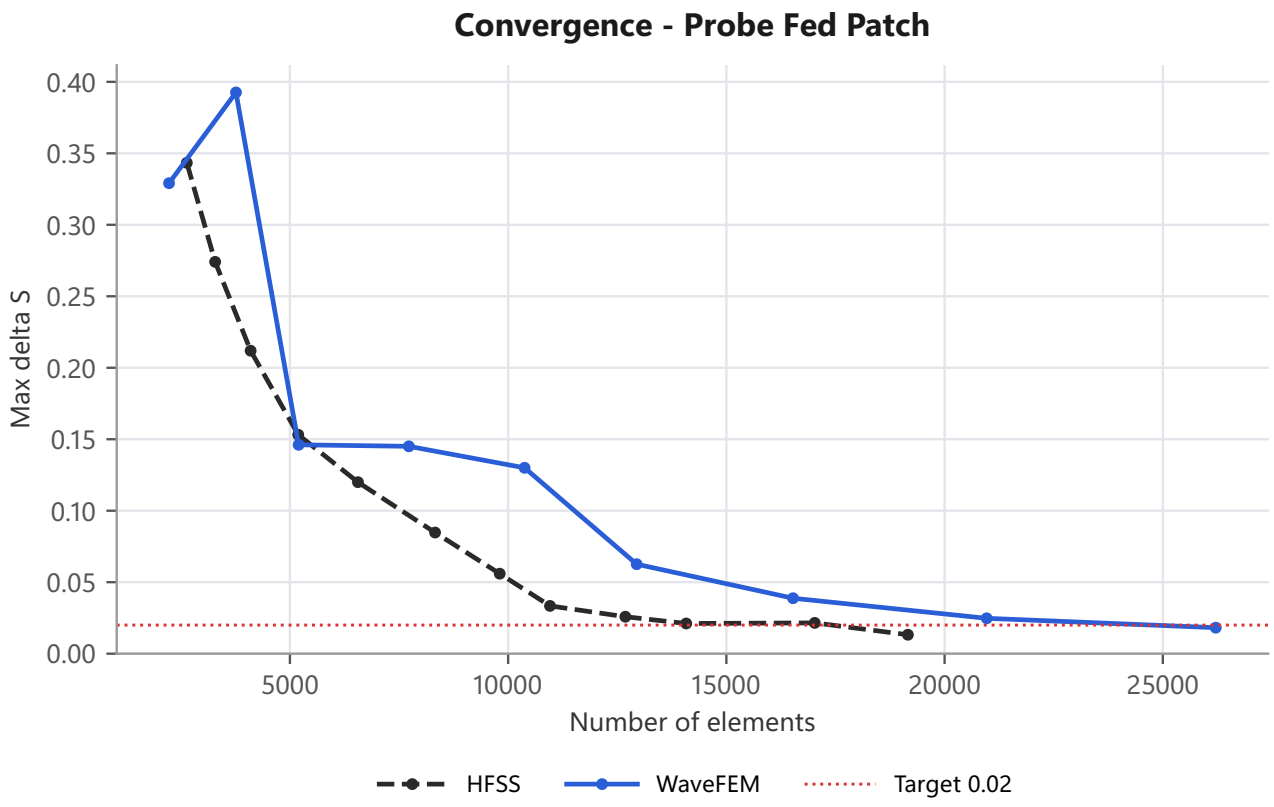


Figure 38 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

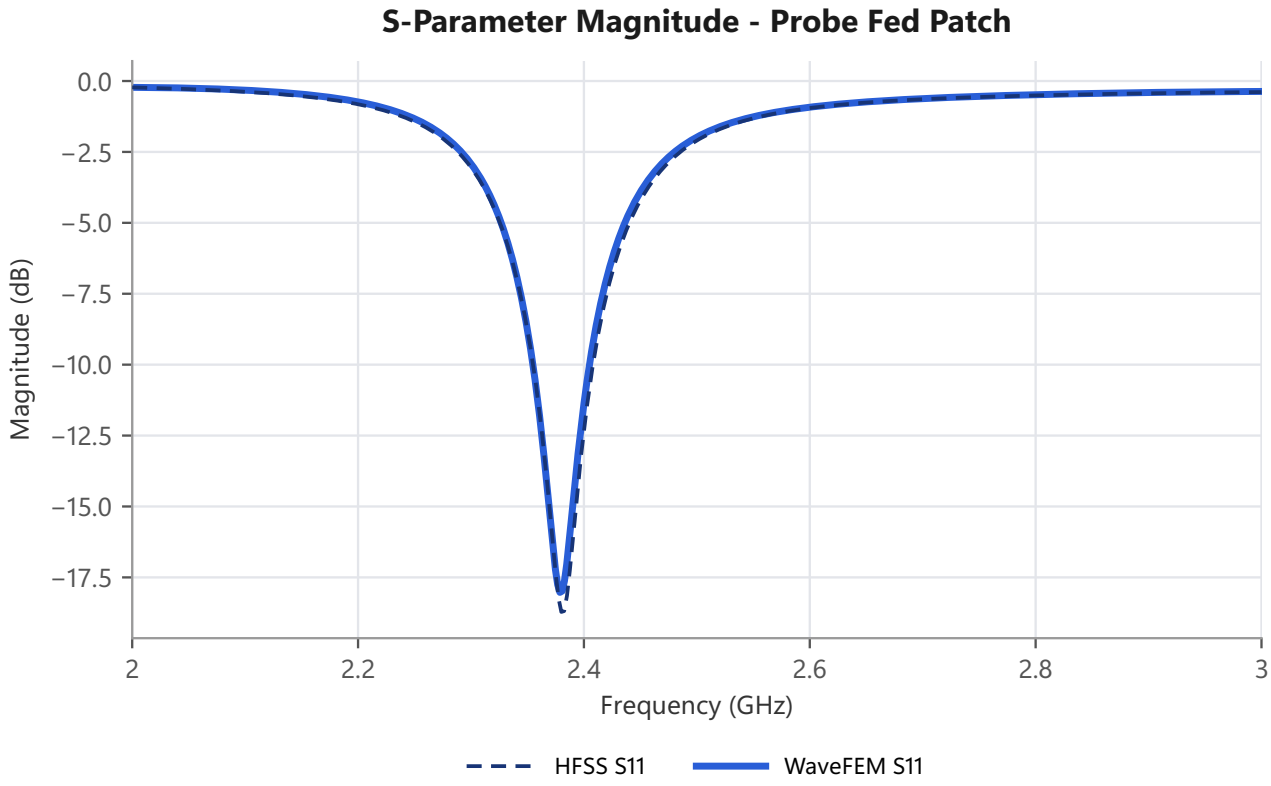


Figure 39 — S-parameter magnitude.

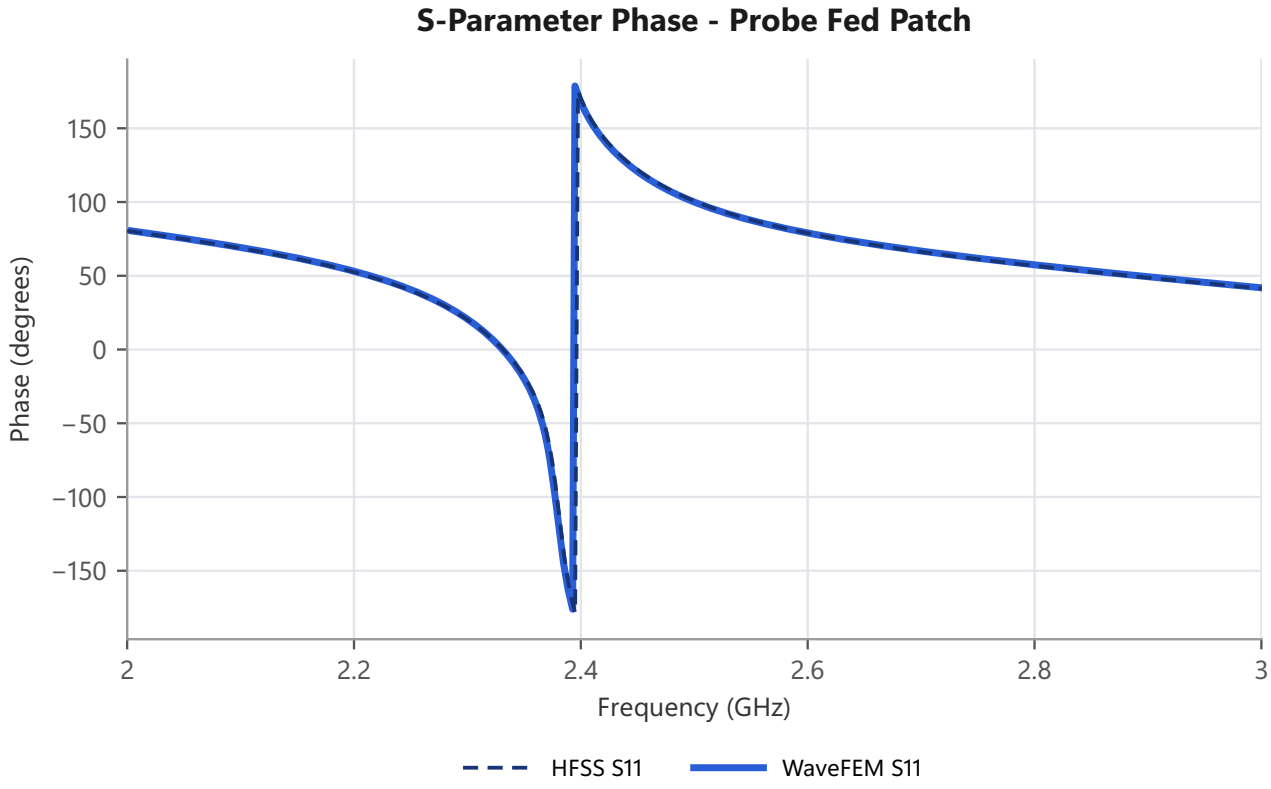


Figure 40 — S-parameter phase.

Far Field

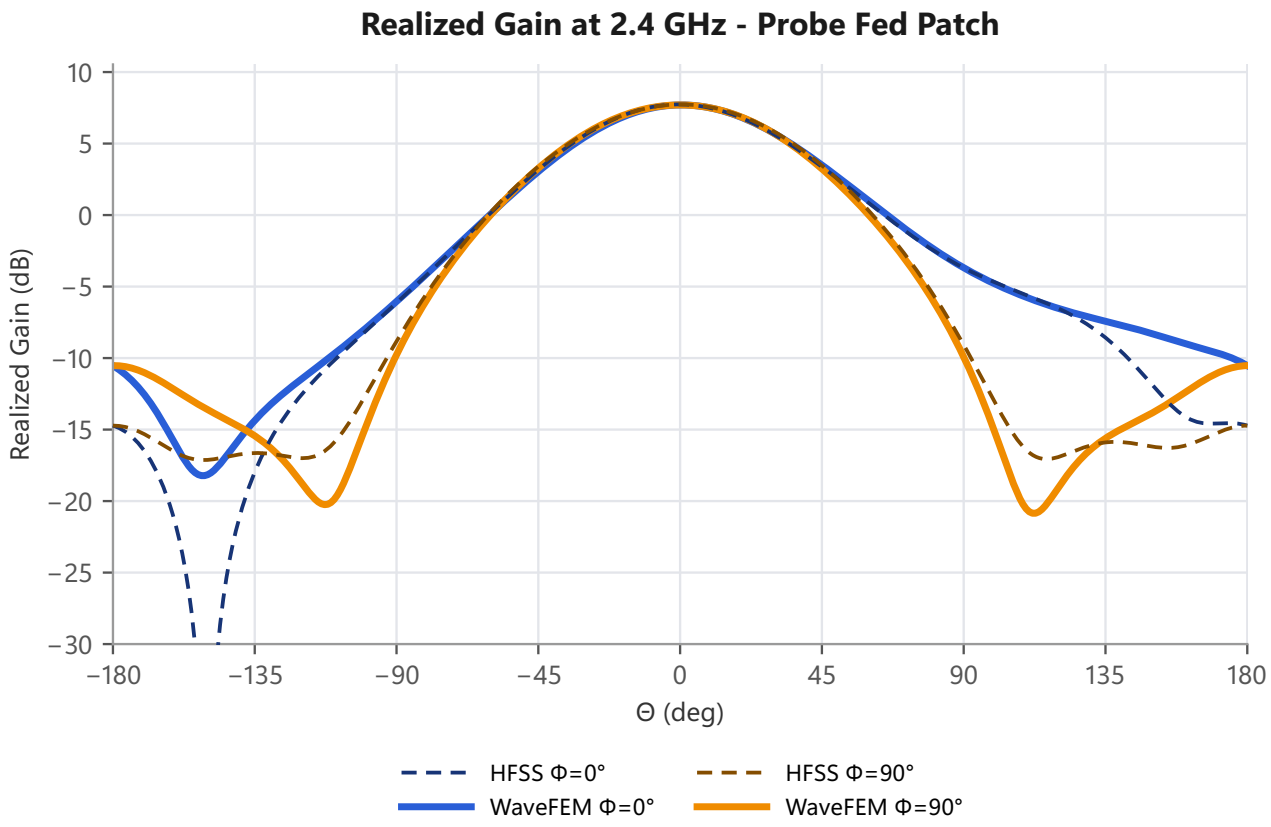


Figure 41 — Total realized gain cuts at 2.4 GHz.

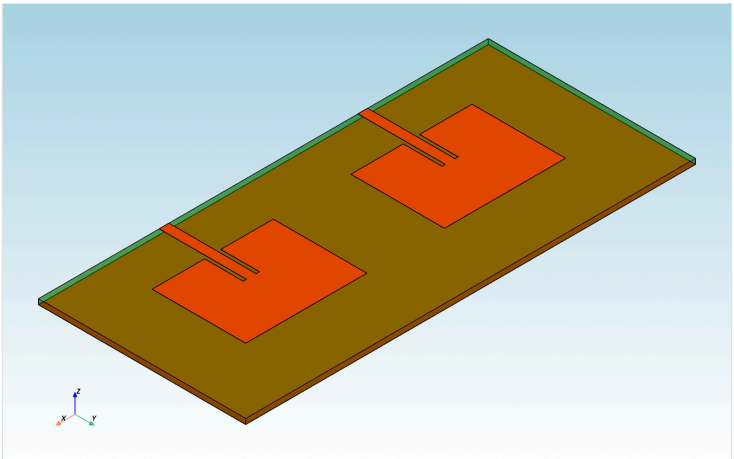
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Two Element Patch Array

Simulation Setup

Refinement frequency	2.45 GHz
Delta S	0.02
Frequency range	2 - 3 GHz
Excitation	port 1: 1 W, 0°; port 2: 1 W, 0°

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:54	00:02:16
Adaptive meshing	00:01:05	00:01:27
Frequency sweep	00:00:29	00:00:47
Adaptive passes	15	18
Final elements	73,331	42,337
Matrix size	415,834	253,708

Convergence

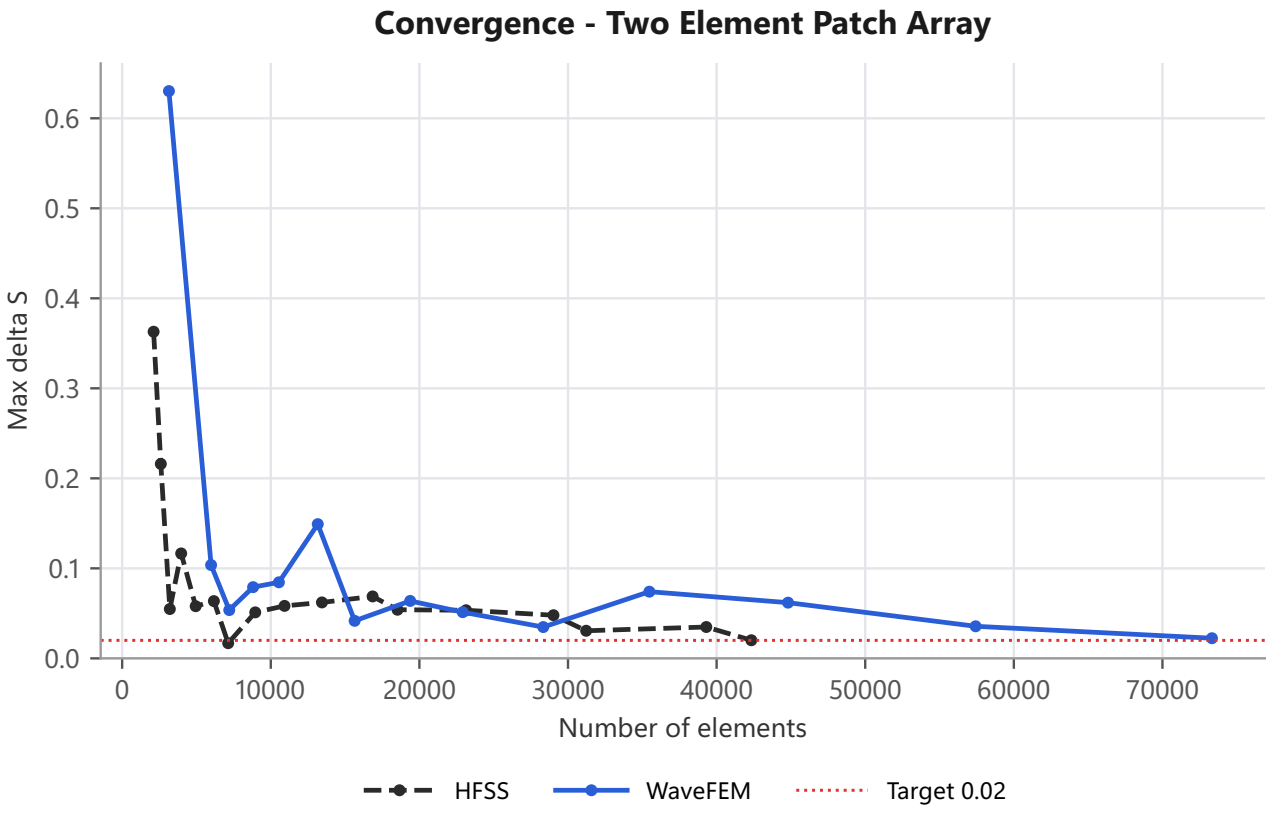


Figure 42 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

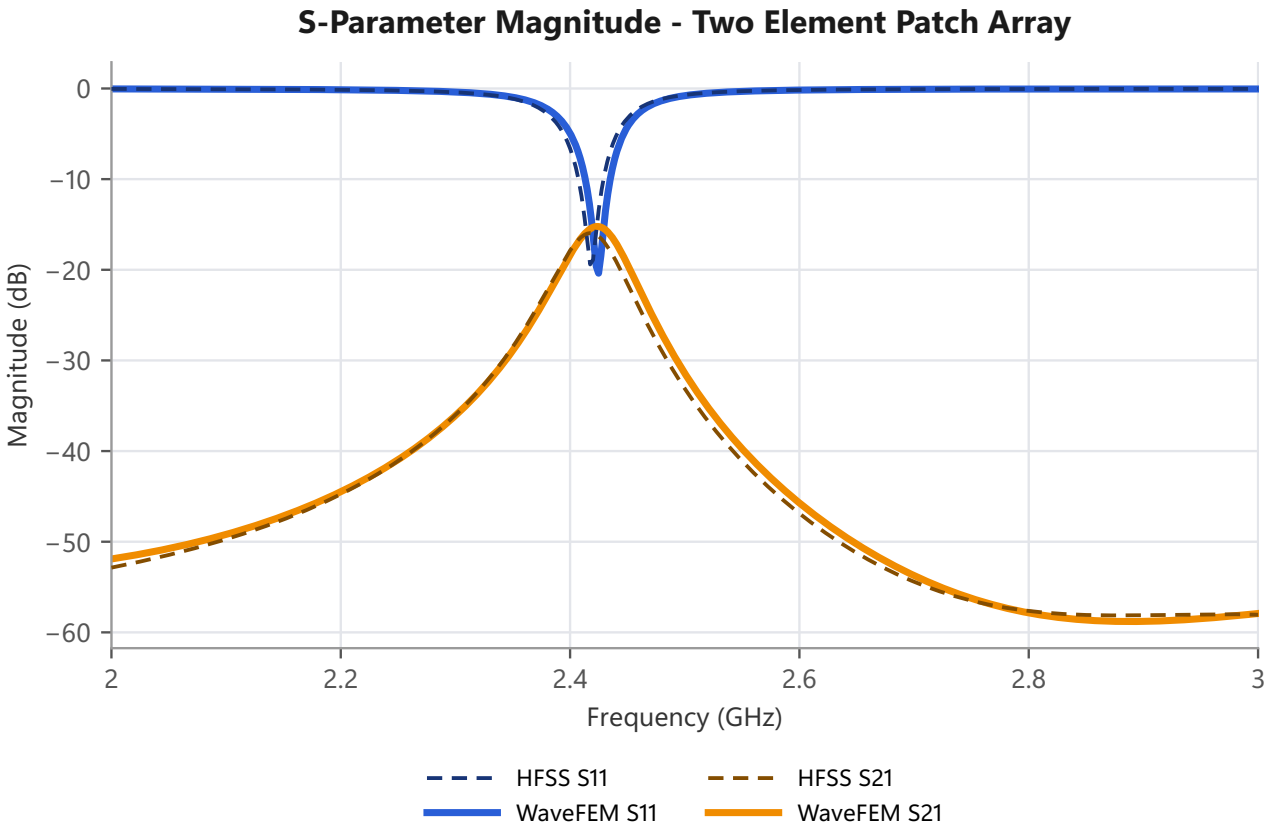


Figure 43 — S-parameter magnitude.

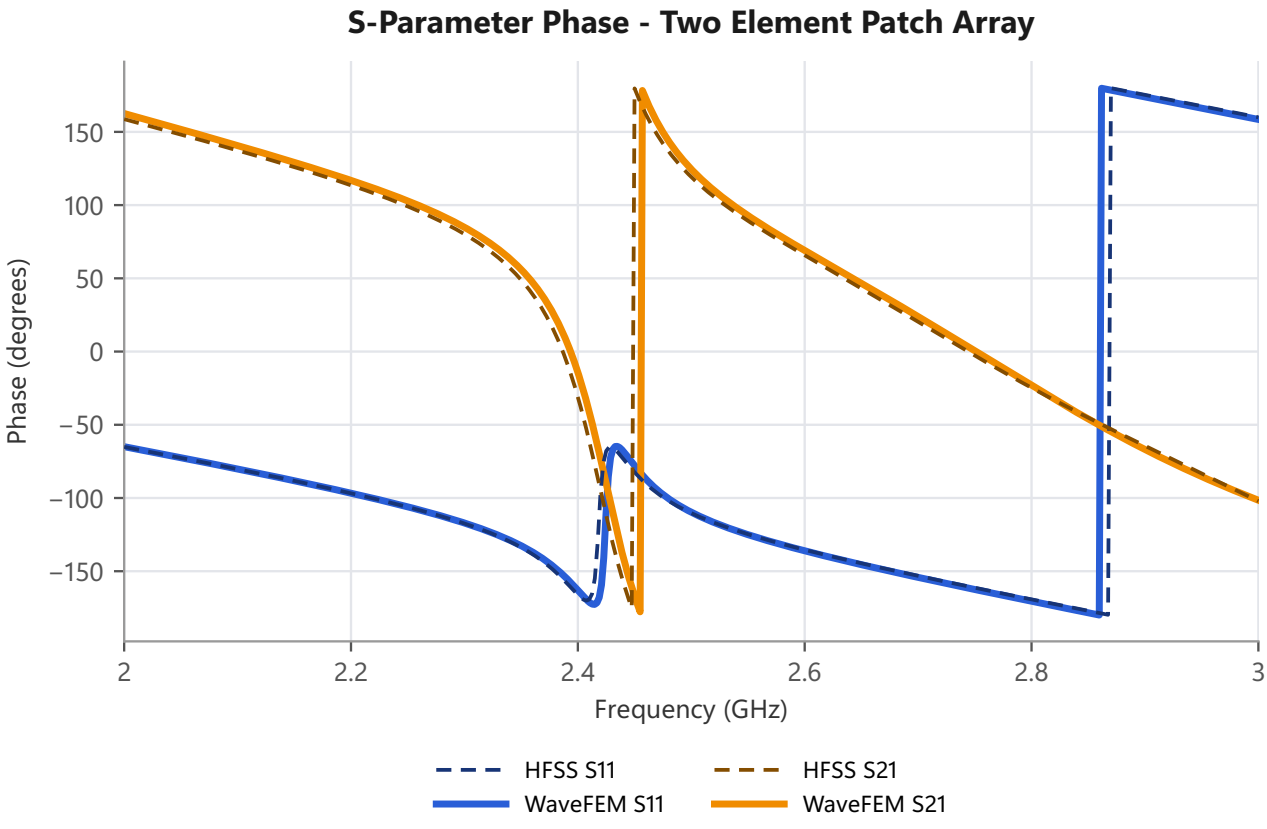


Figure 44 — S-parameter phase.

Far Field

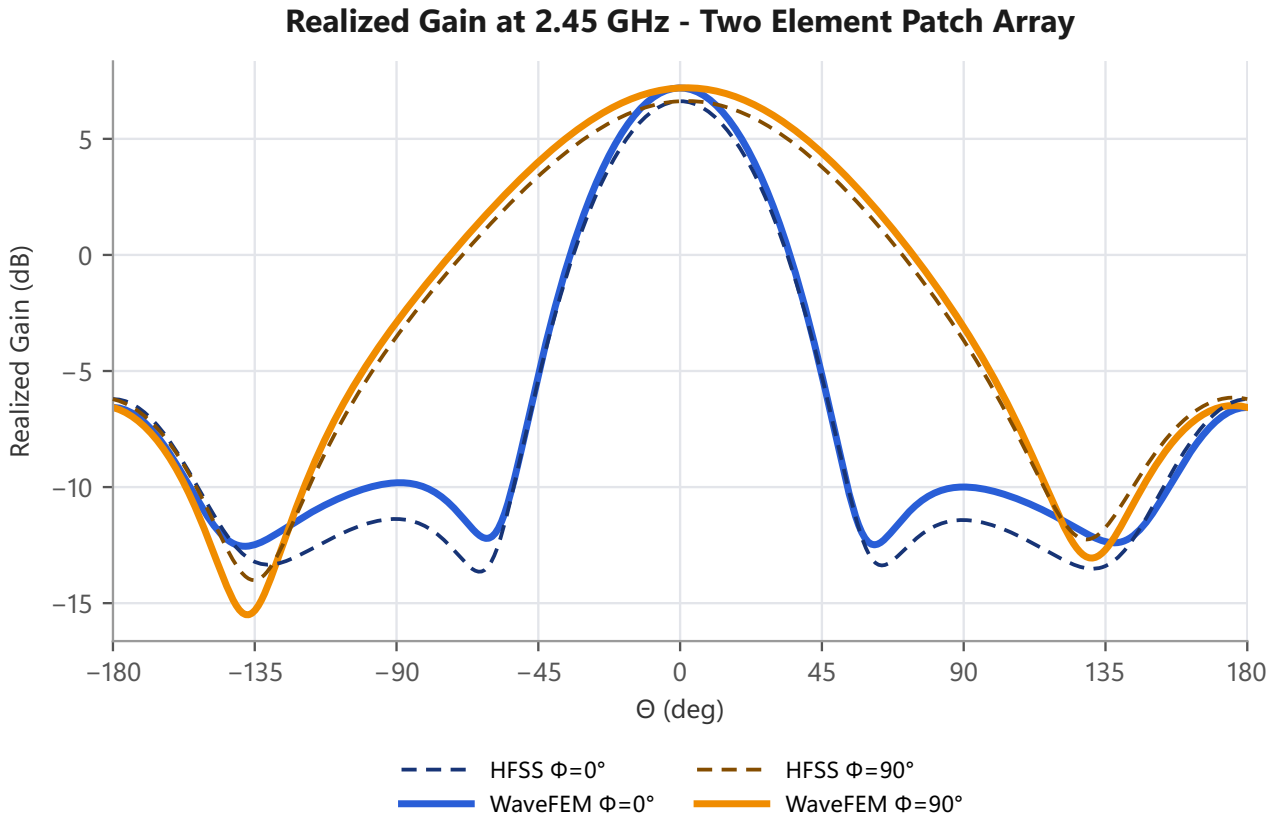


Figure 45 — Total realized gain cuts at 2.45 GHz.

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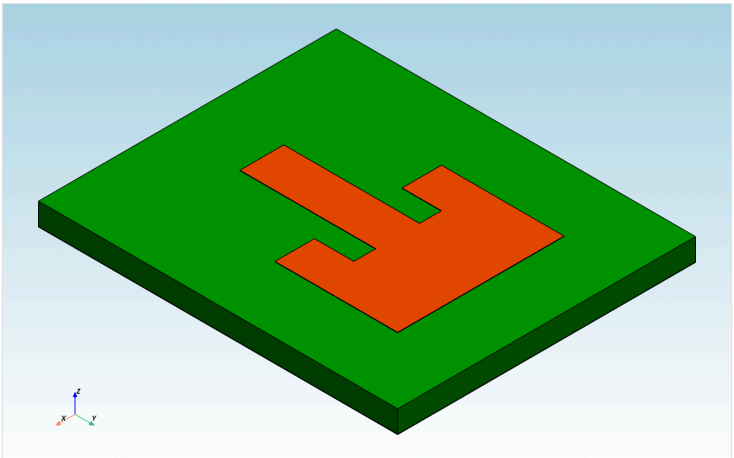
Inset Fed Patch

An inset fed microstrip patch on a 5 mm FR4 substrate (e_r=4.4), resonant at 2.41 GHz and fed by a lumped port at the end of the feed line.

Simulation Setup

Refinement frequency	2.4 GHz
Delta S	0.02
Frequency range	2 - 3 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:31	00:01:26
Adaptive meshing	00:00:20	00:00:45
Frequency sweep	00:00:08	00:00:39
Adaptive passes	13	14
Final elements	23,642	24,003
Matrix size	136,664	147,219

Convergence

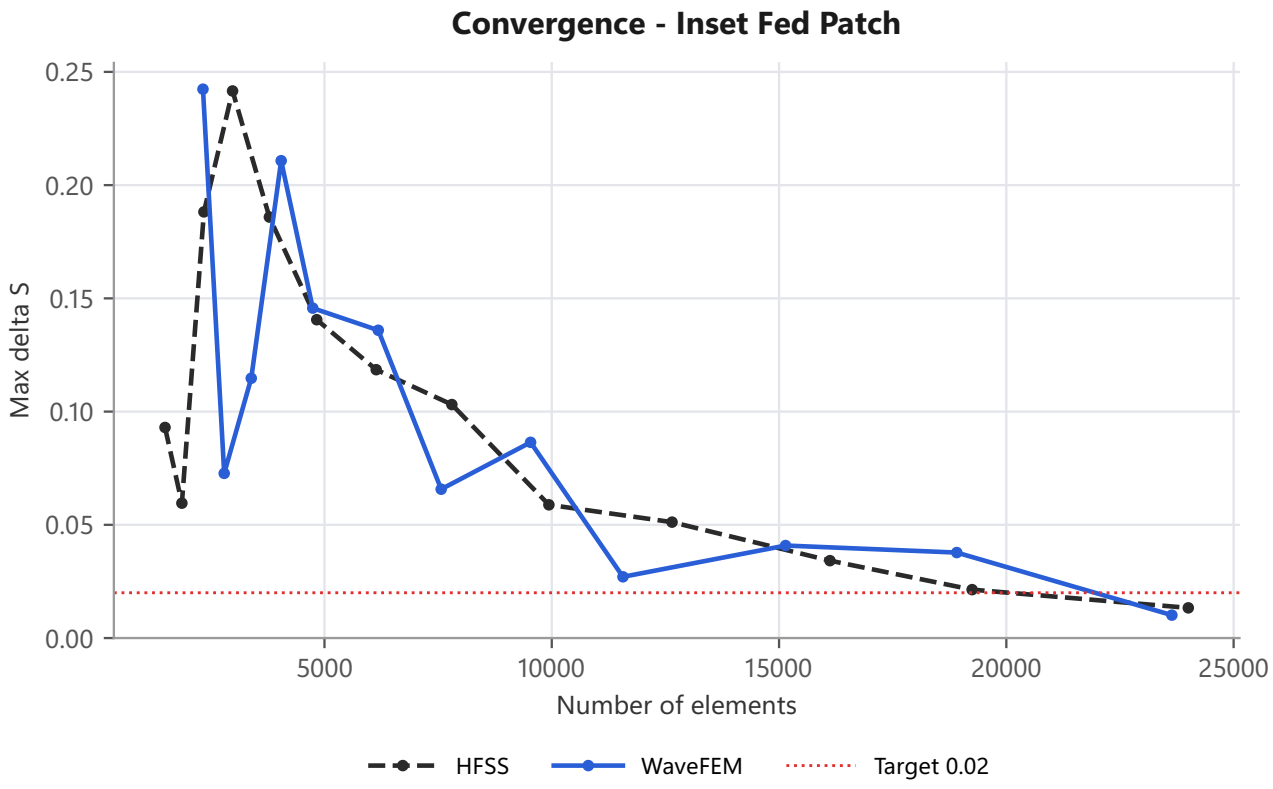


Figure 46 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

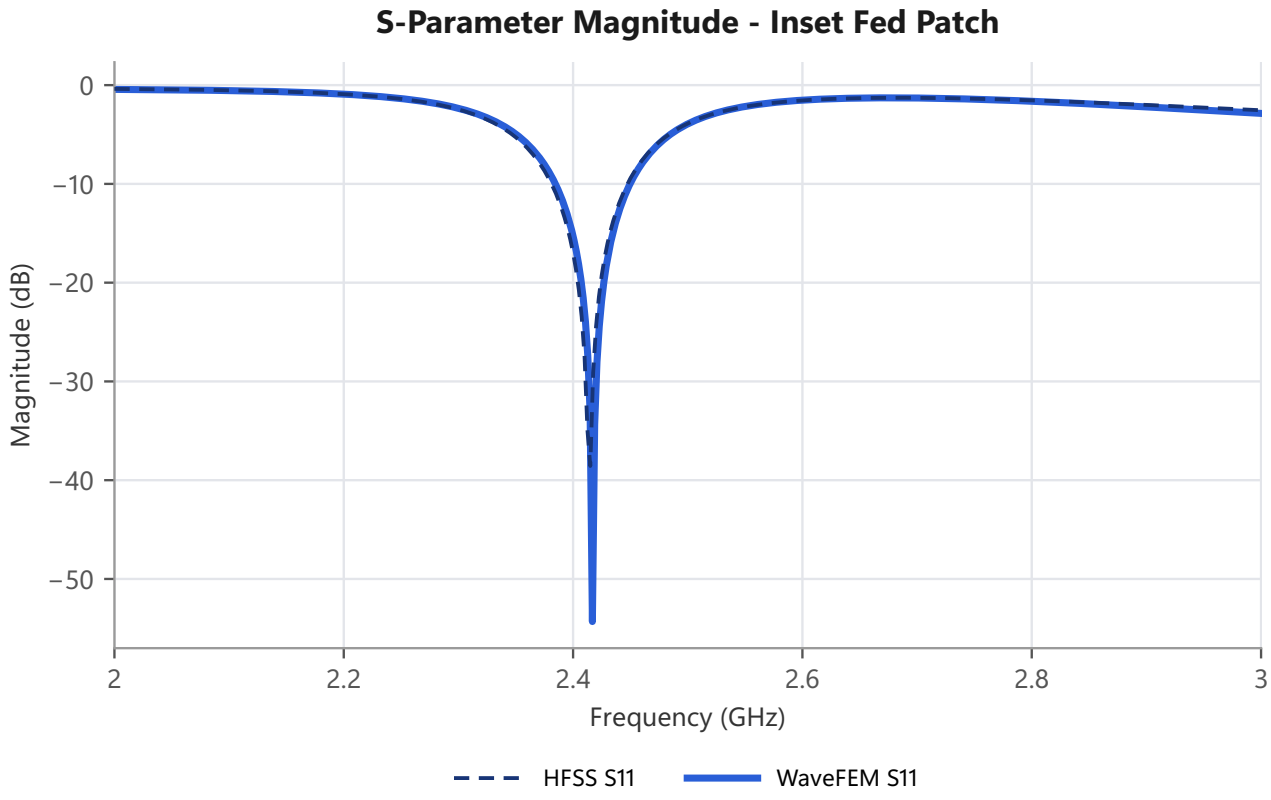


Figure 47 — S-parameter magnitude.

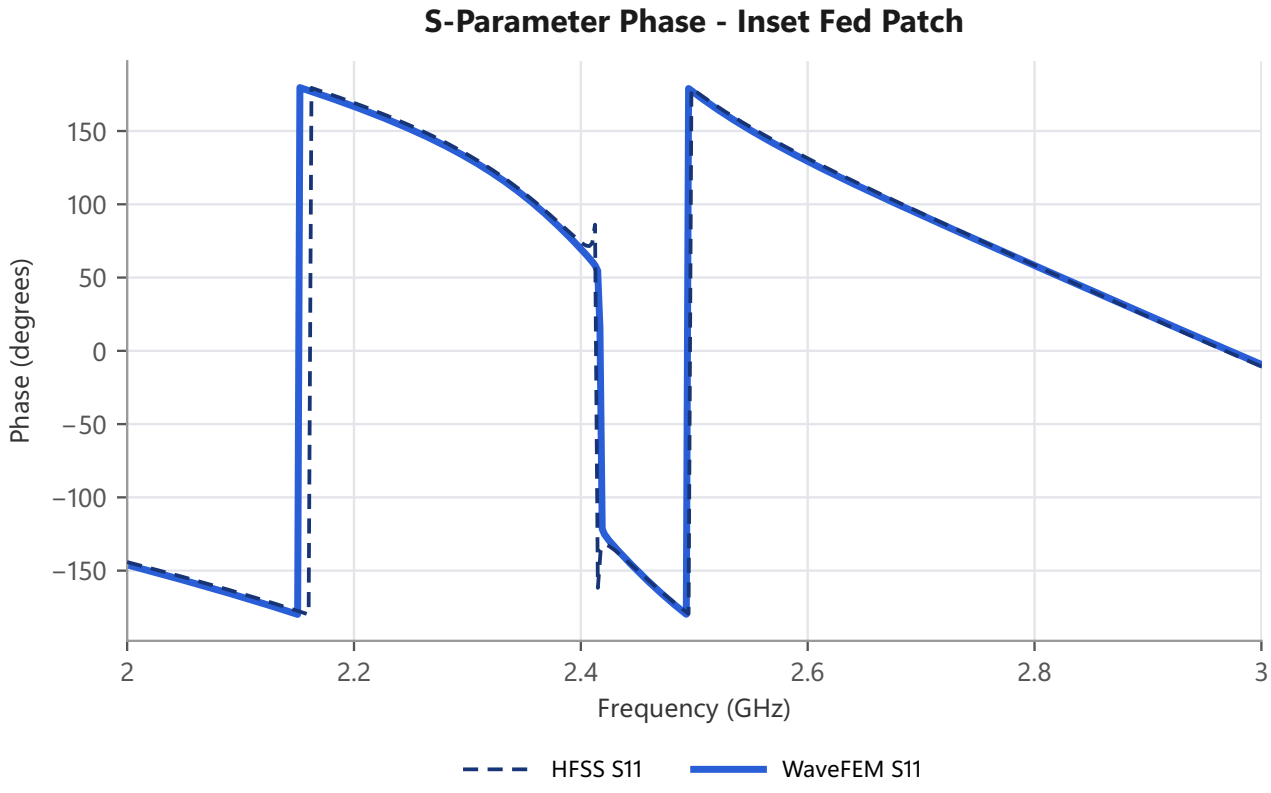


Figure 48 — S-parameter phase.

Far Field

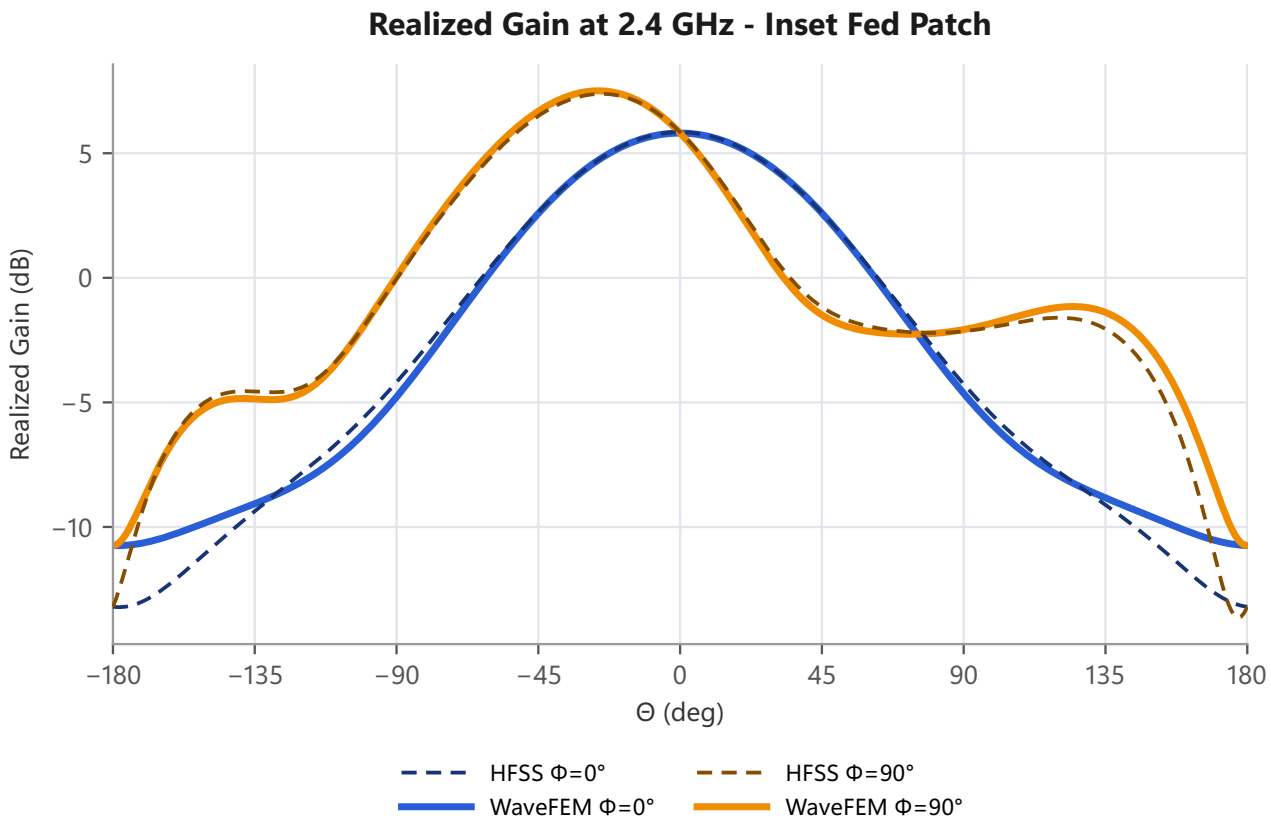


Figure 49 — Total realized gain cuts at 2.4 GHz.

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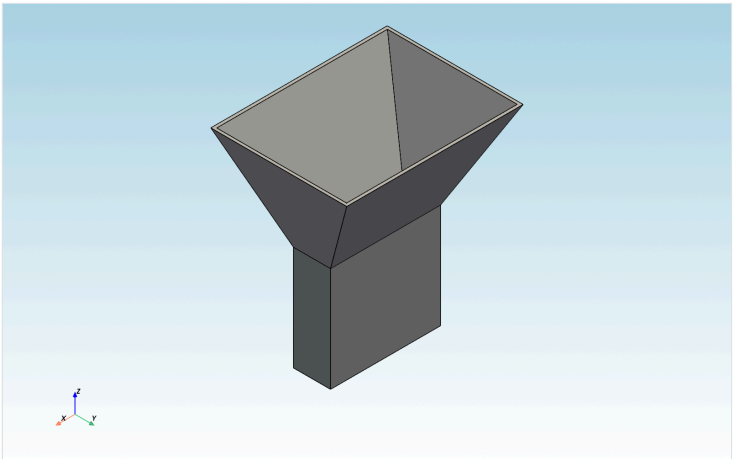
Pyramidal Horn

A pyramidal horn fed by a 62.3 x 18.9 mm rectangular waveguide.

Simulation Setup

Refinement frequency	6 GHz
Delta S	0.02
Frequency range	3 - 6 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:10	00:00:36
Adaptive meshing	00:00:03	00:00:05
Frequency sweep	00:00:07	00:00:28
Adaptive passes	3	2
Final elements	7,991	5,248
Matrix size	49,660	33,387

Convergence

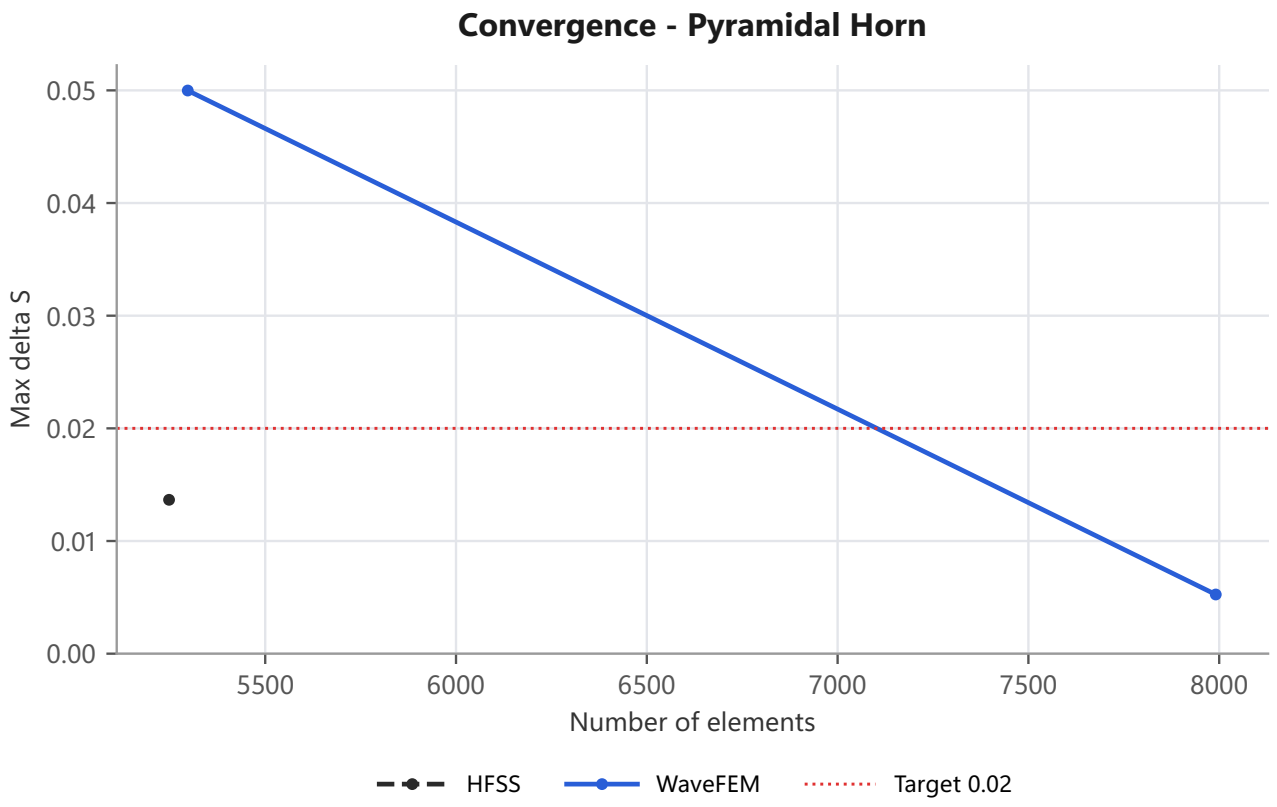


Figure 50 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

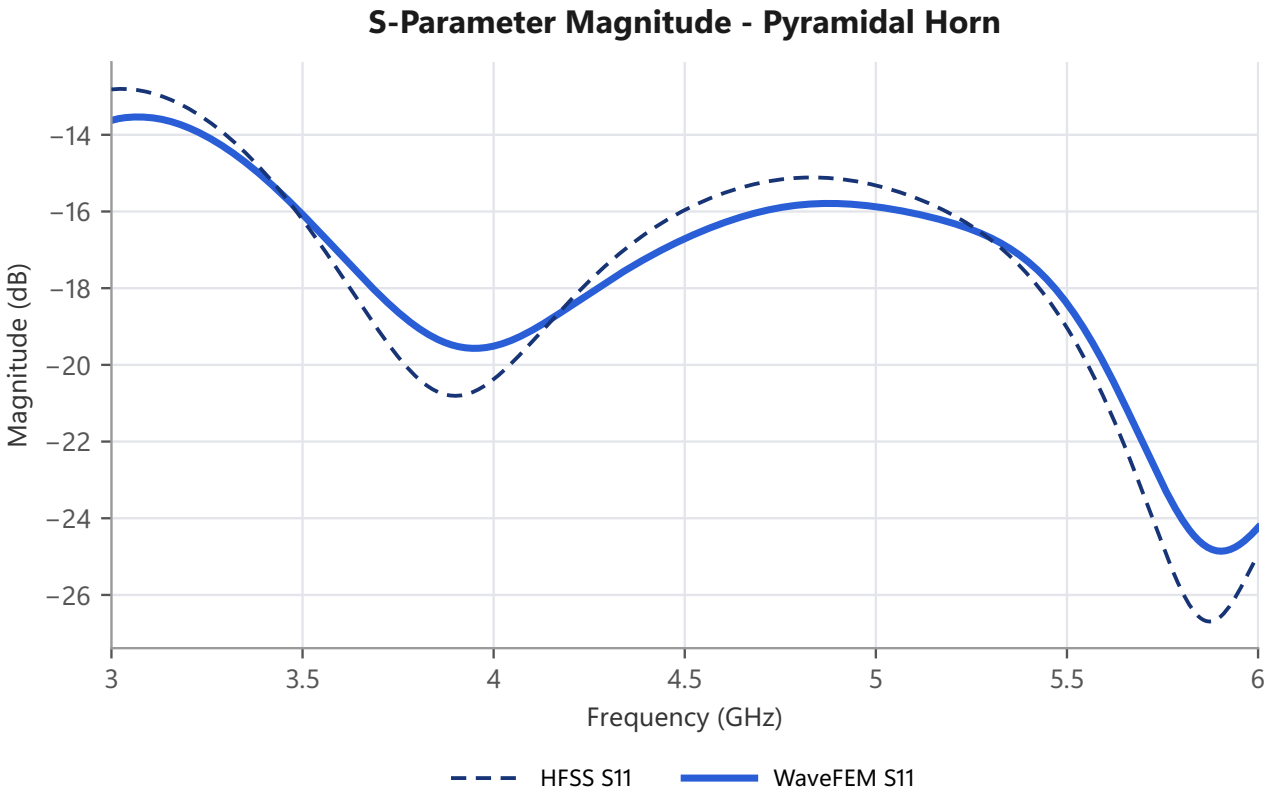


Figure 51 — S-parameter magnitude.

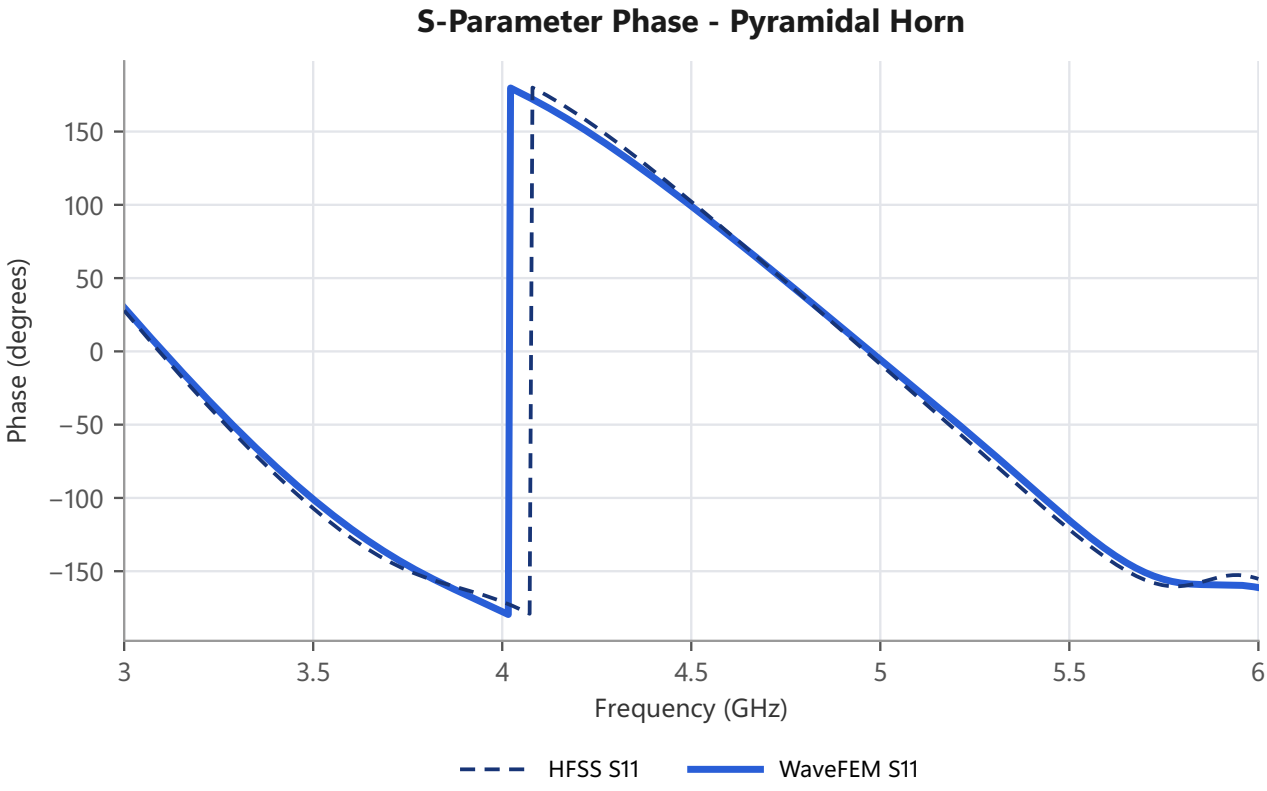


Figure 52 — S-parameter phase.

Far Field

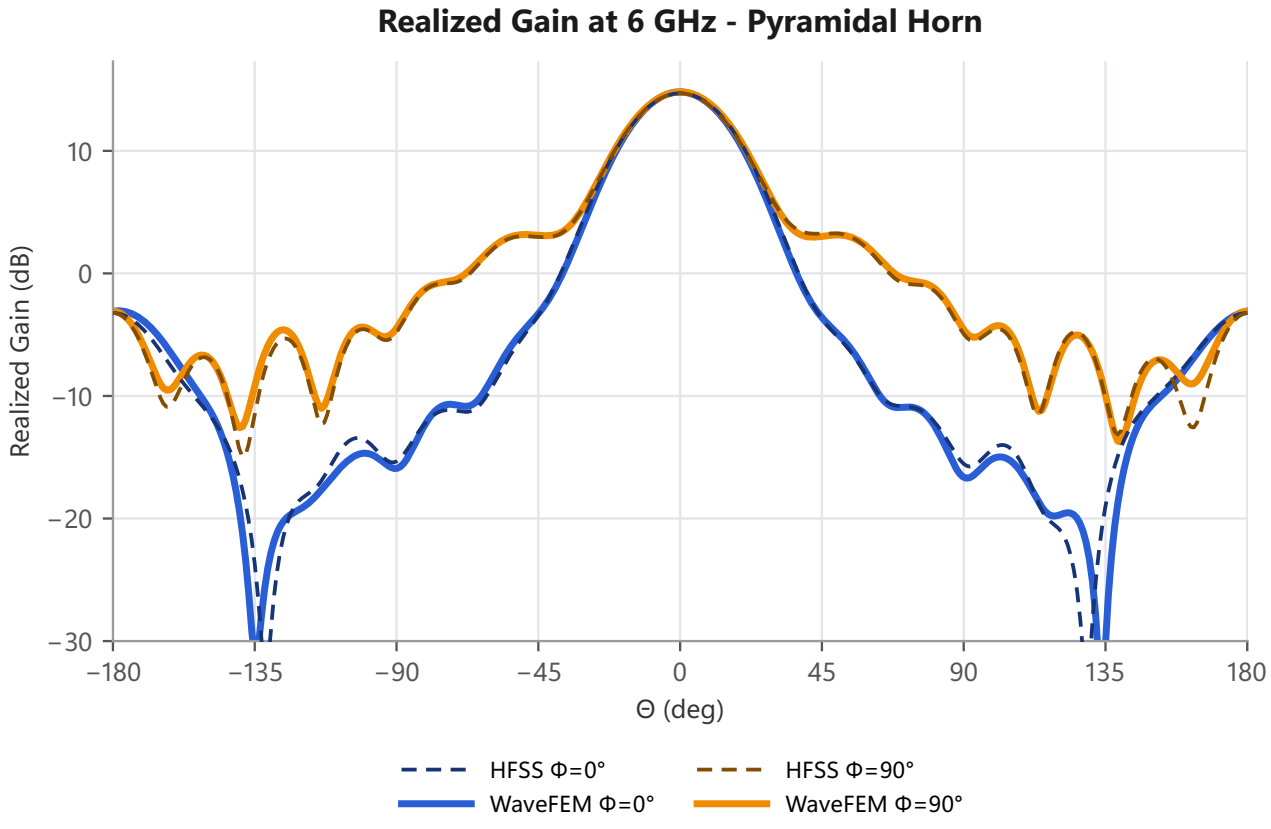


Figure 53 — Total realized gain cuts at 6 GHz.

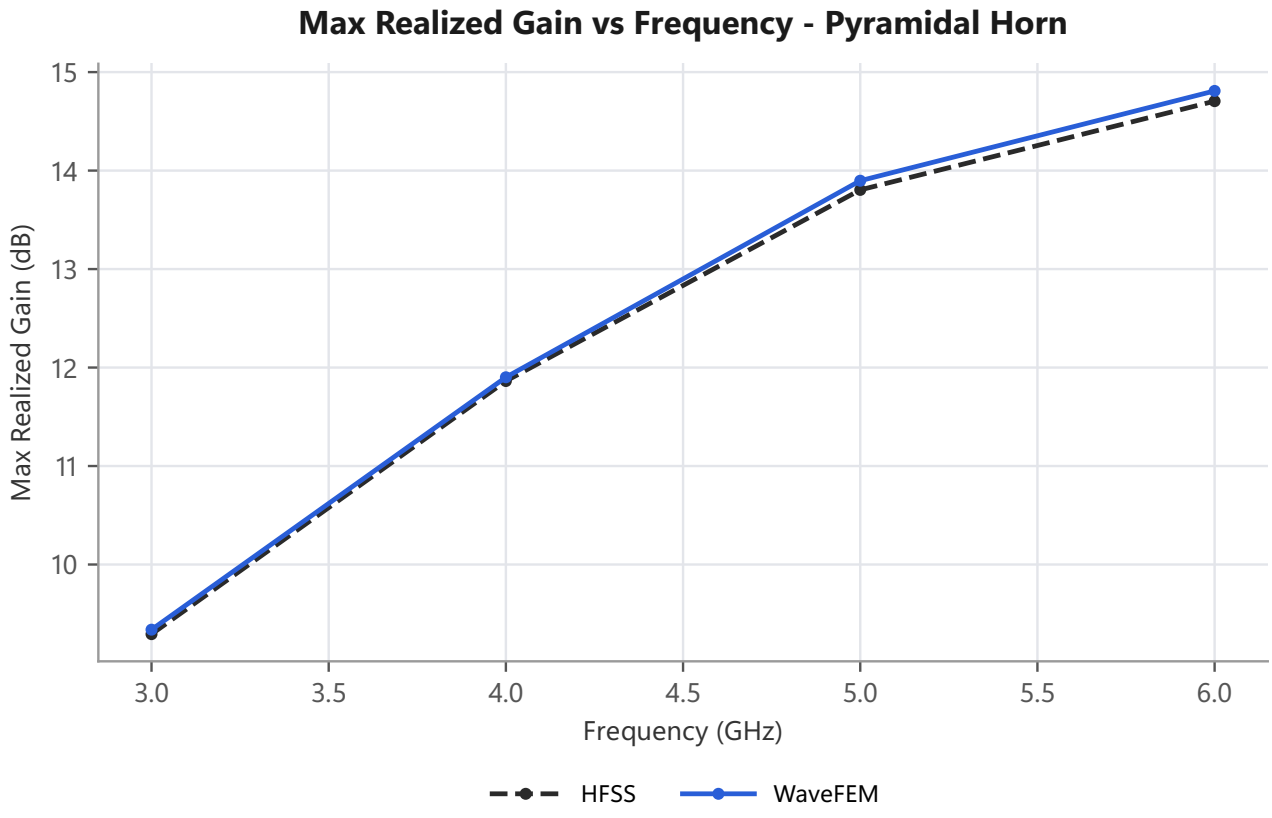


Figure 54 — Maximum realized gain against frequency.

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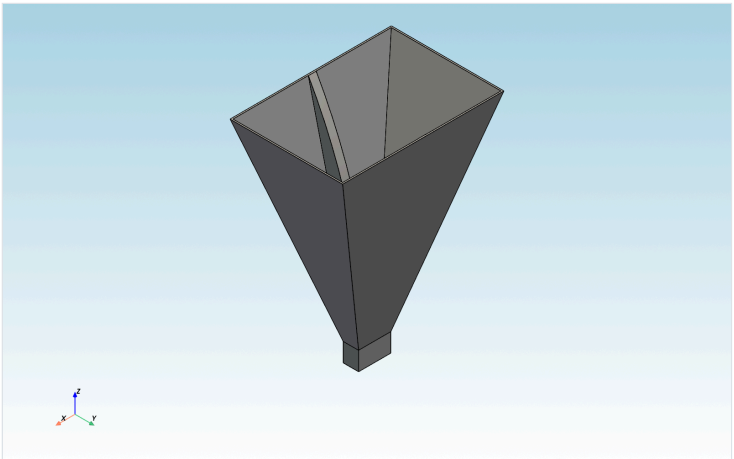
Exponential Horn

A horn with an exponential flare, fed by a 42.3 x 18.2 mm rectangular waveguide.

Simulation Setup

Refinement frequency	5 GHz
Delta S	0.02
Frequency range	4 - 5 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:11	00:01:16
Adaptive meshing	00:00:05	00:00:19
Frequency sweep	00:00:05	00:00:49
Adaptive passes	2	3
Final elements	14,197	23,138
Matrix size	89,458	145,281

Convergence

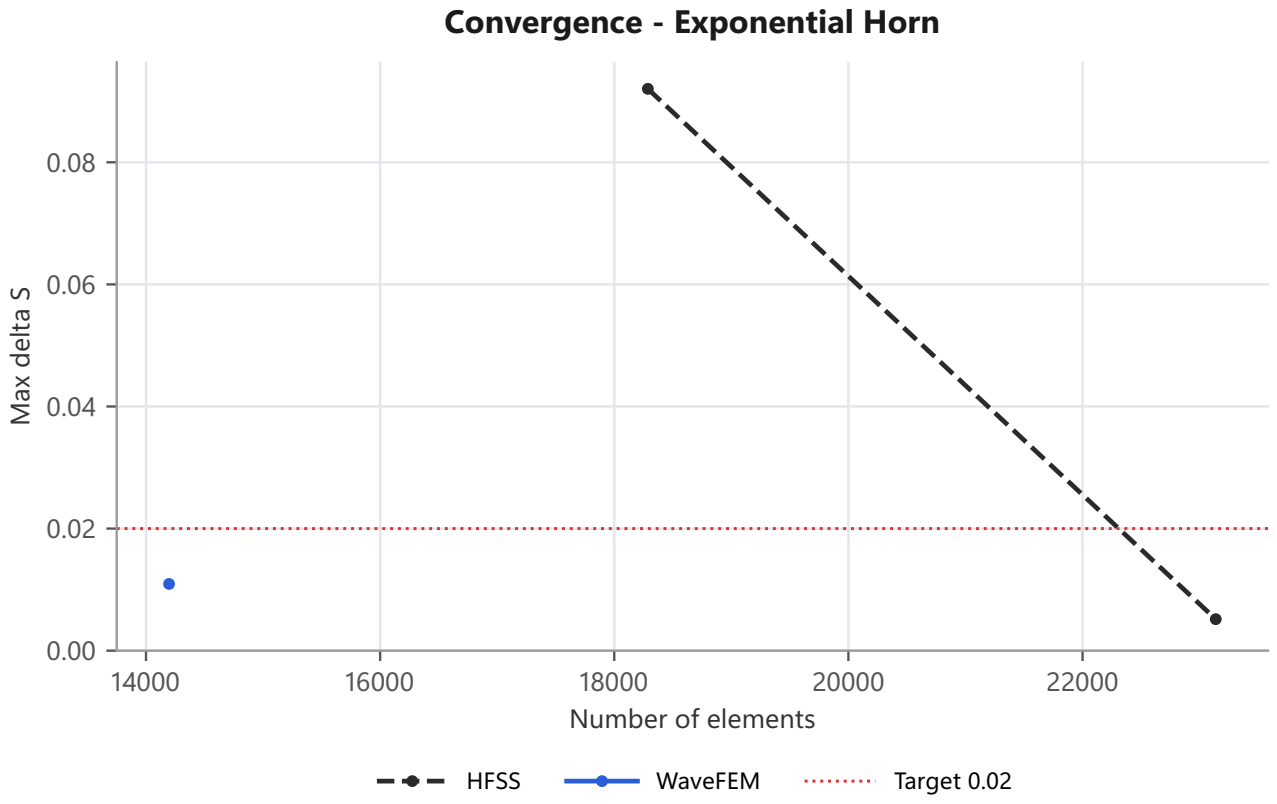


Figure 55 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

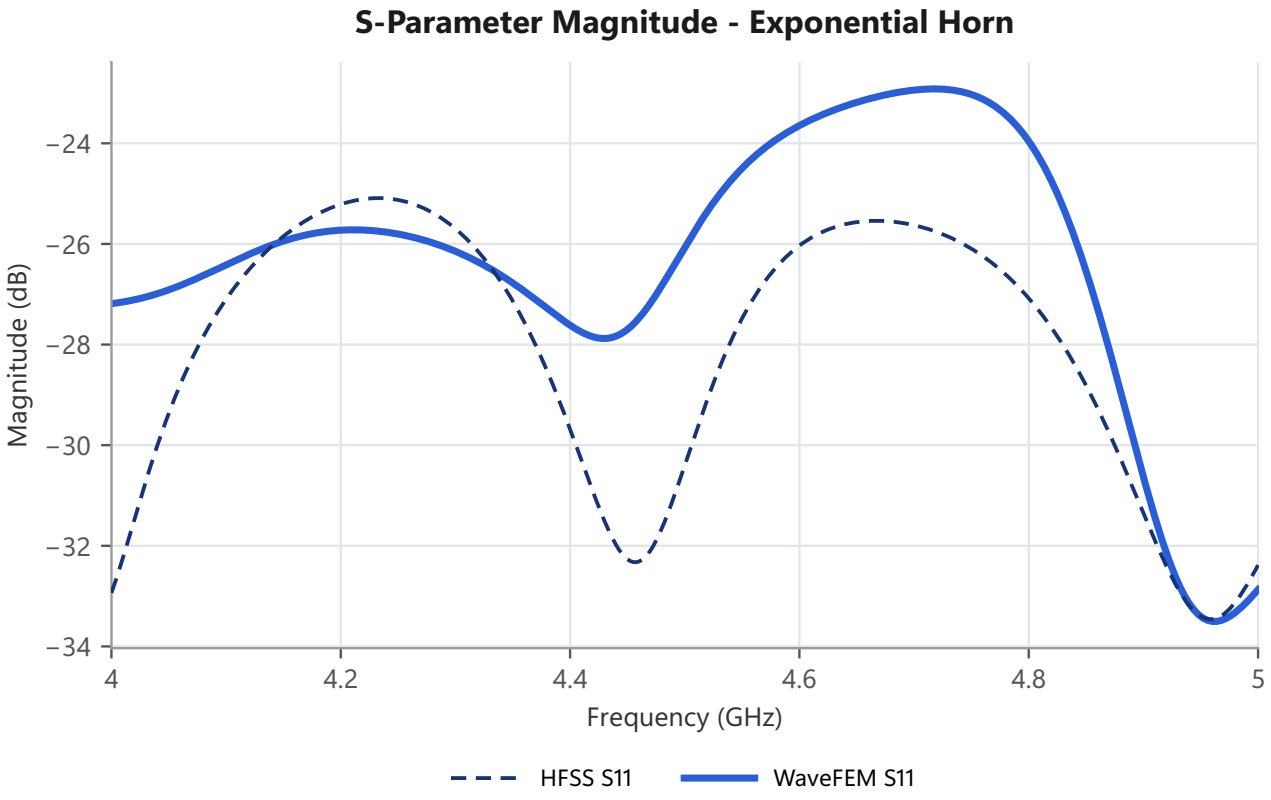


Figure 56 — S-parameter magnitude.

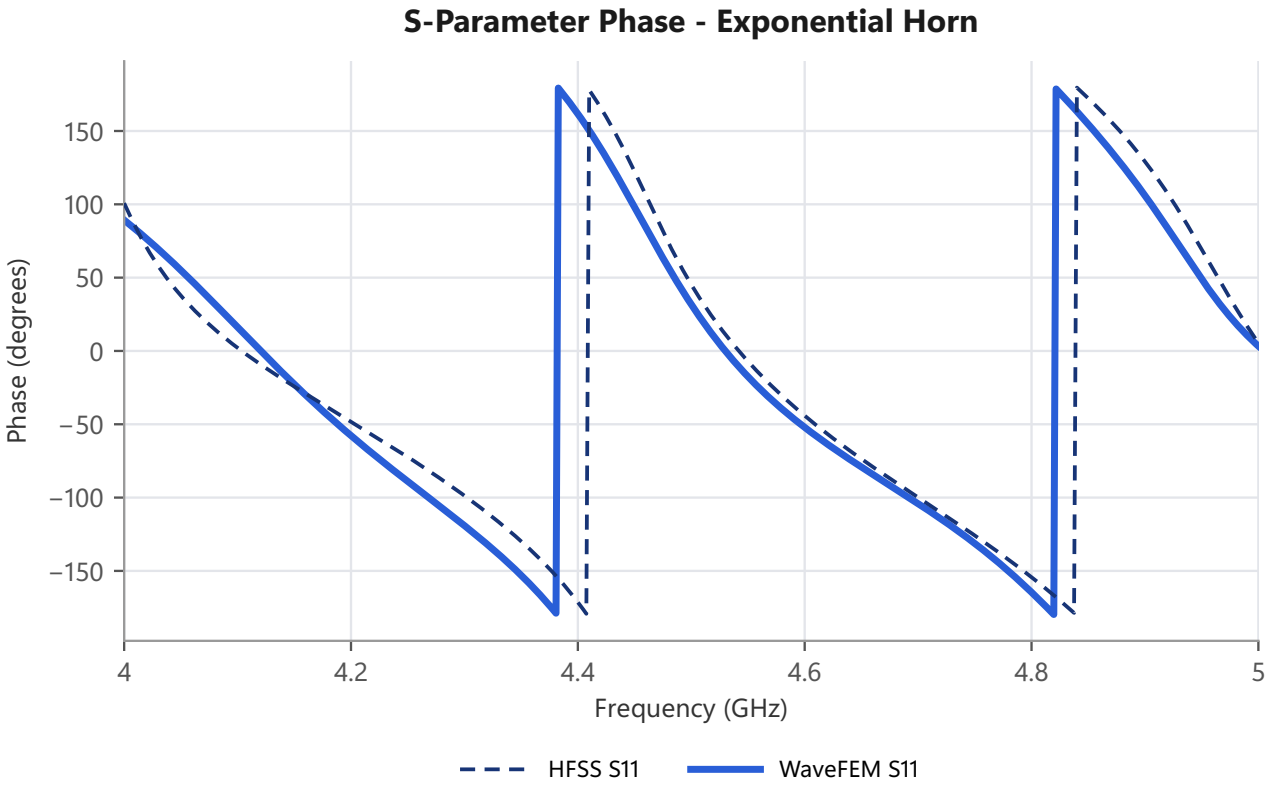


Figure 57 — S-parameter phase.

Far Field

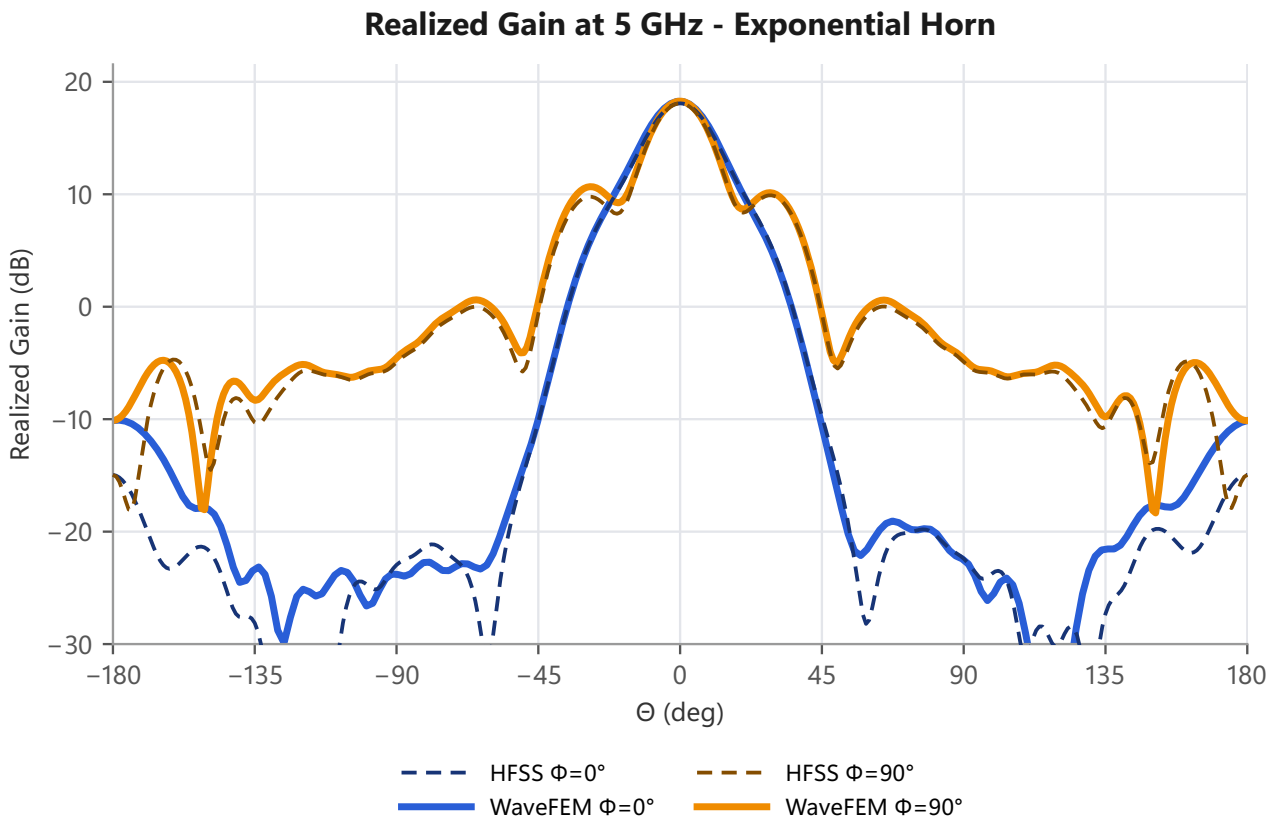


Figure 58 — Total realized gain cuts at 5 GHz.

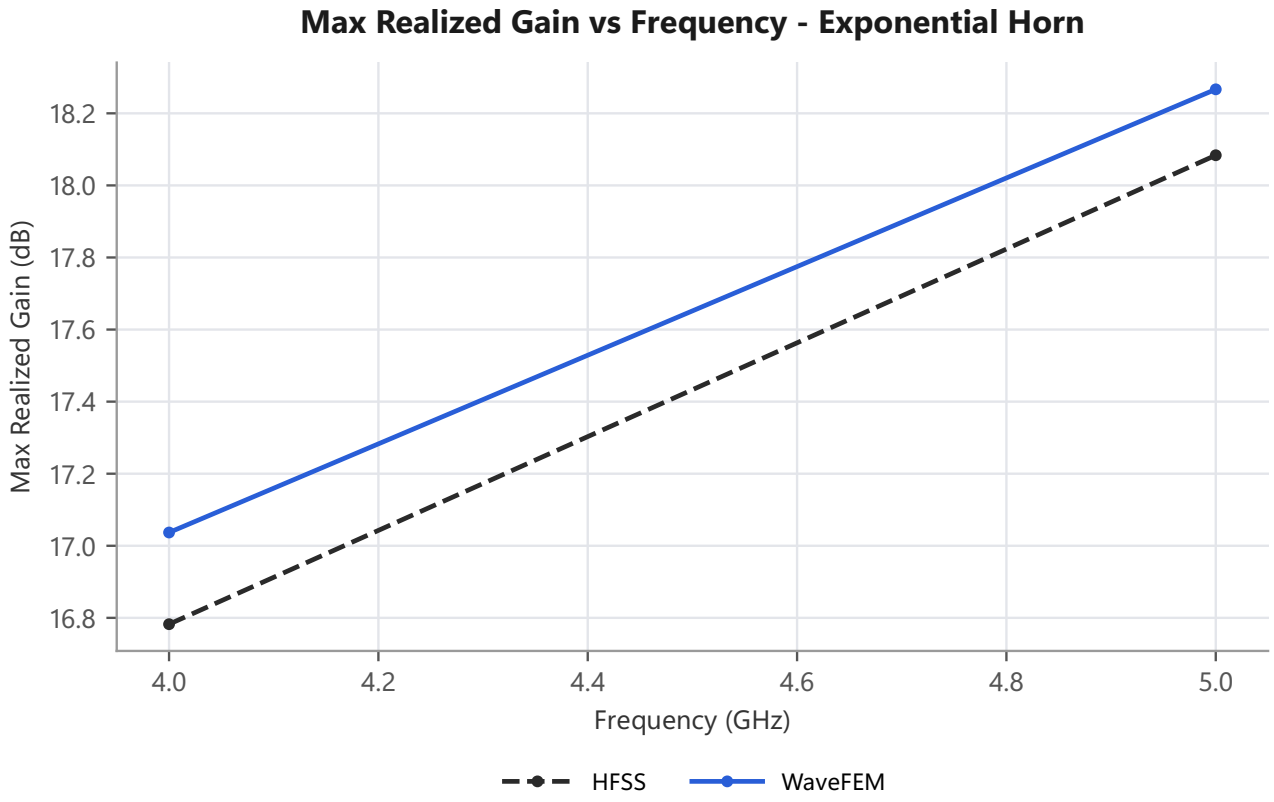


Figure 59 — Maximum realized gain against frequency.

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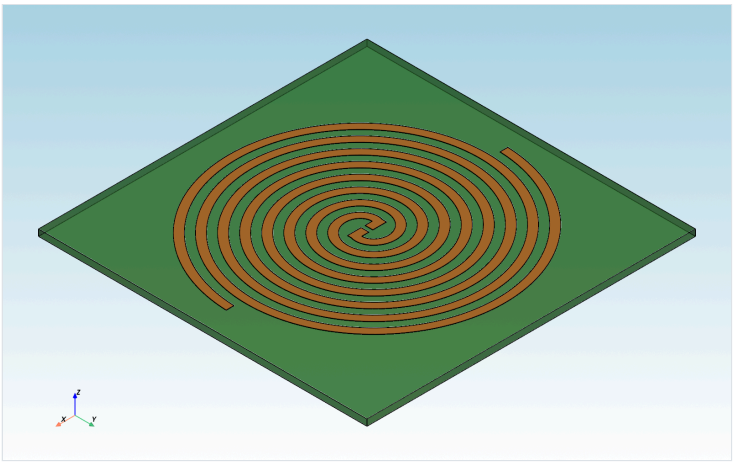
Spiral Antenna

A two-arm Archimedean spiral (4 turns, 1 mm traces) in copper on 0.813 mm RO4003C, fed by a 120 ohm lumped port at the centre gap. Wideband: S11 below -10 dB over most of 2-8 GHz.

Simulation Setup

Refinement frequency	8 GHz
Delta S	0.02
Frequency range	2 - 8 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:04:21	00:05:24
Adaptive meshing	00:00:53	00:00:38
Frequency sweep	00:03:24	00:04:31
Adaptive passes	5	4
Final elements	40,711	36,454
Matrix size	275,390	242,613

Convergence

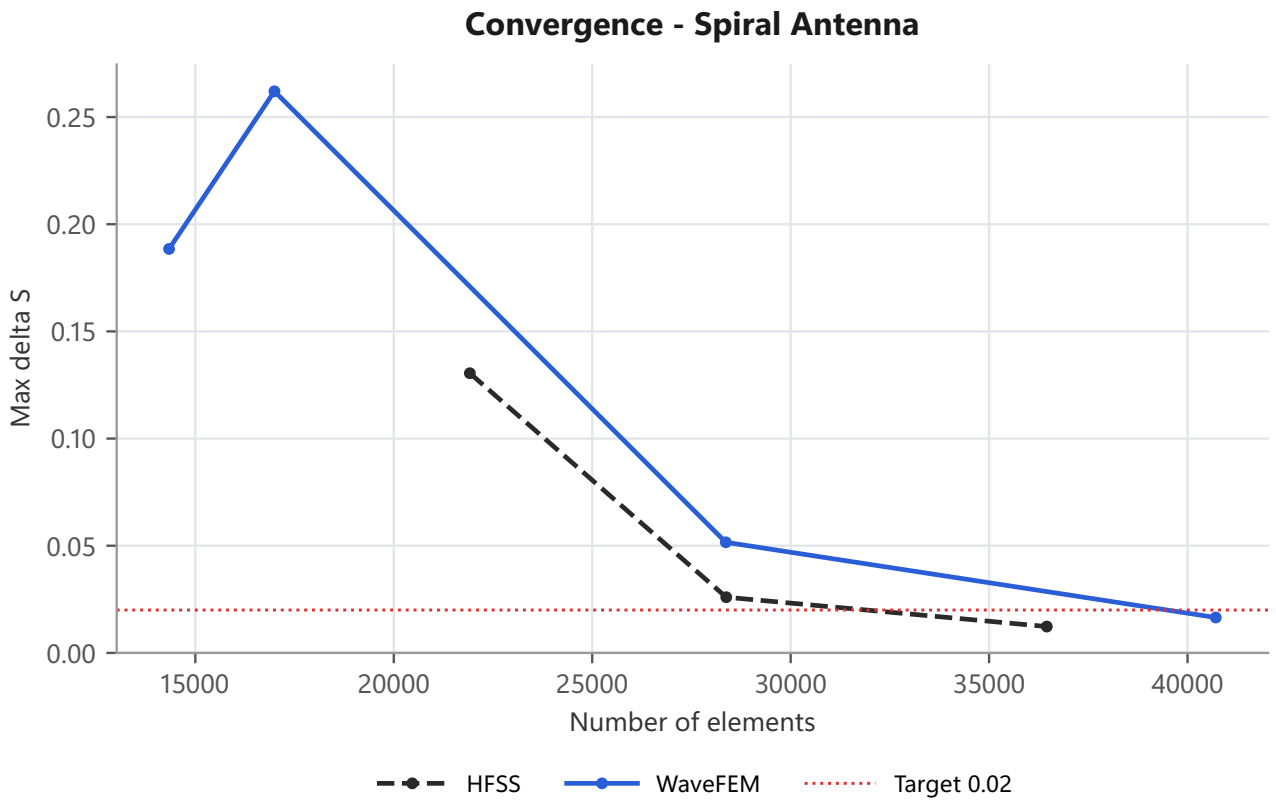


Figure 60 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

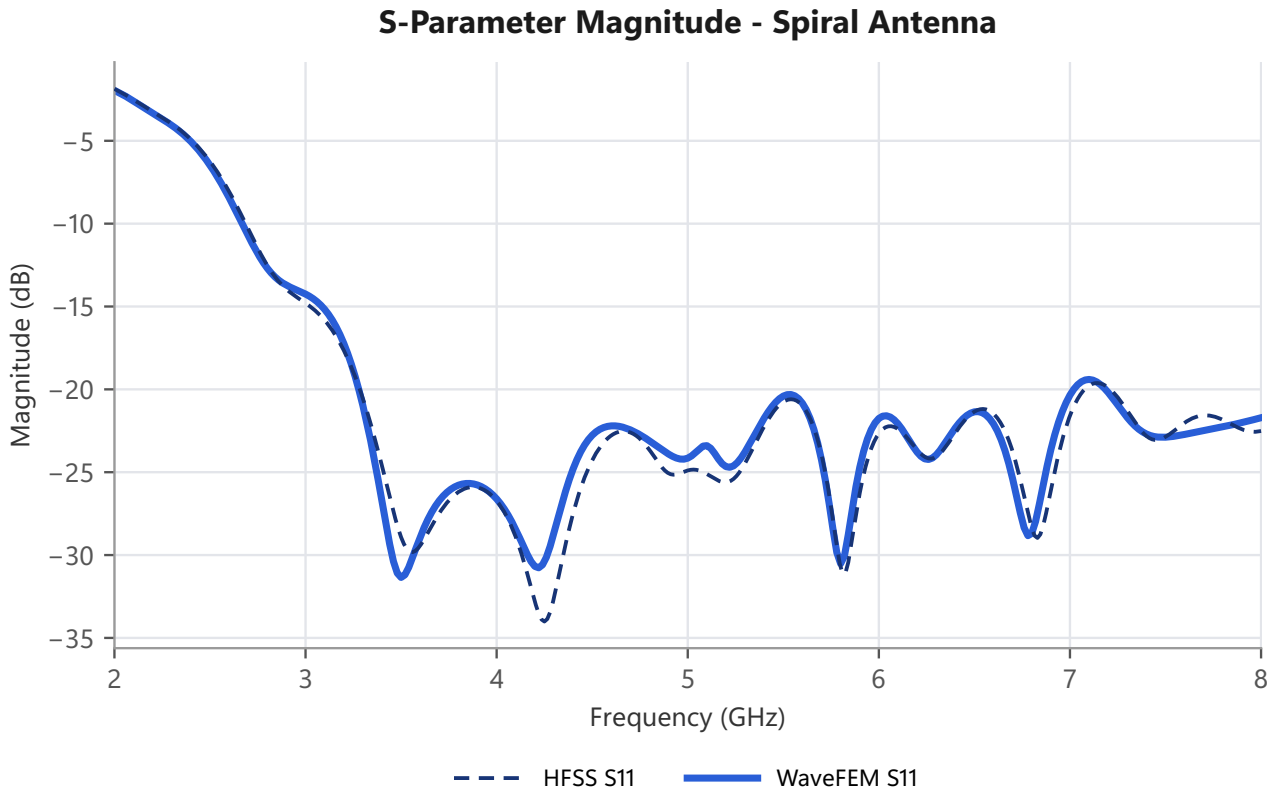


Figure 61 — S-parameter magnitude.

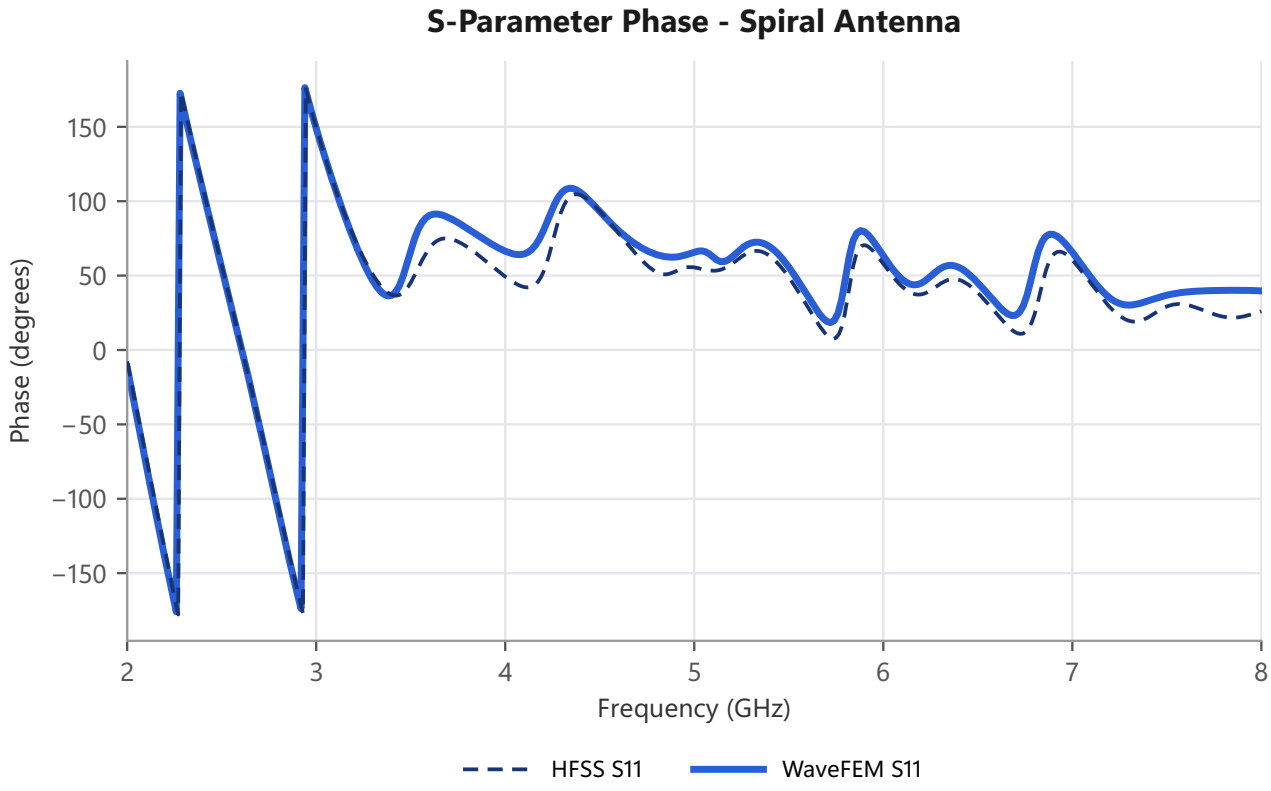


Figure 62 — S-parameter phase.

Far Field

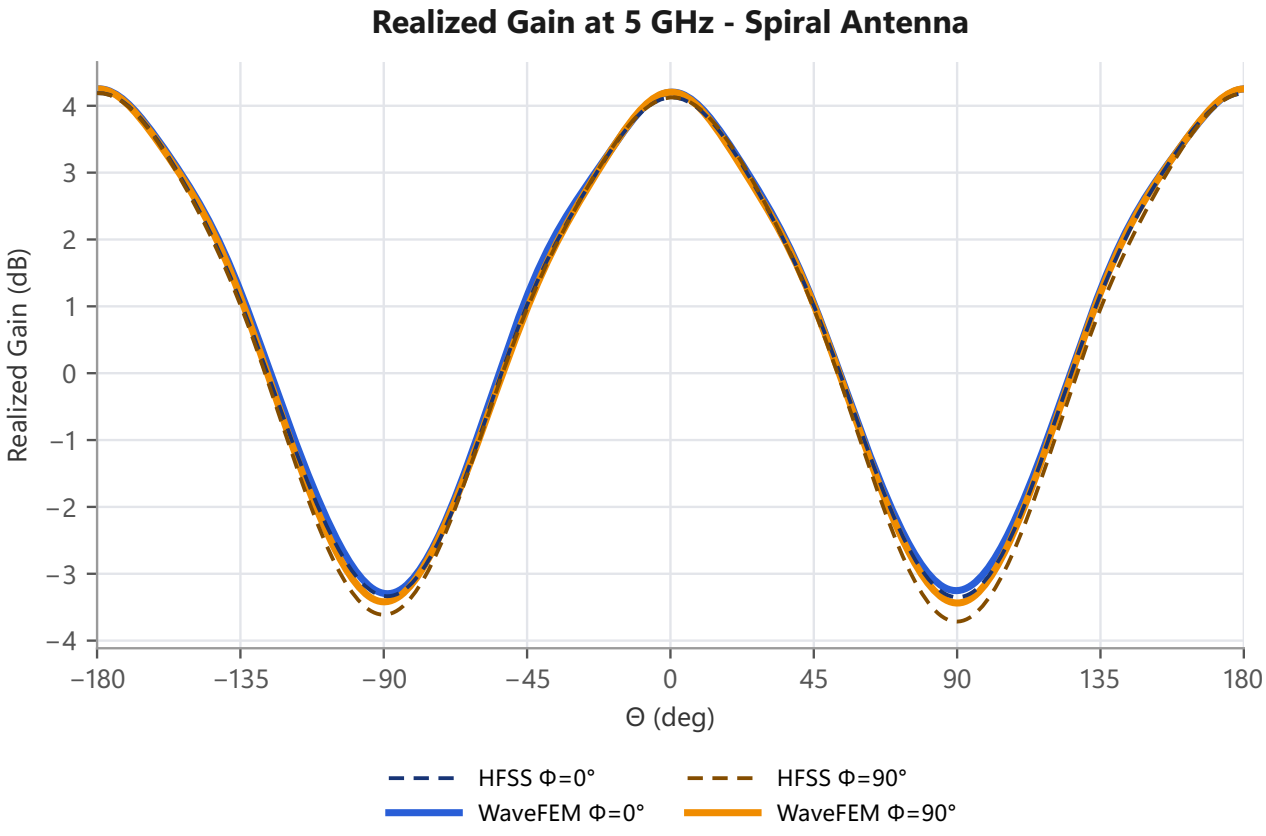


Figure 63 — Total realized gain cuts at 5 GHz.

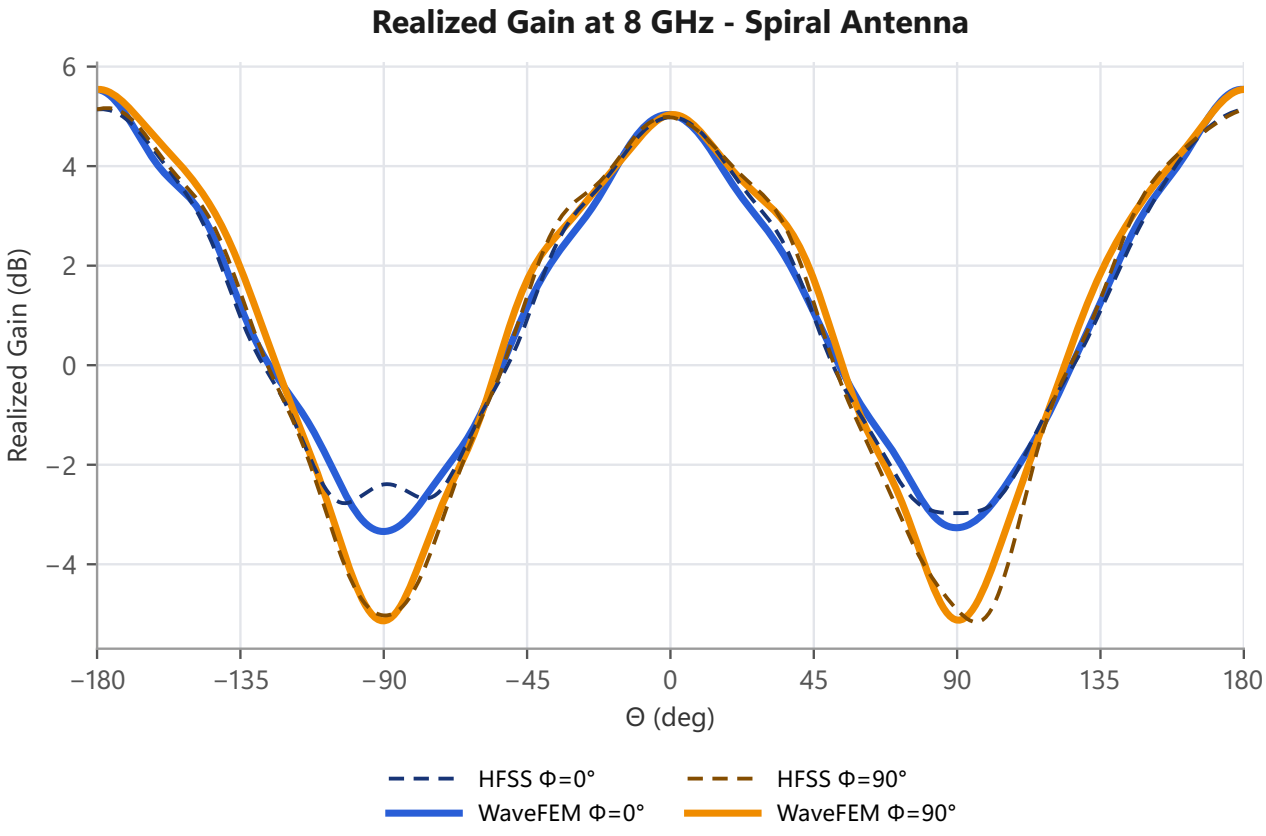


Figure 64 — Total realized gain cuts at 8 GHz.

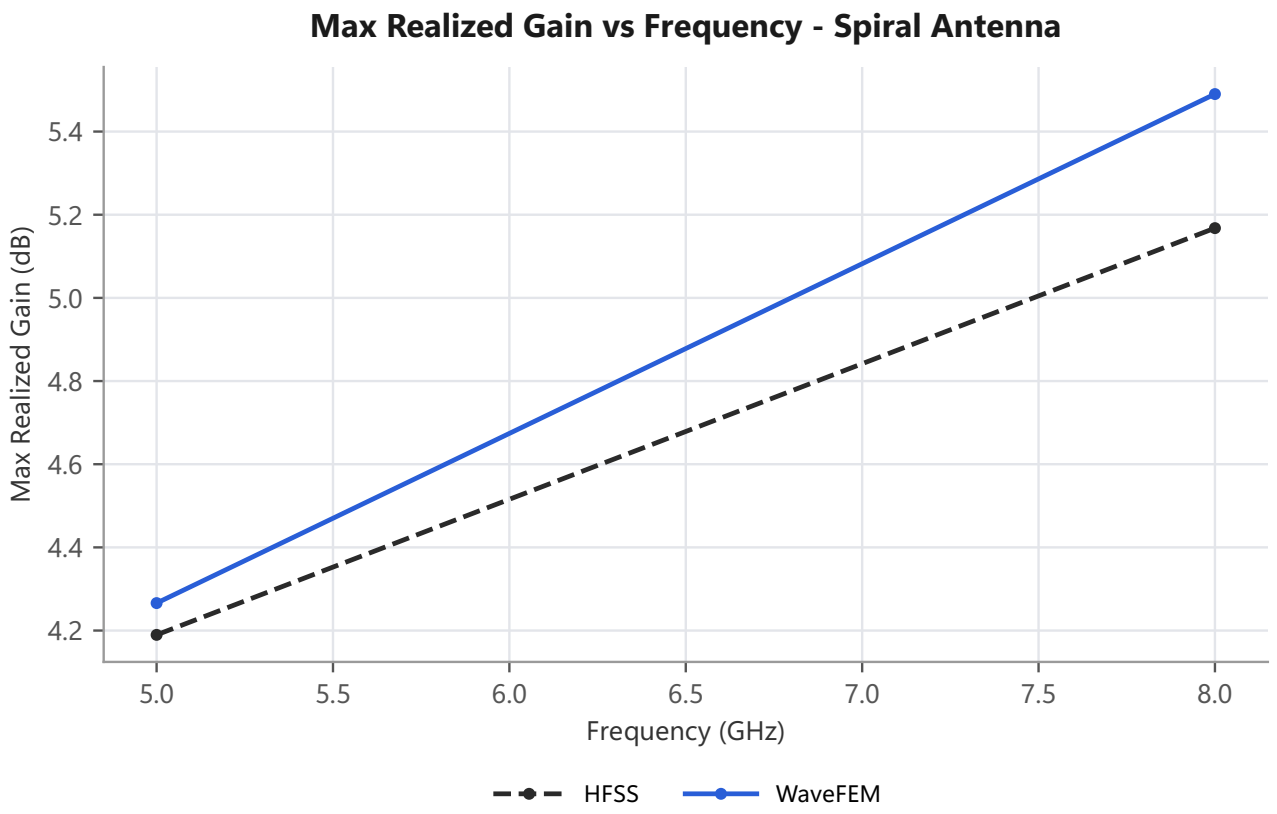


Figure 65 — Maximum realized gain against frequency.

PLANAR

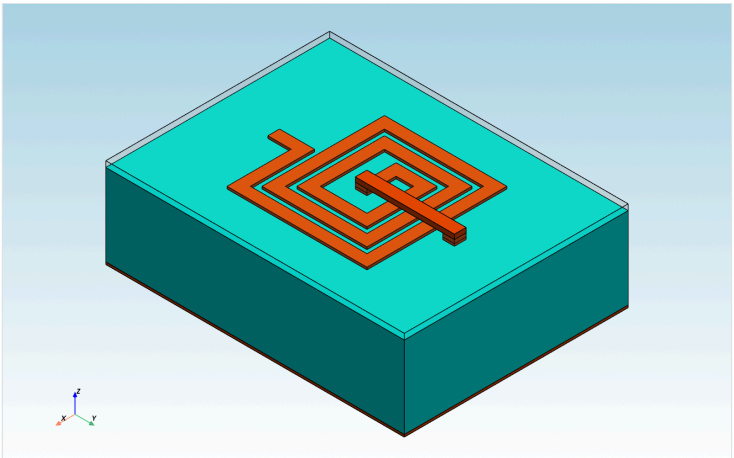
Spiral Inductor

A three-turn square spiral inductor (10 µm lines, 5 µm gaps) with an air bridge on a second metal layer, on a 70 µm substrate over ground.

Simulation Setup

Refinement frequency	70 GHz
Delta S	0.02
Frequency range	1 - 70 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:16	00:00:49
Adaptive meshing	00:00:09	00:00:26
Frequency sweep	00:00:05	00:00:21
Adaptive passes	7	10
Final elements	8,423	8,894
Matrix size	58,470	58,410

Convergence

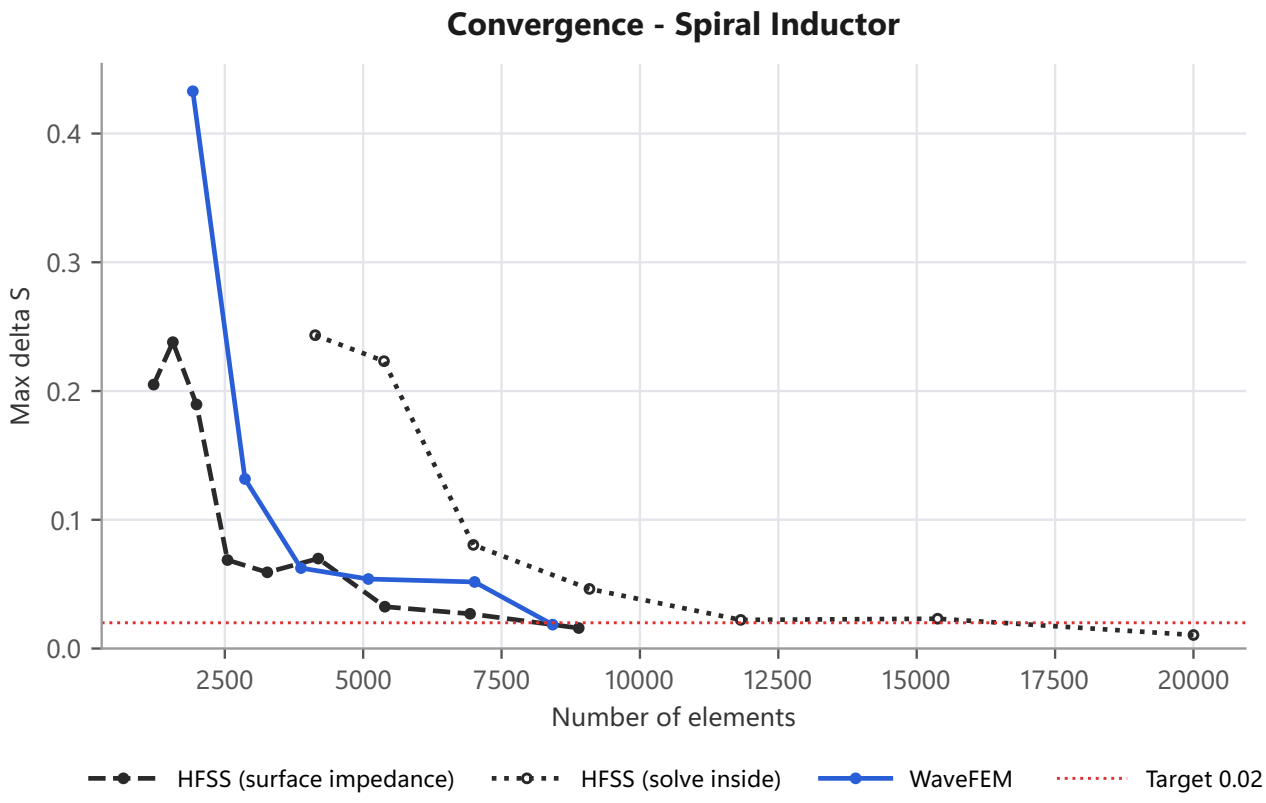


Figure 66 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

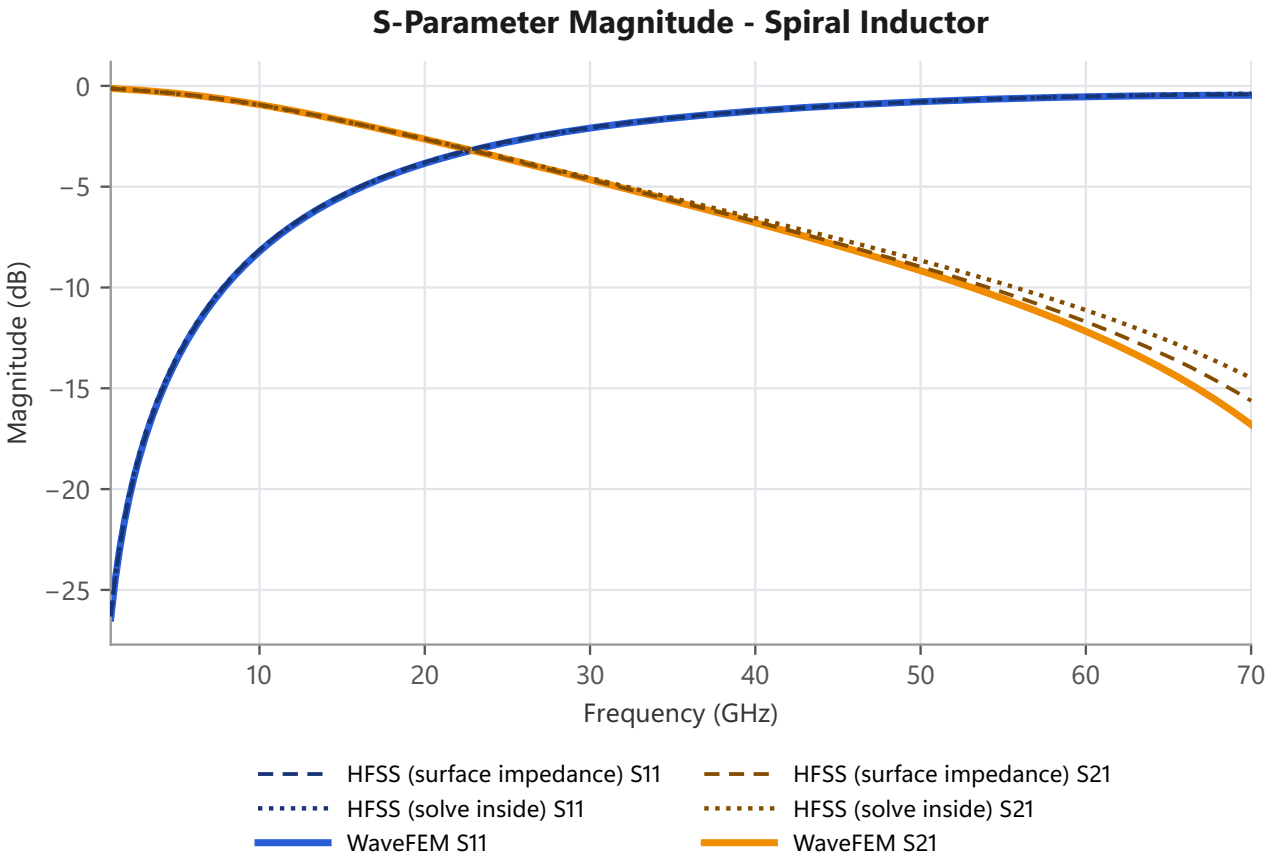


Figure 67 — S-parameter magnitude.

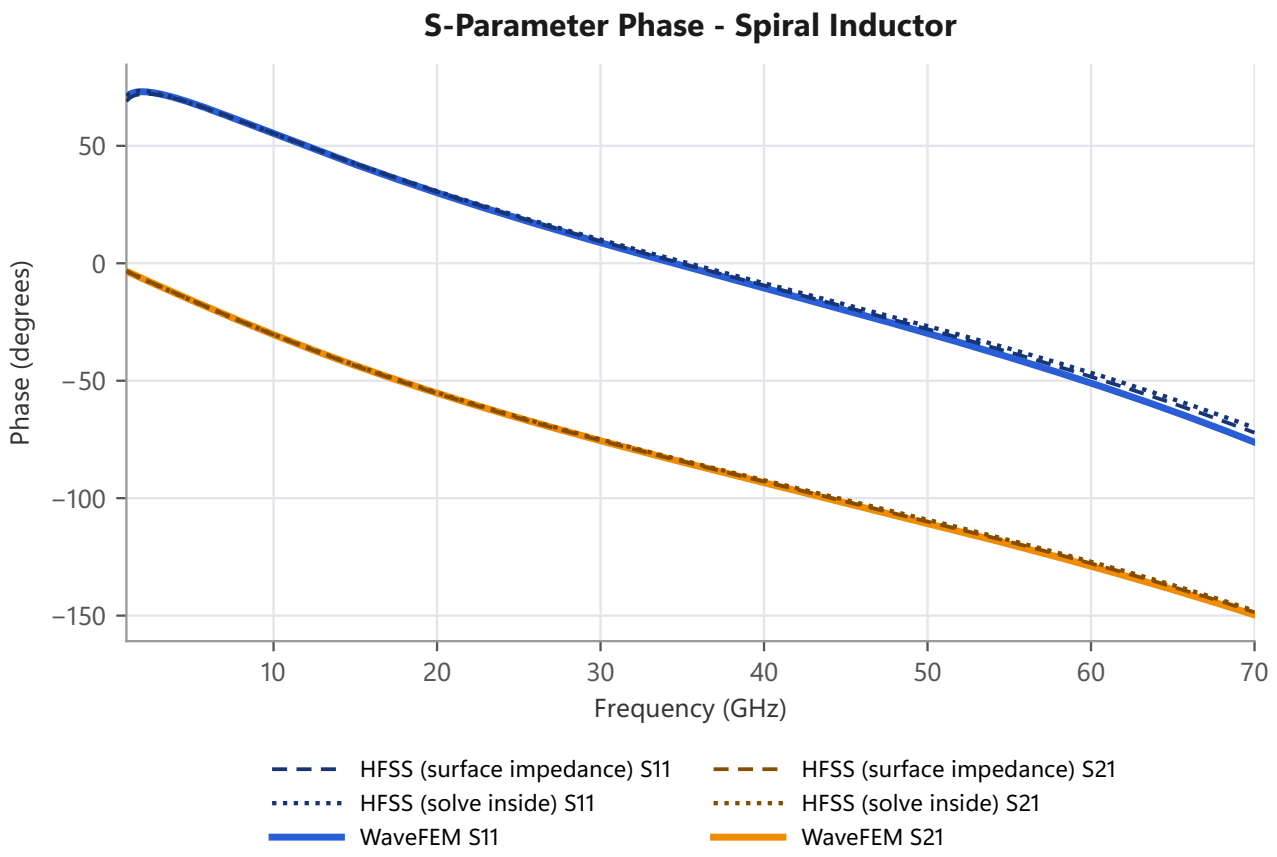


Figure 68 — S-parameter phase.

Inductance and Q

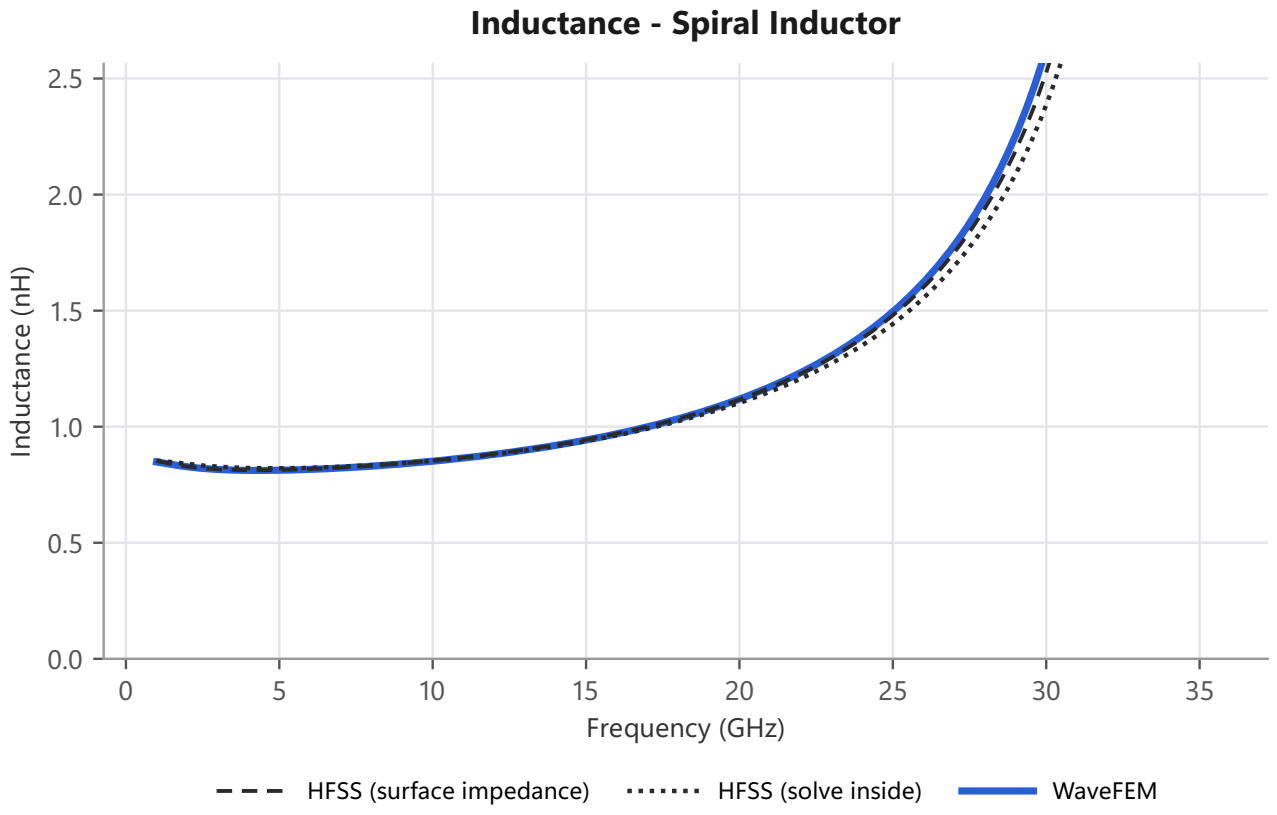


Figure 69 — Inductance Im(Z) / ω, Z = 1 / Y11 (port 2 grounded).

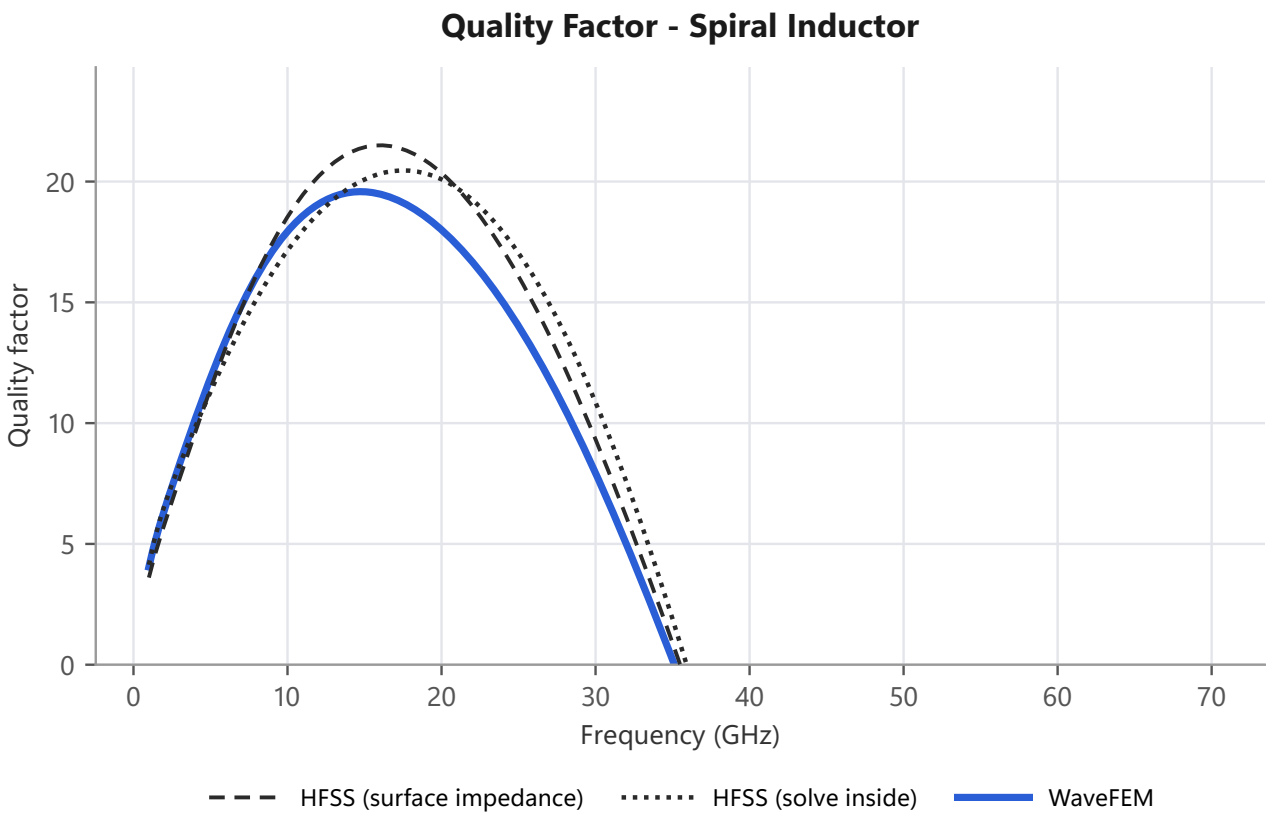


Figure 70 — Quality factor Im(Z) / Re(Z), Z = 1 / Y11 (port 2 grounded). It falls to zero at the self-resonance.

TRANSMISSION LINE

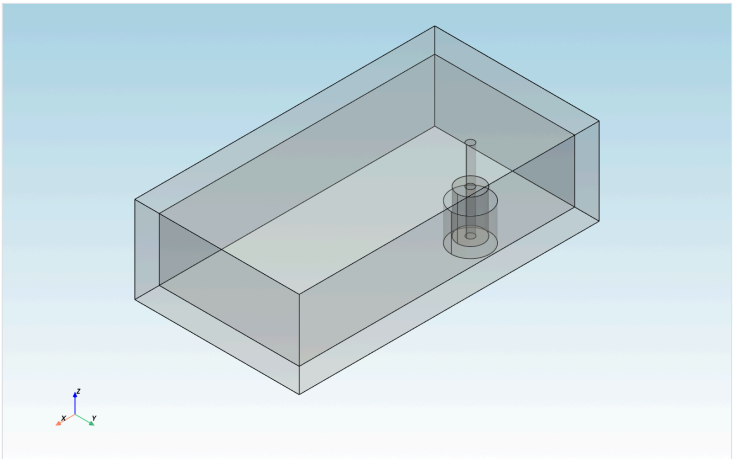
Coax to Waveguide Adapter

A transition between a coaxial cable and a rectangular waveguide, optimised for 9-10 GHz

Simulation Setup

Refinement frequency	10 GHz
Delta S	0.02
Frequency range	8 - 13 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:05	00:00:38
Adaptive meshing	00:00:02	00:00:15
Frequency sweep	00:00:02	00:00:19
Adaptive passes	4	6
Final elements	3,486	3,732
Matrix size	19,308	21,270

Convergence

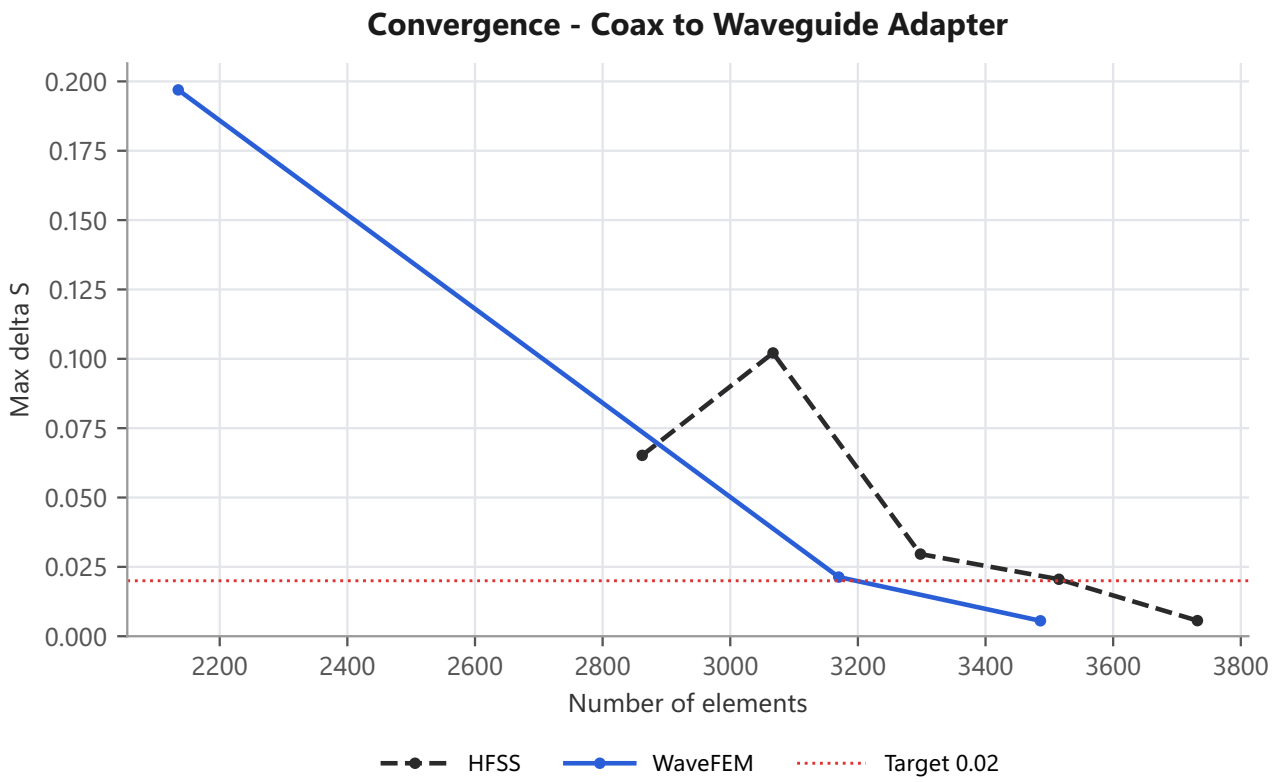


Figure 71 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

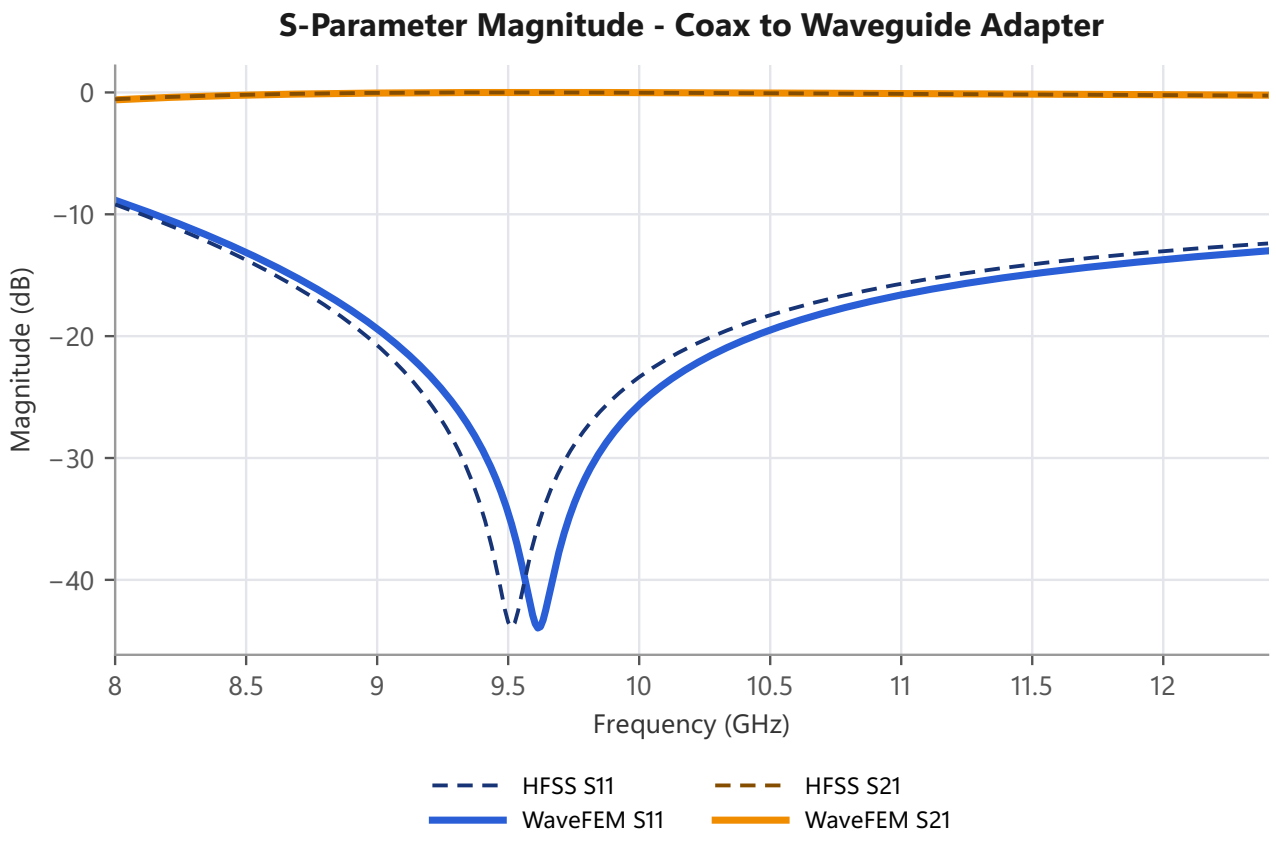


Figure 72 — S-parameter magnitude.

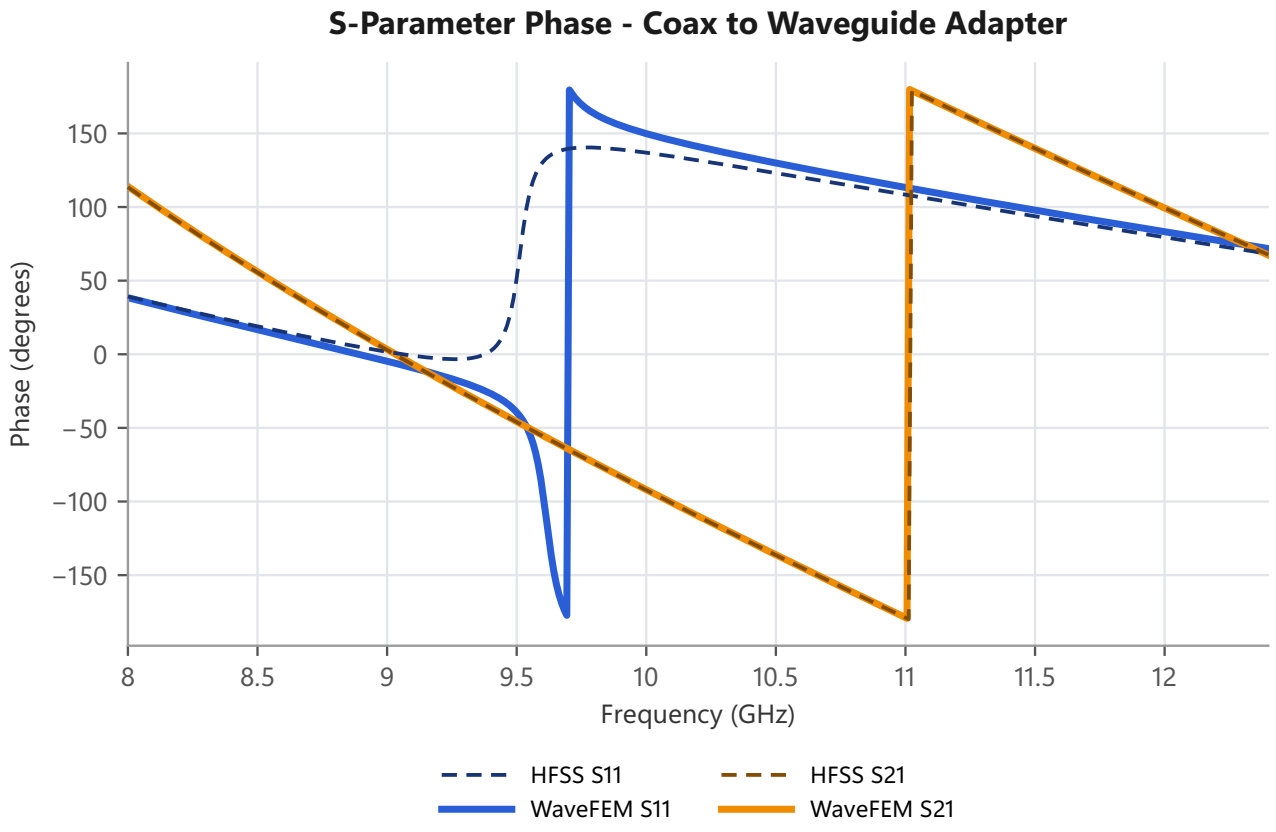


Figure 73 — S-parameter phase.

Last updated 2026-11-02

TRANSMISSION LINE

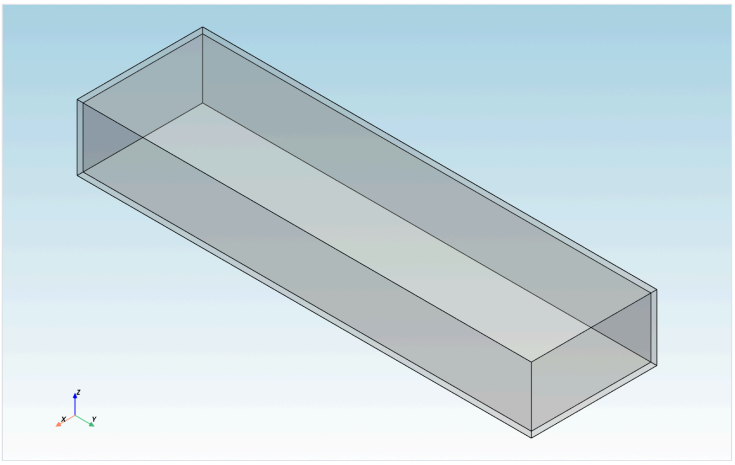
Waveguide

A straight WR159 waveguide, 150mm long

Simulation Setup

Refinement frequency	10 GHz
Delta S	0.02
Frequency range	2 - 10 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:10	00:00:45
Adaptive meshing	00:00:02	00:00:07
Frequency sweep	00:00:06	00:00:34
Adaptive passes	3	3
Final elements	4,909	5,915
Matrix size	34,928	41,448

Convergence

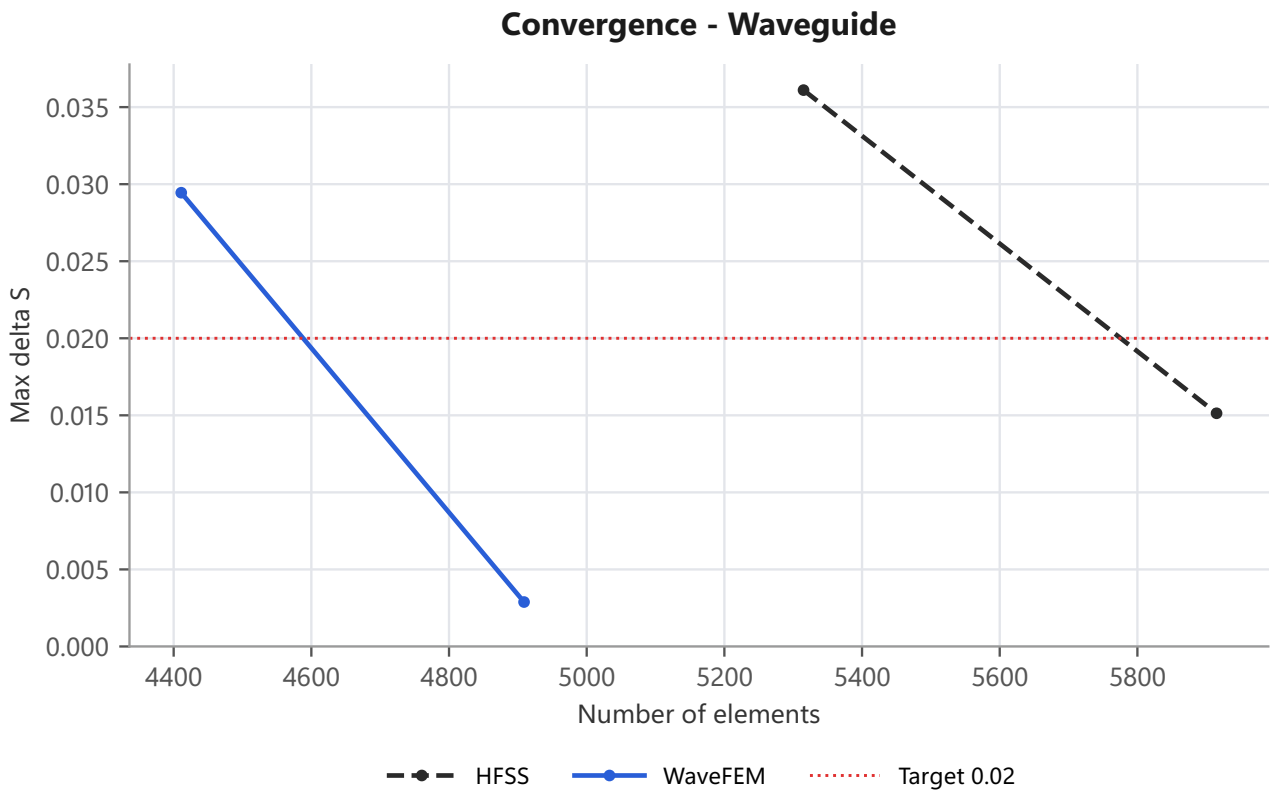


Figure 74 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

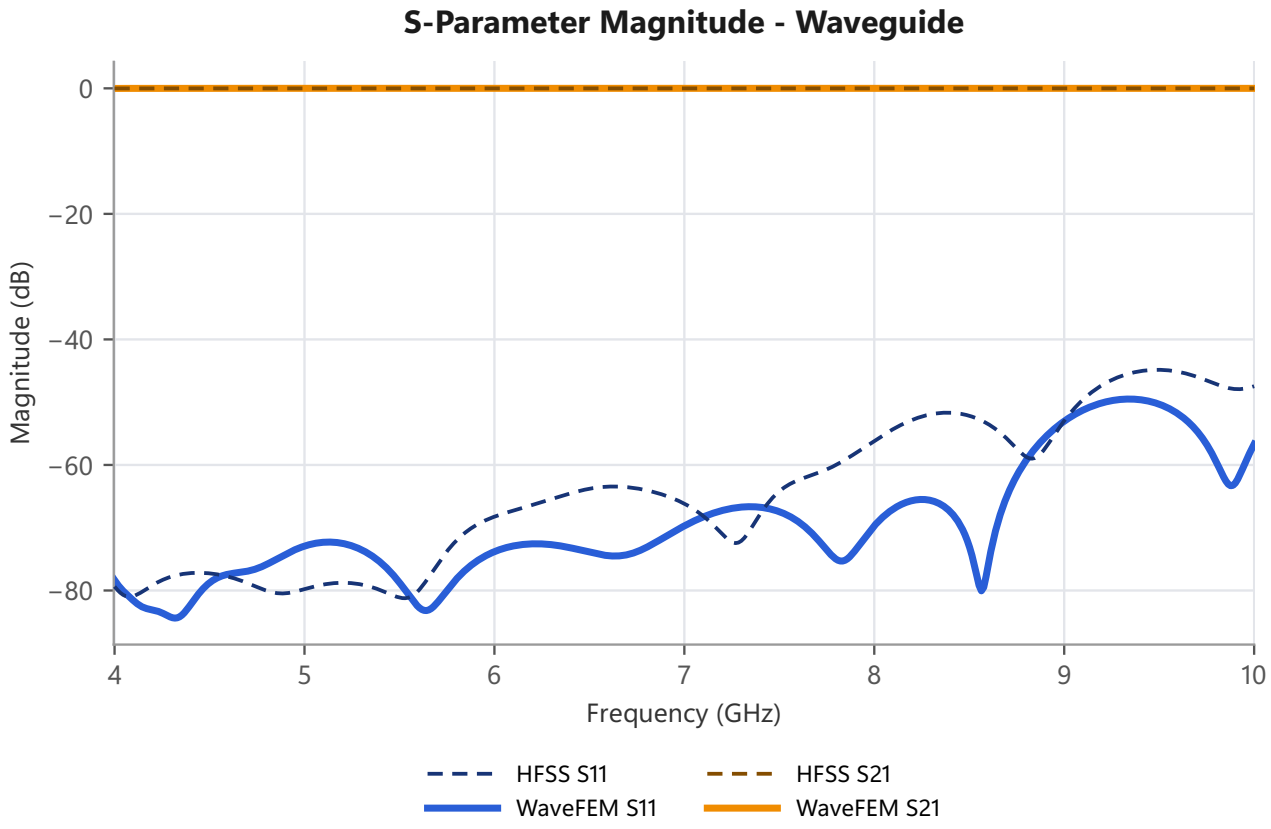


Figure 75 — S-parameter magnitude.

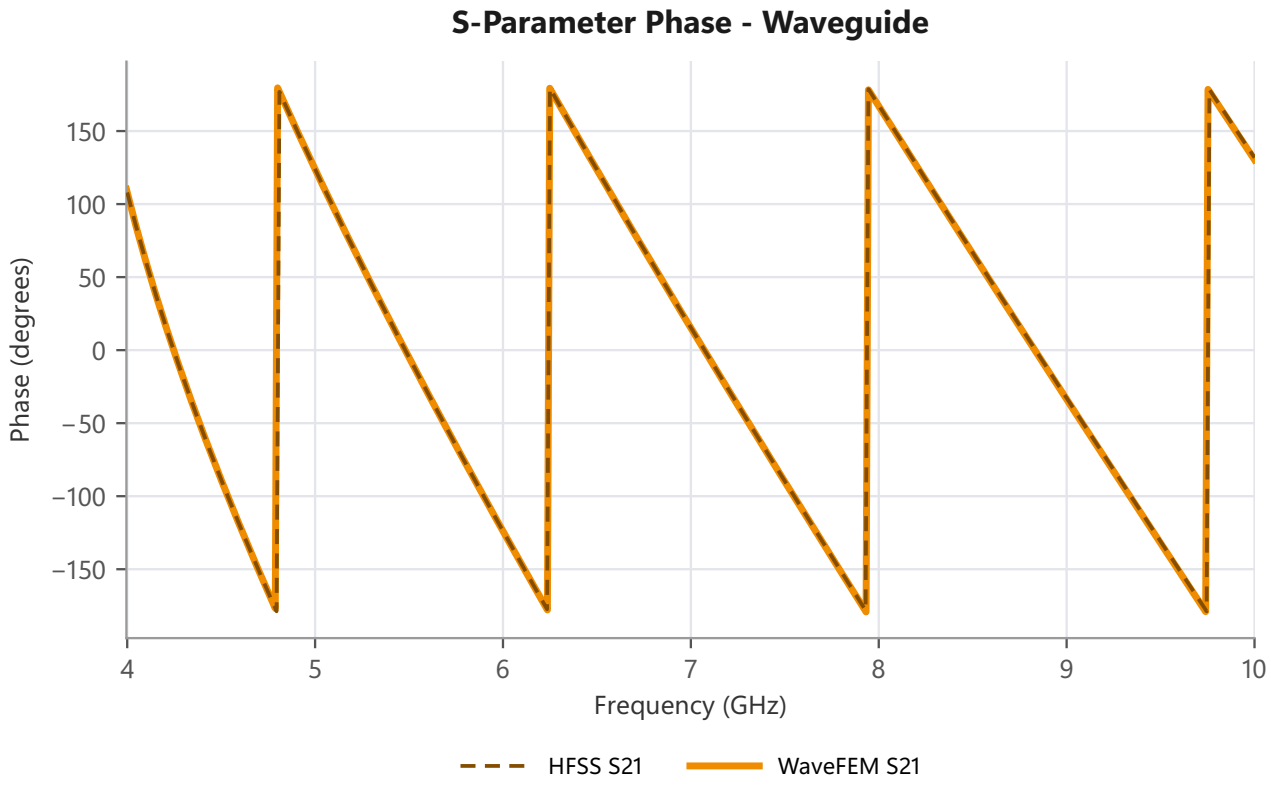


Figure 76 — S-parameter phase.

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